

Inflation and Innovation*

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Abstract

Innovation raises productivity, yet it can also raise inflation when markets are incomplete. Exploiting state-level R&D tax credits as an instrument, we find that innovation raises local inflation, primarily in non-tradables. We rationalize this in a multi-region model where displative innovation shocks reallocate output among agents; the concentrated gains from innovation raise local prices even as productivity rises. Non-innovative households bear the costs of innovation in the form of higher local prices. The calibrated model implies that an innovation shock raising output by 1 percent reduces welfare by approximately 7 percent. The model generates a realistic equity and value premium with modest risk aversion. Since local growth stocks hedge local inflation, households are willing to tilt their portfolios toward local growth stocks despite their low average returns.

*We thank the editor (Anna Pavlova); two anonymous referees; and seminar participants at MIT Sloan and the University of Michigan for helpful comments and suggestions.

Introduction

Over the last four decades, inflation in the United States has exhibited substantial heterogeneity across regions. For instance, Massachusetts has experienced average inflation of 3.6% per year, compared with 2.8% in Illinois. What explains such persistent regional differences in inflation? One possibility is differences in productivity growth. But this only deepens the puzzle: as Figure 1 shows, Massachusetts also experienced faster productivity growth (1.7% per year) than Illinois (1.2%), even though standard logic suggests that faster productivity growth should put downward pressure on local prices. Another possibility is regional demand shocks tied to the business cycle, which can raise both inflation and output in the presence of nominal rigidities. Yet controlling for local measures of economic slack, such as unemployment, does little to alter the positive correlation between inflation and productivity, echoing the findings in [Hazell, Herreño, Nakamura, and Steinsson \(2022\)](#).

We propose an explanation for this positive relation between inflation and productivity that builds on the insight of [Balassa \(1964\)](#) and [Samuelson \(1964\)](#): local innovation raises local wealth, which increases demand for inelastically supplied non-tradable goods and pushes up their price. In the classic Balassa-Samuelson framework, this wealth effect operates through wages: productivity gains in the tradable sector bid up wages economy-wide, raising the cost of non-tradables. As Figure 2 shows, the data are consistent with this broad logic: regions with more innovation also experience a higher cost of living. This raises three questions.

First, is the link between innovation and inflation causal? Using changes in state-level R&D tax credits as a plausibly exogenous source of variation in local innovation, we find that innovation raises local inflation, especially for non-tradable goods. Second, through what channel does innovation raise local wealth? Not through broad-based wage gains: innovation does not raise wages generally, and median wages, if anything, decline. Instead, the gains are concentrated among top earners, pointing to imperfect risk sharing both across and within regions. Third, what are the welfare consequences once these concentrated benefits and diffuse costs are taken into account? To answer this question, we calibrate a multi-region model with incomplete markets using our causal estimates. In the model, an aggregate innovation shock that raises output by 1 percent reduces welfare by roughly 7 percent in consumption-equivalent terms.

We begin by estimating the impact of innovation on the cost of living. Identifying this relation entails two challenges. First, the inflationary effects of innovation are likely to be confounded by monetary policy responses and aggregate demand shocks. Hence, we exploit cross-state variation in relative inflation within the United States. Different U.S. states share a common monetary policy but exhibit significant heterogeneity in innovation intensity. Second, both the local level of innovation and the local price level are likely to be endogenous to local economic conditions—causality can also run from local price appreciation to entrepreneurship ([Fan, Liu, Hu, and Zang, 2023](#); [Kerr, Kerr, and Nanda, 2022](#)). Hence, we use the change in state level R&D tax credit policy

as a plausibly exogenous source of variation in local innovation. Using these tax credit changes, we first confirm that they positively impact local innovation measures based on patents. Our innovation measures recognize that not all patents are equally valuable; we construct innovation indicators that weigh patents based on their forward citations (Hall, Jaffe, and Trajtenberg, 2005); the stock market reaction of innovating firms (Kogan, Papanikolaou, Seru, and Stoffman, 2017); or their novelty and impact based on their textual similarity to prior and subsequent patents (Kelly, Papanikolaou, Seru, and Taddy, 2021).

Our reduced-form estimates indicate that a one-standard-deviation increase in R&D tax credits leads, over the following five years, to a 1.3 percentage point increase in local inflation and a 0.6 percentage point increase in labor productivity growth. These price increases are concentrated in, though not limited to, non-tradable goods such as housing and services, and are robust to alternative measures of inflation and to varying scales of geographical aggregation. We obtain qualitatively similar results using data from EU member countries, and we show that local IPO activity—a proxy for the realization of illiquid innovation wealth—is also positively associated with local price levels.

Turning to the channels through which innovation raises demand for local goods, we find that innovation does not raise wages broadly—median wages, if anything, decline. Tracking individual households, we find no evidence that average or median earnings growth increases either, ruling out composition effects. Innovation weakly raises skilled wages and has some effect on average wages once equity compensation is taken into account, but the dominant effect is on top income inequality: a one-standard-deviation increase in R&D tax credits leads to a 1.8 to 1.9 percent increase in the top 1% income share alongside a 2.7 percentage point *decline* in median wages—the opposite of the broad-based wage gains predicted by the standard wage channel.

We also rule out two alternative explanations for the link between innovation and local prices. First, innovation-driven migration: if innovation hubs attract workers who bid up local prices, we would expect a positive migration response, but population growth is in fact negatively related to R&D tax credit increases, suggesting that rising costs of living push some households out. Second, portfolio home bias: if local households overweight local stocks (Becker, Ivkovic, and Weisbenner, 2011), local productivity gains could raise local wealth mechanically, but returns to local incumbent firms are weakly *negatively* correlated with local inflation, which is inconsistent with this channel. By contrast, returns on a local growth-minus-value portfolio are *positively* correlated with local inflation, consistent with the model of Kogan, Papanikolaou, and Stoffman (2020) in which growth firms provide a hedge against innovation-driven displacement.

We formalize the mechanism of concentrated benefits but diffuse costs of innovation in a multi-region model with incomplete markets. Firms in each region produce tradable and non-tradable goods and are subject to two types of productivity shocks: neutral shocks that raise output for all firms, and displacive innovation shocks that reallocate output from incumbent

to new firms, capturing the idea of creative destruction (Aghion and Howitt, 1992; Gârleanu, Panageas, Papanikolaou, and Yu, 2016). Crucially, innovators cannot sell claims on their future projects, so these displacive shocks redistribute wealth toward a small set of local households. This concentration of wealth raises local demand for non-tradables and hence local prices—even as aggregate output rises. The calibrated model matches the joint dynamics of local inflation, consumption, and inequality as well as the equity and value premiums. It also predicts that households tilt toward local growth stocks to hedge displacement risk, rationalizing the home bias documented by Coval and Moskowitz (1999)—a pattern consistent with the empirical evidence above.

Our paper contributes to several literatures. First, we provide causal evidence that innovation is a significant driver of regional inflation heterogeneity, contributing to the literature on the geographic determinants of inflation (Babb and Detmeister, 2017; Hooper, Mishkin, and Sufi, 2020; Fitzgerald, Jones, Kulish, and Nicolini, 2020; Beraja, Hurst, and Ospina, 2019; McLeay and Tenreyro, 2018; Hazell et al., 2022). While Hazell et al. (2022) document substantial cross-state inflation heterogeneity and a flat regional Phillips curve, we identify innovation as a causal driver of these differences and provide a mechanism—concentrated wealth under incomplete markets—that explains why more innovative regions have persistently higher inflation. Beracha, He, Wintoki, and Xi (2022) documents a positive correlation between local innovation and house price growth, attributing it to innovation-driven migration; our results indicate that innovation has a causal impact on the local cost of living beyond housing. Our analysis also shows that local inequality and inflation share a common cause—displacive innovation shocks—complementing the evidence in Aghion, Akcigit, Bergeaud, Blundell, and Hemous (2018).

Second, by documenting that innovation raises both nominal income inequality and local prices, we identify a channel through which innovation amplifies *real* income inequality, contributing to the literature on the distributional consequences of innovation and trade (Jaravel, 2018; Faber and Fally, 2021; Fajgelbaum, Grossman, and Helpman, 2011; Fajgelbaum and Khandelwal, 2016; Fajgelbaum and Gaubert, 2020; Dingel, 2016; Couture, Gaubert, Handbury, and Hurst, 2019).

Third, our results provide direct empirical support for the theoretical prediction that innovation shocks are displacive and that investors demand compensation for bearing displacement risk. The idea that households may want to hedge against high-innovation states is not new (Papanikolaou, 2011; Gârleanu, Kogan, and Panageas, 2012; Kogan et al., 2020). However, these papers are largely silent on the *diffuse costs* of innovation: in Kogan et al. (2020), for instance, non-innovating households do not experience any declines in real consumption; their utility declines due to a keeping-up-with-the-Joneses effect. Our contribution is to document these costs empirically—showing that innovation raises the local cost of living without broad-based wage gains—and to quantify the resulting welfare loss. By contrast, in our model, non-innovating households experience declines in real consumption as innovators increase their demand for non-tradable

goods and hence their price, similar to [DeMarzo, Kaniel, and Kremer \(2004, 2007\)](#). This real channel is quantitatively powerful: the calibrated model generates a value premium of 4.6% (data: 3.5%) with $\gamma = 6$ and no relative consumption preferences, whereas [Kogan et al. \(2020\)](#) require $\gamma \approx 57$ and an 80% weight on relative consumption to achieve comparable magnitudes.

Finally, we offer a hedging-based rationale for regional portfolio home bias: investors tilt toward local non-tradable and growth firms because these firms help hedge local inflation risk, a prediction that is consistent with the patterns in [Coval and Moskowitz \(1999\)](#) but that information-based explanations ([Huberman, 2015](#); [Ivkovic and Weisbenner, 2005](#); [Van Nieuwerburgh and Veldkamp, 2009](#); [Grinblatt and Keloharju, 2001](#)) do not generate (see also [Korniotis and Kumar \(2013\)](#); [Pirinsky and Wang \(2006\)](#); [Garcia and Norli \(2012\)](#); [Tuzel and Zhang \(2017\)](#); [Parsons, Sabbatucci, and Titman \(2020\)](#); [John, Knyazeva, and Knyazeva \(2011\)](#) on the relation between local economic conditions and regional stock returns).

1 Inflation and the Cost of Living

This section presents the empirical analysis. We first describe the data, then document the positive correlation between innovation and local inflation and provide causal evidence using R&D tax credits as an instrumental variable. Next, we investigate the mechanisms through which innovation raises local demand—including its effects on wages, inequality, migration, and local stock returns—and rule out alternative channels. We conclude with a summary of the main empirical takeaways.

1.1 Data Construction

We perform our empirical analysis primarily at the state level within the U.S. Our sample covers the 1978 to 2017 period, constrained by the availability of local inflation data. Figure [A.3](#) shows the 32 states with inflation data available; we exclude Alaska and Hawaii due to the limited number of public firms. Below, we briefly discuss the construction of the central variables and refer the reader to Appendix [A](#) for additional details.

Inflation

Our starting point in creating state-level inflation indices is [Hazell et al. \(2022\)](#), who construct state-level price indexes for the U.S. using the data that the Bureau of Labor Statistics (BLS) collects to compute the aggregate CPI indices. One important shortcoming of this data is that it does not include the rent prices used to construct the shelter component of CPI, which accounts for roughly one-third of the expenditure weight (32.9% as of December 2021, comprising owner’s equivalent rent, rent of primary residence, and lodging away from home). Therefore, we augment the data with house price data from the Federal Housing Finance Agency (FHFA), which produces nominal state-level price indices for all states. We construct our headline inflation measure as a weighted

average of housing price appreciation and the non-housing inflation measure of [Hazell et al. \(2022\)](#), using a fixed weight on housing equal to 32.9%. This approach mirrors the BLS methodology prior to 1983, when housing-service inflation was computed as a weighted average of changes in housing prices and mortgage costs; after 1983, the BLS switched to using rent inflation—primarily owner-occupied “rental equivalence”—as a proxy. Since interest-rate changes can directly affect housing prices, our estimation strategy includes state and time-fixed effects to mitigate any resulting bias. In Section 6, we show that our results are robust to using post-1983 rent-based MSA-level CPI data.

Innovation

We use patents to measure innovation at the state-year level. We assign a patent to a given year based on its grant date and assign it to a state based on the location of the assignee(s) listed on the patent. To associate a patent with a firm (assignee), we use the mapping from patents to Compustat firms provided by [Kogan et al. \(2017\)](#). We then combine this information with the firm’s headquarters information from COMPUSTAT to link patents to states. Importantly, we also allow for innovation to spill over to neighboring states. Specifically, we apply the innovation impact of a patent assigned to state i to a collection of states, including state i itself and its neighboring states (i.e., those sharing a border with state i). In what follows, we refer to this collection of neighboring states, including state i , as the vicinity of state i , denoted $\{i\}$.

Next, we construct three measures of innovation flows that consider differences in patent quality. First, we adjust the number of patents based on their number of forward citations. Second, we adjust patents for their economic value, according to [Kogan et al. \(2017\)](#). Breakthrough patents are defined as being in the top 20% based on citation or economic value. Lastly, we use the patent breakthrough characterization of [Kelly et al. \(2021\)](#), who use textual analysis to identify significant novel patents. The breakthrough patents are defined as being in the top 20% of the distribution in terms of their backward and forward similarity, using a five-year window. We scale the number of breakthrough patents of these three measures by population each year.

We obtain other state-level economic indicators, such as GDP, employment, and population, from the Bureau of Economic Analysis (BEA tables SAGDP and SAINC). The real GDP is calculated by deflating state-level GDP by the GDP deflator (BEA account code: A191RD). State-level productivity is calculated as total real GDP divided by employment. The state-level real median income data is from the U.S. Census Bureau, retrieved from FRED; the income series is adjusted by the Census Bureau using the aggregate CPI.

R&D Tax Credits

Data on state R&D tax credit changes comes from [Wilson \(2009\)](#) and [Bloom, Schankerman, and Van Reenen \(2013\)](#). [Wilson \(2009\)](#) quantifies the rate of each of these credits. There is variation across states in implementing R&D tax credits, including decisions on whether the credit covers all

qualifying expenditures or is incremental, based on prior expenses. In addition, for the incremental tax credits, the basis of previous expenditures could be fixed in time or a moving average of recent activity. We calculate the rate of R&D tax credit following the method outlined in [Wilson \(2009\)](#). See Appendix [A.2](#) for details.

1.2 Innovation, Productivity, and Inflation

A stylized fact

We begin by documenting a positive correlation between productivity and inflation across U.S. states. Figure [1](#) shows that the unconditional cross-state correlation between average productivity growth and average inflation is positive. For reference, the cross-state standard deviation of relative inflation is 2.3 percent at the annual horizon. To control for unobserved heterogeneity, we estimate:

$$\pi_{i,t,t+s} = \beta g_{i,t,t+s} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t} \quad (1)$$

where $\pi_{i,t,t+s}$ is the growth in the (log) price level in state i from t to $t + s$ and $g_{i,t,t+s}$ is labor productivity growth (real output per worker) over the same horizon. Our benchmark uses $s = 5$ years. The control vector $\mathbf{X}$ includes lagged productivity and price levels, state and year fixed effects, and the local output gap (unemployment). Standard errors are robust to serial correlation and cross-sectional dependence using the [Driscoll and Kraay \(1998\)](#) procedure with an adjustment for small sample bias.

Table [1](#) shows that a one-standard-deviation increase in productivity growth is associated with approximately a 1.5 percentage point higher cumulative inflation at the five-year horizon. This relation is largely invariant to controlling for lagged levels or the output gap (column 5), and Figure [3](#) shows it increases with the horizon. The insensitivity to unemployment echoes [Hazell et al. \(2022\)](#) on the flatness of the Phillips curve at the state level and argues against a pure demand-side explanation.

Innovation and inflation

We next assess how much local innovation helps explain this positive correlation. Specifically, we re-estimate equation [\(1\)](#), replacing labor productivity with measures of local innovation:

$$\pi_{i,t,t+5} = \beta \ln(\text{innov}_{\{i\},t,t+5}) + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}. \quad (2)$$

Here, $\ln(\text{innov}_{\{i\},t,t+s})$ denotes the flow of innovation between year t and $t + s$ in the vicinity of state i . When $\text{innov} = 0$, we replace $\ln(\text{innov})$ with the sample minimum and include a zero-innovation dummy so as not to drop these observations. The controls include one lag of the log price level, as well as state and year fixed effects and, in some specifications, the unemployment

rate. These estimates are descriptive and do not by themselves establish causality; we turn to an IV strategy below for causal identification. The results are robust to using innovation measures based only on activity within the state, excluding neighboring states (Table A.1 in the Online Appendix), although the estimated magnitudes are smaller. Clustering by state and year, rather than using the Driscoll and Kraay (1998) adjustment, yields qualitatively similar results (Appendix Table A.2).

Table 2 shows that the coefficient on innovation is positive and statistically significant across all innovation measures and specifications. Quantitatively, a one-standard-deviation increase in local innovation is associated with a 3 to 4 percentage point increase in the local price level over the subsequent five years. In terms of magnitudes, innovation also has substantially greater explanatory power than labor productivity. Appendix Figure A.1 shows that innovation explains 15.7% of the cross-state variation in long-run inflation, compared with 3.8% for productivity. A similar gap remains after residualizing with respect to state and year fixed effects: 14.0% versus 4.4%.

Inflation components

Decomposing the response by inflation category, we see in Figure A.2 that inflation in non-tradable goods responds significantly more than in tradable goods, and housing inflation responds more than non-housing inflation—though housing inflation is also more volatile (see, e.g., Stock and Watson, 2019; Hazell et al., 2022). After normalizing each component to unit standard deviation, innovation accounts for a comparable fraction of the variation in housing and non-housing inflation (approximately 0.2 standard deviations per one-standard-deviation innovation increase, based on the KPSS measure).

Causal identification

Both innovation and inflation are endogenous to local conditions. To identify the causal effect, we follow Bloom et al. (2013) and use changes in state-level R&D tax credits in the vicinity of state i as an instrument for local innovation. Shocks to R&D tax credits affect the user cost of R&D capital and hence innovation intensity—consistent with Akcigit, Grigsby, Nicholas, and Stantcheva (2022), who demonstrate that changes in corporate taxes have significant effects on both the quantity and quality of innovation. The identifying assumption is that changes in state-level R&D tax credits affect local inflation only through their effect on local innovation intensity, conditional on state and year fixed effects. A potential concern is that tax policy changes may be endogenous to local economic factors, but Bloom et al. (2013) argue this is unlikely, given the absence of a significant statistical relation between previous fluctuations in a state’s R&D spending or GDP and subsequent changes in R&D tax policy. Moreover, our inclusion of state and year fixed effects absorbs permanent state-level differences and common time shocks, so identification comes from within-state, year-to-year changes in the tax credit rate relative to other states.

For a given state i at time t , the R&D tax credit in the vicinity of state i is the GDP-weighted average of R&D tax credits across neighboring states:

$$RDTC_{\{i\},t} = \sum_{s \in \{i\}} w_{s,i,t} RDTC_{s,t}; \quad w_{s,i,t} = \frac{Y_{s,t-1}}{\sum_{j \in \{i\}} Y_{j,t-1}} \quad (3)$$

Table A.3 shows that the R&D tax credit is a strong predictor of local innovation intensity. A one-standard-deviation increase in the R&D tax instrument leads to a 0.2 to 0.3 standard deviation increase in local innovation measures, with first-stage Kleibergen-Paap F-statistics ranging from 30 to 60—well above conventional weak-instrument thresholds—confirming that the instrument has substantial explanatory power for local innovation.

We then estimate the reduced-form effect of R&D tax credits on productivity and inflation:

$$y_{i,t,t+5} = \beta RDTC_{\{i\},t} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}. \quad (4)$$

The dependent variables are equal to the growth in output per worker and the growth in the (log) price level in state i from t to $t + 5$.

Table 3 shows that a one-standard-deviation increase in R&D tax credits leads to a 0.6 pp increase in productivity (Columns 1–2) and a 1.2 to 1.3 pp increase in cumulative headline inflation over the next five years (Columns 3–4). In brief, innovation raises local productivity as well as local inflation.

The remaining columns of Table 3 decompose the inflation response by component. We see that the causal effect of innovation on inflation is significantly larger for non-tradable goods (1.7 pp) than for tradable goods (0.6 pp), and larger for housing (2.7 pp) than for non-housing inflation (0.6 pp). Figure 5 confirms that, after normalizing each component to unit standard deviation, a one-standard-deviation increase in the R&D tax instrument leads to a 0.1 to 0.15 standard deviation increase across all inflation components. In Section 6, we confirm that these findings are robust to using post-1983 rent-based MSA-level CPI data and are replicated in a very different institutional setting—the Eurozone—where cross-country innovation differences generate quantitatively similar effects on local prices.

1.3 Mechanisms

Having provided evidence that innovation raises both local productivity and local prices, we now investigate through which channel innovation raises local demand for non-tradable goods. We consider three candidate mechanisms: (i) broad-based wage increases, as in the standard Balassa-Samuelson framework; (ii) migration-induced demand; and (iii) portfolio home bias in local equity markets. We present evidence on each and conclude that the data point instead to a concentrated-wealth mechanism operating through incomplete markets.

Local Wages and Income

Innovation can raise local prices if it increases the relative wealth of households in the innovative region. The standard [Balassa \(1964\)](#); [Samuelson \(1964\)](#) channel operates through broad-based wage increases: higher tradable-sector productivity raises wages across sectors (assuming intra-regional labor mobility), boosting demand for non-tradables. Under incomplete markets, however, a second channel can arise: if the gains from innovation are concentrated among entrepreneurs and key talent rather than shared broadly across workers, local wealth and demand for non-tradables can rise even without broad-based wage gains—and local inequality should increase. To distinguish the channels behind the increases in local demand, we next examine the response of average and median wages to innovation.

Our main source of data is the Panel Study of Income Dynamics (PSID), which provides individual-level labor outcomes from the core sample (ages 20–59). Because the PSID becomes biennial after 1998, we interpolate missing state-level values between adjacent survey years (see [Appendix A](#) for details on the sample construction). For each state i and year t , we construct the state-level median and mean wage earnings as the mean and median wage earnings over the core PSID sample in the state in year t . We complement this with aggregate wage and salary data from the BEA and median income from the US Census Bureau. One key difference is that the BEA data includes equity-based compensation, which is not captured in the PSID data. We compute growth rates in mean and median wage earnings over t to $t + 5$.

We re-estimate equation (4), substituting the dependent variable with nominal wage growth over t to $t + 5$. [Table 4](#) shows that a one-standard-deviation increase in R&D tax credits leads to a 2.7 percentage point decline in median wages; average wages in the PSID and median income from the US Census Bureau show statistically insignificant declines. This pattern is largely robust across different datasets: replicating the analysis using the PSID full sample and the BLS Occupational Employment and Wage Statistics, we find that median and average wages are again weakly negatively related to changes in R&D tax credits. ([Appendix Table A.5](#)). The pattern is similar if we focus on the median and average individual earnings growth from the PSID ([Appendix Table A.6](#))—as opposed to the growth in median or average wages.

One important exception to the above is that [columns 5 and 6 of Table 4](#) show that average BEA earnings increase by a statistically significant 0.6 to 0.7 percentage points in response to a one-standard-deviation increase in the R&D tax credit. The fact that mean BEA earnings increase, while wages in the PSID decline, suggests that any increase in earnings is likely driven by equity-based compensation and is likely concentrated to top earners. Indeed, we see in [columns 9 and 10 of the table](#) that innovation increases top income inequality: a one-standard-deviation increase in R&D tax credits leads to a 1.8 to 1.9 percent increase in the share of income accruing to the top 1%. In addition, in [Appendix Table A.4](#), we examine separately the response of low- and high-skill wages to innovation. We find some evidence that high-skill wages respond positively to innovation, while

low-skill wages respond negatively. However, the difference in the response of low- and high-skill wages is significant for median but not average wages.

In sum, we find that innovation increases local wealth, but the wealth gains are likely highly concentrated at the top of the income distribution. This suggests that the standard broad-based Balassa-Samuelson wage channel cannot account for our findings. Instead, the evidence points to a concentrated-wealth mechanism: the gains from innovation are highly concentrated in the top of the income distribution, consistent with [Hsu and Ziedonis \(2008\)](#); [Haeussler, Harhoff, and Mueller \(2014\)](#); [Aghion et al. \(2018\)](#); [Kogan, Papanikolaou, Schmidt, and Song \(2020\)](#).

Migration

An alternative channel through which innovation can increase the local demand for non-tradable goods is through migration. Innovation hubs tend to attract skilled workers ([Moretti, 2012](#)), so one might expect a positive migration response that bids up demand for local non-tradable goods. However, as we see in columns 11 and 12 of Table 4, population growth shows a negative response to increases in the R&D tax credit. One interpretation of this negative coefficient is that increases in the cost of living resulting from higher innovation activity lead some households to leave the state, consistent with the general-equilibrium spatial sorting mechanisms discussed in [Moretti \(2012\)](#). This net outflow does not rule out compositional changes—for instance, a simultaneous inflow of high-skill workers and outflow of lower-skill households—but it does suggest that migration is unlikely to be the primary channel through which innovation raises local prices.

Local Stock Market

Last, another channel through which innovation could raise local wealth is portfolio home bias: if households overweight local stocks ([Becker et al., 2011](#)), an increase in the productivity of local firms could generate a positive wealth effect that bids up local non-tradable prices. However, the last column of Table 1 shows that the correlation between local stock returns and local inflation is, in fact, *negative*. This finding is inconsistent with a broad equity-wealth channel and suggests that the aggregate local stock market does not mediate the link between innovation and local prices. In Section 4.1, we show that while the aggregate local market does not comove with inflation, a local growth-minus-value portfolio—which isolates displacement risk—does, consistent with our model’s predictions.

1.4 Summary

The empirical results in this section establish three facts. First, innovation raises local inflation, with the effect concentrated in non-tradable goods and housing. Second, the wealth gains from innovation are concentrated at the top of the income distribution: average earnings—inclusive of

equity-based compensation—rise but median wages decline, and top income inequality increases. Third, the local stock market portfolio is negatively correlated with local inflation, ruling out portfolio home bias as the primary channel.

Together, these findings point to a mechanism in which innovation generates concentrated wealth gains for innovators and key talent, raising local demand for non-tradable goods and hence local prices. The model in the next section formalizes this intuition: increases in productivity lead to local wealth effects, as in [Balassa \(1964\)](#) and [Samuelson \(1964\)](#), but in our setting these gains are concentrated in a few households—the innovators.

2 The Model

In this section, we develop a model with incomplete financial markets that generates a positive correlation between productivity, innovation, and the local price level. The core intuition can be stated simply. Relative inflation across regions reflects differences in marginal utility growth. Under complete markets, marginal utility growth equals the negative of aggregate consumption growth, so a region that produces more should have lower prices. Under incomplete markets, however, displacement shocks generate cross-sectional dispersion in consumption growth within a region: a few innovators capture the bulk of the gains while everyone else is displaced. Because marginal utility is convex, this within-region dispersion raises average marginal utility and hence the local price level—even as aggregate output increases.

An important difference with [Balassa \(1964\)](#) and [Samuelson \(1964\)](#) is in the mechanism through which innovation raises local demand for non-tradeable goods. In the classic Balassa-Samuelson framework, productivity gains in the tradable sector raise wages economy-wide, which in turn raises the cost of non-tradable goods; the mechanism requires broad-based wage gains and productivity shocks originating in the tradable sector. In our model, neither condition is necessary: innovation raises the wealth of a small set of local households—the innovators—who bid up the price of non-tradable goods. This mechanism operates even when productivity gains occur in the non-tradable sector and even when median wages are unaffected, consistent with our empirical findings.

We first develop a simple model that delivers the intuition for the primary mechanism of the paper. The model has multiple regions that produce tradable and non-tradable goods. To simplify the exposition, we assume flexible prices, and no fiscal or monetary policy. Given the absence of nominal rigidities, the mechanism we propose operates through the real price of non-tradable goods. Markets are exogenously incomplete: a share of the benefits of innovation from local firms accrues only to local investors. This assumption can be justified if part of the value of new ideas accrues to the innovators, who cannot sell claims on their future projects (see, e.g. [Kogan et al., 2020](#); [Crouzet, Eberly, Eisefeldt, and Papanikolaou, 2022](#), for a discussion). Since the gains from

innovation are not shared equally across households, innovation will raise the wealth of some local households, who can afford to spend more on local goods, leading to price increases.

2.1 Model Setup

The economy consists of two regions, which we label as 'home' (H) and 'foreign' (F), and two goods: the tradable good (T) and the non-tradable good (NT). Both types of goods are produced in both regions. While the tradable good can be moved across regions and consumed in either region, the non-tradable good must be consumed in the same region where it is produced. Time is discrete and is indexed by t . Given that the model is symmetric, we focus on solving the problem of the domestic household, with the foreign household solving an identical problem. We use stars to denote variables associated with the foreign region.

Production

Each region contains a continuum of productive units (projects), which one may think of as individual product lines or establishments. There are two types of projects: those producing tradable goods (T) and those producing non-tradable goods (NT). Each project i produces a fraction $a_{S,t,s}^i$ of aggregate sectoral output $Y_{S,t}$, where s denotes the project's creation date and $S \in \{T, NT\}$.

The model features creative destruction, following [Gârleanu et al. \(2016\)](#). Each period, a new cohort of projects arrives and displaces a fraction of incumbent output. Specifically, the total output of newly created non-tradable projects at time t equals

$$\int_{i \in [0,1]} Y_{NT,t,t}^i = (1 - e^{-u_{NT,t}}) Y_{NT,t}, \quad (5)$$

where $u_{NT,t} \geq 0$ is a random displacement shock. Incumbent projects see their output share shrink by a factor of $e^{-u_{NT,t}}$ (that is, if project i was created at $s < t$, its share evolves as $a_{NT,t,s}^i = a_{NT,s,s}^i e^{-\sum_{n=s+1}^t u_{NT,n}}$). Tradable-good projects are subject to an analogous displacement process with shock $u_{T,t}$.

Households

Each region contains a unit measure of households, indexed by $h \in [0, 1]$. At time zero, households have equal endowments consisting of all projects in existence at that time. Households maximize expected utility of consumption,

$$U_{h,t} = E_t \left[\sum_{s=t}^{\infty} \beta^{s-t} \ln C_{h,s} \right]. \quad (6)$$

To prevent degenerate wealth distributions, we introduce overlapping generations: each period a mass ζ of households exits and is replaced by newborns. The wealth of exiting households is partly redistributed to newborns (fraction $1 - \theta$, which acts as a redistributive tax on the wealth of the deceased) and partly returned to survivors via annuity contracts (fraction θ); see Appendix C for details.

Households in the home region consume a composite C_t of tradable $C_{T,t}$ and non-tradable $C_{NT,t}$ goods,

$$C_t = (C_{T,t})^\alpha (C_{NT,t})^{1-\alpha} \quad (7)$$

The parameter α captures the share of tradable goods in the household's utility.

Last, we normalize the price of the tradable good (the numeraire good) to one; hence,

$$p_{T,t} = 1 \quad (8)$$

where $p_{T,t}$ is the price of tradable goods in both regions. We denote the price of non-tradable goods in the home and foreign region as $p_{NT,t}$ and $p_{NT,t}^*$, respectively.

Aggregate Output

The aggregate output of each of the two types of goods $S \in \{NT, T\}$ in each region $r \in [H, F]$ evolves exogenously according to

$$\Delta \ln Y_{S,t+1}^r = \mu + \varepsilon_{S,t+1}^r + \delta u_{S,t+1}^r \quad (9)$$

Each output process is driven by two shocks, ε and u . The first shock, ε , affects the output (and dividends) of all projects symmetrically. The second shock, u , is the displacive shock in the model: it reallocates market share from existing projects to new projects (recall equation (5)). We allow this shock to affect aggregate output—motivated by the standard endogenous growth models—and parameterize its impact on output by $\delta \in (0, 1)$. For the purpose of the main analysis, we treat the processes $Y_{S,t}^r$ as reduced-form endowments and abstract from explicit factor markets; Section 2.7 and Appendix C.6 show how these endowments can be microfounded in a standard two-region production economy with locally supplied land and mobile labor.

In sum, displacement shocks serve two functions in our model. First, positive displacement shocks increase overall output. Second, they reallocate market share from incumbent firms to new entrants. As such, our specification is a reduced-form description of a process of innovation and creative destruction as in expanding varieties models (e.g., Romer, 1987, 1990) or as in models with creative destruction (see e.g., Aghion and Howitt, 1992).

Households in our model can trade a complete set of securities contingent on aggregate shocks. In particular, they can trade equity claims on existing firms and riskless, zero-net-supply bonds in both regions. There is no asset market segmentation. Consumers can trade state-contingent claims on the displacement shock u and the neutral shock ε .

However, financial markets in our model are incomplete along an important dimension related to how new projects are created and how the existing households participate in the resulting wealth creation. In each period, households innovate with some probability. Successful innovation leads to creating a new project. Importantly, households cannot share this risk ex-ante; that is, they cannot sell claims against *their own* future endowment of new projects, as in [Kogan et al. \(2020\)](#). As a result, a shock to the relative profitability of new projects u leads to the redistribution of wealth from the owners of existing projects to the new entrepreneurs. This market incompleteness is the key aspect of the model, as it introduces a wedge between aggregate consumption growth and the marginal utility of the average investor.

From time zero onward, new projects arrive in the economy each period, with productivity proportional to $a_{NT,t}^i$. Property rights to these new projects are assigned partly to the inventors and partly to existing firms. Specifically, we assume that a fraction $\eta \in (0, 1)$ of the total value of new projects is allocated to a measure $\zeta \in (0, 1)$ of the population—inventors or entrepreneurs, and established firms (and therefore all households) own the residual fraction $1 - \eta$. To facilitate aggregation, we assume that the likelihood of any household becoming an inventor in any given period is proportional to its wealth entering the period. To highlight the model's central mechanism, we first examine the case where $\eta = 1$ and subsequently relax this assumption when calibrating the model to the data.

2.2 Equilibrium

Our definition of equilibrium is standard. An equilibrium is a set of price processes, consumption choices, and asset allocations such that (a) consumers maximize expected utility over consumption and asset choices subject to their dynamic budget constraint, (b) all asset and goods markets clear.

Markets are incomplete; hence households' marginal utilities are not equalized across states through risk sharing. However, because all agents within a region are solving the same optimization problem, we can represent their collective demand for consumption and financial assets using a representative agent construct, which has a state-dependent utility over the region-level aggregate consumption C_t , and controls the entire region's wealth.

We can describe equilibrium allocations using a central-planner formulation with stochastic Pareto-Negishi utility weight λ_t , defined as the time- t weight on the foreign region relative to the home region (without loss of generality, we normalize the home weight to one, so $\lambda_t \equiv \Lambda_t^* / \Lambda_t$).

where Λ_t, Λ_t^* are the underlying Pareto weights):

$$\max_{\{C_{NT,t}, C_{NT,t}^*, C_{T,t}, C_{T,t}^*\}} \sum_t \beta^t (\ln C_t + \lambda_t \ln C_t^*), \quad (10)$$

subject to the aggregate resource constraints

$$C_{T,t} + C_{T,t}^* = Y_{T,t} + Y_{T,t}^*, \quad C_{NT,t} = Y_{NT,t}, \text{ and } C_{NT,t}^* = Y_{NT,t}^*, \quad (11)$$

along with the consumption aggregator in (7). Tradable goods are shared across regions, while non-tradable goods must be consumed where they are produced.

2.3 Relative prices and inflation

We can define the price level for a region as the ratio of consumer expenditures to real consumption,

$$P_t \equiv \frac{C_{T,t} + P_{NT,t} C_{NT,t}}{C_t}. \quad (12)$$

As shown in Appendix C.2, combining the planner's FOCs with the Cobb–Douglas demand system implies

$$\frac{P_t}{P_t^*} = \frac{C_t^*}{C_t} \frac{1}{\lambda_t} \quad (13)$$

So the relative price level e_t across the two regions is then equal to

$$e_t \equiv \ln \left(\frac{P_t}{P_t^*} \right) = \ln C_t^* - \ln C_t - \ln \lambda_t. \quad (14)$$

Equation (14) highlights the critical element of our model. In complete markets, λ_t is a constant, and equation (14) reduces to a Backus-Smith condition applied to regional price levels. Relative to the complete-market setting, the time-varying term λ_t reflects the wealth redistribution created by innovation shocks under incomplete markets.

Given Cobb–Douglas preferences, the Negishi weight also equals the ratio of consumption expenditures (and, with log utility, the ratio of wealth) across regions: $\lambda_t = C_t^*/C_t = W_t^*/W_t$, where C_t denotes nominal consumption expenditures (see Appendix C.2). We define relative inflation π_{t+1} as the growth rate of e_t . Since the stochastic discount factor (SDF) in each region equals β times the growth in local marginal utility, relative inflation is the difference in log SDFs:

$$\pi_{t+1} \equiv \Delta e_{t+1} = \ln M_{t,t+1} - \ln M_{t,t+1}^* = \Delta \ln C_{t+1}^* - \Delta \ln C_{t+1} - \Delta \ln \lambda_{t+1}. \quad (15)$$

Equivalently, relative inflation reflects differences in marginal utility growth across regions. In our model, λ_t varies over time in response to displacement shocks due to market incompleteness,

driving a wedge between aggregate consumption growth and the SDF.

2.4 The role of displacement risk

Displacement risk introduces a wedge between aggregate consumption growth and the stochastic discount factor (SDF). Recall that the SDF is proportional to the growth in local marginal utility, averaged across all households in the region. Because of incomplete markets, households' consumption growth is not perfectly correlated. Some households receive profitable new projects, while others do not.

At each point in time, we partition households into those that receive profitable new projects (fraction ζ) and those that do not (fraction $(1 - \zeta)$). Households have a constant consumption-to-wealth ratio, so their consumption is proportional to the market value of their portfolios. The equilibrium SDF is therefore:

$$M_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-1} \left(\underbrace{(1 - \zeta) b_{t+1}^{-1}}_{\text{displacement}} + \underbrace{\zeta \left(\frac{1 - (1 - \zeta) b_{t+1}}{\zeta} \right)^{-1}}_{\text{receiving new projects}} \right) \quad (16)$$

where C_t, C_{t+1} are the aggregate consumption bundles in the home region. b_{t+1} is a key variable in our model: it corresponds to the wealth share of households that do not receive shares in new projects in their home region.

Importantly, the average growth rate in marginal utility is not equal to the growth rate in log (aggregate) consumption. Agents who receive profitable new projects have lower marginal utility than those who do not. The resulting dispersion in consumption growth, combined with the convexity of marginal utility, drives a Jensen's inequality wedge in the SDF, in the spirit of [Constantinides and Duffie \(1996\)](#).

If we denote the aggregate effect of displacement at home region as Ω_{t+1} :

$$\Omega_{t+1} = (1 - \zeta) b_{t+1}^{-1} + \zeta \left(\frac{1 - (1 - \zeta) b_{t+1}}{\zeta} \right)^{-1}, \quad (17)$$

then we can express the dynamics of λ as

$$\Delta \ln \lambda_{t+1} = \ln \Omega_{t+1}^* - \ln \Omega_{t+1}. \quad (18)$$

That is, the Negishi weight shifts toward whichever region experiences less displacement—reflecting the wealth redistribution that displacement shocks create under incomplete markets.

To gain intuition, consider the limiting case $\zeta \rightarrow 0$ (a small fraction of households receive new

projects). The SDF simplifies to

$$M_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-1} b_{t+1}^{-1}. \quad (19)$$

Any shock that reallocates wealth from incumbents to new projects (lowering b_{t+1}) increases the SDF—equivalently, raises the marginal utility of the representative household. The intuition is that, because marginal utility is convex in consumption, greater dispersion in consumption growth raises average marginal utility and therefore the SDF, similar to the mechanism in [Constantinides and Duffie \(1996\)](#).

2.5 Two illustrative special cases

We next discuss two special cases that build intuition for the general result. We maintain the assumption that all new projects accrue to innovating households ($\eta = 1$).

Displacement shocks in the tradable-good sector

The cleanest case arises when displacement shocks originate only in the tradable-good sector ($u_{NT,t} = u_{NT,t}^* = 0$, $u_{T,t} = u_t$, $u_{T,t}^* = u_t^*$). Because the supply of non-tradable goods is unaffected, a positive displacement shock raises local inflation unambiguously through its effect on local wealth and demand. In particular,

$$\pi_{t+1} = (1 - \alpha) \left(-\Delta \ln \lambda_{t+1} - \varepsilon_{NT,t+1} + \varepsilon_{NT,t+1}^* \right). \quad (20)$$

Relative inflation depends on the change in the Negishi weight (reflecting wealth redistribution) and on neutral non-tradable supply shocks. In the limiting case of $\zeta \rightarrow 0$, we can approximate relative inflation around the stochastic steady state as

$$\pi_{t+1} \approx (1 - \alpha) \left(\frac{\alpha pd_{T,s}}{\alpha pd_{T,s} + (1 - \alpha) pd_{NT,s}} (u_{t+1} - u_{t+1}^*) + \varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} \right), \quad (21)$$

where $pd_{T,s}$ and $pd_{NT,s}$ are the steady-state price-dividend ratios in each sector (equal across regions by symmetry). A positive home displacement shock reallocates wealth toward the home region, lowering λ_t and increasing demand for domestic non-tradable goods, with no offsetting supply effect.

Displacement shocks only in the non-tradable good sector

When displacement shocks affect only the non-tradable sector ($u_T = 0$, $u_{NT,t} = u_t$, $u_{NT,t}^* = u_t^*$, with $\text{corr}(\varepsilon_{T,t}, \varepsilon_{T,t}^*) = 1$), the wealth shares of non-innovating households are

$$b_{t+1} = \frac{\alpha p d_T + (1 - \alpha) p d_{NT} e^{-u_{t+1}}}{\alpha p d_T + (1 - \alpha) p d_{NT}}, \quad b_{t+1}^* = \frac{\alpha p d_T + (1 - \alpha) p d_{NT} e^{-u_{t+1}^*}}{\alpha p d_T + (1 - \alpha) p d_{NT}}. \quad (22)$$

A larger displacement shock u lowers b , meaning non-innovating households retain a smaller share of total wealth. In the limit $\zeta \rightarrow 0$, relative inflation becomes

$$\pi_{t+1} = (1 - \alpha) \left(\left(\frac{(1 - \alpha) p d_{NT}}{\alpha p d_T + (1 - \alpha) p d_{NT}} - \delta \right) (u_{t+1} - u_{t+1}^*) + \varepsilon_{t+1}^* - \varepsilon_{t+1} \right). \quad (23)$$

Unlike the T-sector case, the effect of displacement shocks on inflation is now ambiguous. A positive u -shock simultaneously increases the supply and the demand for non-tradable goods. The demand (wealth) effect dominates as long as the valuation of the NT sector is sufficiently large:

$$\delta < \frac{(1 - \alpha) p d_{NT}}{\alpha p d_T + (1 - \alpha) p d_{NT}}. \quad (24)$$

This analysis highlights a key difference from the Balassa-Samuelson hypothesis. The Balassa-Samuelson mechanism relies on broad wage pass-through from sectoral productivity to local non-tradable prices. Our mechanism relies on incomplete sharing of innovation rents, so local prices rise because gains are concentrated in a subset of households, not because wages rise broadly. As a result, our mechanism does not require productivity shocks to occur in the tradable sector; innovations in non-tradable sectors, such as services, can also raise local prices.

2.6 The general case

We conclude our discussion of the model mechanism by focusing on the case in which displacement shocks affect projects in both the tradable and non-tradable good sectors (that is, $u_{T,t} = u_{NT,t} = u_t$ and $u_{T,t}^* = u_{NT,t}^* = u_t^*$). In this case, the wealth share of households that do not receive shares in new projects in their home region is given by the following expression:

$$b_{t+1} = 1 - \frac{y_{T,t+1} (1 + \lambda_{t+1}) \alpha p d_{T,t+1} + (1 - \alpha) p d_{NT,t+1}}{\alpha \left(y_{T,t+1} p d_{T,t+1} + y_{T,t+1}^* p d_{T,t+1}^* \right) + (1 - \alpha) \left(\frac{1}{1 + \lambda_{t+1}} p d_{NT,t+1} + \frac{\lambda_{t+1}}{1 + \lambda_{t+1}} p d_{NT,t+1}^* \right)} (1 - e^{-u_{t+1}}), \quad (25)$$

where $y_{T,t+1} \equiv Y_{T,t+1} / \bar{Y}_{T,t+1}$ is the share of the tradable good produced by the home region. Given the discussion in the previous section, we see that a positive displacement shock induces a positive wealth effect in the local region and therefore tends to increase local inflation.

In the limit of $\zeta \rightarrow 0$ and assuming the u shocks are sufficiently small in magnitude, we can approximate the relative inflation around the symmetric steady state as

$$\pi_{t+1} \approx (1 - \alpha) \left((1 - \delta) (u_{t+1} - u_{t+1}^*) + \varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} \right). \quad (26)$$

A region that experiences a larger displacement shock than its counterpart ($u_{t+1} > u_{t+1}^*$) sees higher relative inflation, scaled by $1 - \delta$ to reflect the net of the demand (wealth) and supply effects of innovation.

2.7 Discussion of Modeling Assumptions

Our model features some assumptions that are not standard in the literature.

Innovators

Our notion of innovators is broad and not limited to the inventors of new technologies. It encompasses all agents who appropriate a non-diversified share of the value of innovation: key employees whose human capital is embodied in new intangibles ([Crouzet et al., 2022](#)); workers whose compensation is tied to firm value ([Kogan et al., 2020](#)); angel investors and venture capitalists; corporate executives who finance and implement new projects ([Frydman and Papanikolaou, 2018](#)); and local investors who hold undiversified stakes in innovative firms. The exact process by which innovators capture these rents can take many forms—from equity compensation within existing firms to founding startups partly funded by outside investors—but the key feature is that a share of innovation wealth accrues to a small set of local households rather than being fully diversified.

Incomplete Financial Markets

The assumption that households cannot share rents from innovation ex-ante can be motivated on theoretical grounds. For instance, workers whose compensation is tied to their firm’s innovation success cannot easily hedge that risk in financial markets. More broadly, new ideas are the product of human capital, which is inalienable. [Hart and Moore \(1994\)](#) show that the inalienability of human capital limits the amount of external finance that can be raised by new ventures. It is this friction—not the broad-based wage channel of [Balassa \(1964\)](#) and [Samuelson \(1964\)](#)—that drives the link between innovation and local prices in our model. Our empirical finding that innovation raises top income inequality without raising median wages (Section 1.3) is consistent with this friction playing a quantitatively important role.

Production Structure

Our baseline framework models regional output as an endowment process in which displacive shocks reallocate profits from incumbent to new firms. This is a reduced-form stand-in for the standard process of creative destruction (Aghion and Howitt, 1992). In Appendix C.6, we develop a fully micro-founded spatial model—featuring a quality-ladder structure inspired by Aghion and Howitt (1992), endogenous production, mobile labor, and a fixed supply of land—and show that its aggregate allocations coincide with those of our baseline model.

A natural concern is whether the endowment abstraction omits economically important channels; we think not. In a general spatial equilibrium, non-tradable prices reflect both wage costs and land rents. Our empirical evidence shows that median wages do not respond positively to innovation shocks (Section 1.3), so the classic wage channel is quantitatively muted. The relevant margin is the land rent channel: displacement shocks concentrate wealth among local innovators, whose expenditure on non-tradables bids up local prices. The endowment model captures this wealth effect directly, and the micro-founded model in Appendix C.6 confirms that its aggregate allocations are equivalent.

3 Quantitative Analysis

So far, we have presented a stylized model to highlight the essential mechanism of the paper. To explore the quantitative fit of the model to the data, we extend the model along several dimensions: CRRA preferences, correlated displacement shocks across regions, firm heterogeneity (growth vs. value), and partial sharing of new projects with incumbent firms.

3.1 Setup

To conserve space, we only highlight the differences from the previous formulation.

Preferences

We allow households to have a CRRA utility:

$$U_{i,t} = E_t \left[\sum_{s=t}^{\infty} \beta^{s-t} \frac{C_s^{1-\gamma}}{1-\gamma} \right] \quad (27)$$

where, as before, $C_{i,t} = [(C_{T,i,t})^\alpha (C_{NT,i,t})^{1-\alpha}]$.

Endowment

We allow displacement shocks to be correlated across regions (for instance, due to technology spillovers),

$$u_{t+1} = \rho_u \tilde{u}_{t+1} + (1 - \rho_u) \tilde{u}_{t+1}^*, \quad u_{t+1}^* = \rho_u \tilde{u}_{t+1}^* + (1 - \rho_u) \tilde{u}_{t+1}.$$

Each region's effective shock is a weighted blend of its own and the other region's idiosyncratic shock; ρ_u governs the degree of spillover (higher ρ_u implies more region-specific and less correlated shocks). We impose a common displacement shock to both sectors ($u_{NT,t} = u_{T,t} = u_t$) and impose that regional output is co-integrated:

$$\begin{aligned} \Delta \ln Y_{S,t+1} &= \mu + \varepsilon_{S,t+1} + \delta u_{S,t+1} + \tau(\ln Y_{S,t}^* - \ln Y_{S,t}) \\ \Delta \ln Y_{S,t+1}^* &= \mu + \varepsilon_{S,t+1}^* + \delta u_{S,t+1}^* - \tau(\ln Y_{S,t}^* - \ln Y_{S,t}). \end{aligned} \tag{28}$$

Firms

We relax the assumption that new projects are allocated only to new firms: we now allow a fraction $1 - \eta > 0$ of new projects to be allocated to local incumbent firms. Allowing for $\eta < 1$ weakens the level of market incompleteness since markets would be complete in the case $\eta = 0$. Importantly, how these projects are distributed to local firms is not perfectly symmetric—we assume that firms vary in their investment opportunities.

We allow for two types of firms. Growth firms can receive new projects and trade at higher multiples in equilibrium; we denote them by H . Value firms do not receive new projects and derive their value only from assets in place; we denote them by L . Firms can transition between the two types, H and L , according to the following transition probability matrix:

$$\Sigma = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}. \tag{29}$$

Given (29), there is a constant fraction of high- and low-growth firms in each region.

3.2 Equilibrium

The equilibrium in the full model is largely similar to the simplified model's, although the algebra is somewhat more involved. Given our assumptions, the stochastic discount factor in each region depends on regional consumption and displacement shocks:

$$M_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1 - \zeta)b_{t+1}^{-\gamma} \right) \tag{30}$$

The first factor is the standard consumption-based SDF; the bracketed term captures the displacement wedge: heterogeneous consumption growth within the region raises the SDF above what aggregate consumption alone would imply. As before, relative inflation equals the ratio of stochastic discount factors. We have

$$\begin{aligned}
\pi_{t+1} &= \ln M_{t,t+1} - \ln M_{t,t+1}^* \\
&= \Delta \ln C_{t+1}^* - \Delta \ln C_{t+1} - \Delta \ln \lambda_{t+1} \\
&= \gamma(\Delta \ln C_{t+1}^* - \Delta \ln C_{t+1}) + \ln \Omega_{t+1} - \ln \Omega_{t+1}^*
\end{aligned} \tag{31}$$

Relative inflation is driven by relative consumption growth in the two regions and by the displacement terms $\ln \Omega_{t+1}$, $\ln \Omega_{t+1}^*$, which reflect the degree of wealth redistribution within each region.

3.3 Calibration

In this section, we describe how we calibrate the model parameters. Given our model's degree of non-linearity, we solve for the global solution of the model by discretizing the state space for (λ_t, Z_t) , where λ_t is the stochastic Pareto weight and Z_t collects the relative endowment states. The numerical algorithm iterates jointly on the policy rule for λ_{t+1} and on the associated pricing functions. Appendix C.9 provides the recursive equilibrium mapping and the fixed-point procedure.

To reduce the number of estimated parameters, we impose several restrictions on the dynamics of u shocks: we assume that displacement shocks follow a two-state, i.i.d process: $u \in \{u_l, u_h\}$, with probability density $\{p_l, 1 - p_l\}$. In addition, we assume the displacement shock in the low state is zero. That is, $u_l = 0$. After restricting the evolution of u , the full model has 18 parameters. We set the weight on tradable goods to 0.34, which equals the tradable share used in [Hazell et al. \(2022\)](#). We set the degree of co-integration to 0.1% and the death rate to 2.5%. We set the relative risk aversion coefficient to $\gamma = 6$, within the range commonly used in the asset pricing literature (e.g., [Barro, 2006](#); [Bansal and Yaron, 2004](#); [Mehra and Prescott, 1985](#)). We calibrate firms' transition probabilities p and q directly from the data. In particular, each year, we sort firms into growth and value categories based on the median book-to-market breakpoint at the vicinity of each state. Then we estimate the transition probability over the sample period across all the states.

We calibrate the remaining eleven parameters of the model by minimizing the distance between the model and the data. Specifically, given a vector X of target statistics in the data, we choose parameters to minimize the following objective function,

$$\hat{p} = \arg \min_{p \in \mathcal{P}} \left(X - \frac{1}{S} \sum_{i=1}^S \hat{X}_i(p) \right)' W \left(X - \frac{1}{S} \sum_{i=1}^S \hat{X}_i(p) \right), \tag{32}$$

We minimize the weighted distance between model-simulated moments and their data counterparts. Here $\hat{X}_i(p)$ is the vector of statistics computed in one simulation of the model. The weighting matrix W determines the importance of each statistic to the distance criterion to be minimized. We penalize proportional deviations of the model statistics from their empirical counterparts, so $W = \text{diag}(XX')^{-1}$.

We report our calibration targets in the first column of Table 9. Importantly, we target the passthrough of innovation-driven productivity increases both to local inflation and to local inequality. The empirical moments correspond to the ratio of the reduced-form response of inflation to the reduced-form response of productivity (corresponding to columns four and two in Table 3), and the ratio of the reduced-form response of inequality to the reduced-form response of productivity (corresponding to column ten in Table 4 and column two in Table 3). The equivalent moments in the model correspond to the increase in local inflation and the increase in top income inequality, each relative to the increase in local output, in response to a positive innovation shock (u transitions from u_l to u_h).

The remaining moments include a combination of the first and second moments of aggregate output, consumption, and relative inflation. In addition, we target a set of correlations between these variables. The neutral shock and displacement shock have different implications for inflation. Thus, the correlation between inflation and output and the set of bilateral correlations are informative about the relative magnitude of these two sets of shocks. We also target the correlation between relative consumption growth and relative inflation—the regional equivalent of the Backus–Smith statistic—which disciplines the degree of market incompleteness. In addition, we target the level of top 1% income share, the equity risk premium, the correlation between top 1% income inequality, inflation, and local output. Given that displacement shocks drive the dynamics of inequality, these correlations are informative about the magnitude of displacement shocks in the model. Intuitively, the volatility of output and inflation pins down the overall shock variances; the correlation between inflation and output, and between inflation and inequality, helps separate displacement shocks (u) from neutral shocks (ε), since the two shock types have opposing implications for inflation; and the equity risk premium and inequality level are informative about the degree of market incompleteness (η) and the concentration of innovation gains (ζ). See Appendix B for more details on constructing the moments we target in our estimation.

3.4 Model Fit

Table 8 reports the calibrated parameters. The calibration implies a right-skewed distribution of displacement shocks: the low state ($u_l = 0$) occurs with probability $p_l = 0.825$, while the high state features a large magnitude ($u_h = 0.278$). Major technological advances are thus rare but displace a sizable share of incumbents. The parameter η , which governs the division of surplus between incumbents' shareholders and innovators, is equal to 0.642. This implies that roughly 35% of the

value of new projects accrues to public shareholders. The technology spillover $\rho_u = 0.543$ implies strong comovement in regional quantities and stock returns.

Table 9 shows that the model fits the data well. It produces realistic growth rates of regional consumption and output, matches the level of interest rates, the equity risk premium, the level of top 1% income share, the regional Backus–Smith statistic (0.196 in the model versus 0.207 in the data), and replicates the key correlations between inflation, output and inequality. The high degree of technology spillover generates strong comovement in regional output, consumption, and stock markets, allowing the model to deliver reasonable volatility of relative inflation despite high output volatility. Because the SMM has more moments than parameters, we can test the overidentifying restrictions: the J -statistic is 36.07, with a p -value below 0.01. This is not uncommon for parsimonious structural models, but it suggests that some dimensions of the data are not perfectly captured.

Table 10 reports additional untargeted moments. Although the value premium is not a calibration target, the model generates 4.6%, close to the 3.5% in the data. This is achieved with $\gamma = 6$ and standard CRRA preferences—without relative consumption (“keeping up with the Joneses”) preferences. By contrast, Kogan et al. (2020) require $\gamma \approx 57$ and a relative consumption weight of roughly 80% to generate a comparable value premium. The key difference is that our model features a real channel through which non-innovating households are harmed: displacement shocks concentrate wealth among innovators, who bid up the price of non-tradable goods, eroding the real purchasing power of all other households. This tangible cost of displacement—documented empirically in Section 1.3—substitutes for the high risk aversion and relative consumption preferences needed in frameworks where non-innovators do not experience actual declines in real consumption. Essentially, our model here provides an explicit micro-foundation for the strong relative-consumption preferences inferred under the reduced-form preference specification in Kogan et al. (2020).

The model also generates realistic correlations between innovation and output, inflation, and inequality. Two limitations are worth noting. First, the model cannot simultaneously match the level of top income shares and their sensitivity to innovation shocks. Income inequality in the model is driven by two forces: incumbent households with accumulated wealth who earn significant dividends, and new innovators. Even though displacement shocks are substantial, targeting the level of top income shares leaves insufficient room to match the marginal response of inequality to innovation observed in the data. Second, the model cannot generate the observed difference in price-dividend ratios between tradable (26) and non-tradable (31) firms, since we assume for simplicity that displacement shocks are equally prevalent in both sectors.

3.5 Model mechanisms

We examine how the model’s key quantities respond to the three shocks: the displacement shock u , the neutral T-sector shock ε_T , and the neutral NT-sector shock ε_{NT} . We focus on home-region

shocks; foreign-region responses are symmetric.

Quantities

Figure 6 plots impulse responses for relative inflation, output, consumption expenditure ratios and wealth ratios. The central result is that a positive displacement shock u in the home region raises both local output *and* local inflation—precisely the comovement of productivity growth and rising prices that motivates our paper. Because displacement shocks spill over across regions ($\rho_u > 0$), foreign output also rises, but the inflationary effect is concentrated in the home region where the displacement originates. Neutral shocks, by contrast, raise output without generating inflationary pressure, since their benefits accrue symmetrically to all households.

The wealth dynamics reinforce this asymmetry across regions. A positive u -shock shifts wealth toward the home region: innovators capture gains that are not fully shared with other regions, bidding up local prices and effectively transferring purchasing power from households in other regions through equilibrium portfolio rebalancing. A positive ε_T -shock produces a smaller, more persistent tilt in domestic wealth—a larger local tradable sector amplifies the impact of future displacement shocks. Conversely, a positive ε_{NT} -shock shifts wealth toward the foreign region through equilibrium risk sharing.

Income inequality

The preceding discussion focused on disparities *across* regions; displacement shocks also generate disparities *within* regions. A positive displacement shock implies that a significant share of the value of new projects accrues to a small group of households. The smaller ζ is, the larger the windfall for each innovator, pushing them into the top of the income distribution and raising the top 1% income share (see Appendix C.7 for a detailed explanation of this mechanism).

The left column of Figure 8 illustrates this mechanism in the model. A positive displacement shock generates an immediate spike in the top 1% income share (Panel A), as rents from new projects flow to a small group of innovators. Neutral productivity shocks — whether in the tradable sector (Panel B) or non-tradable sector (Panel C) — leave the income distribution virtually unchanged, since they raise output symmetrically across all existing projects. This pattern is consistent with our evidence in Section 1.3: it is the *displacive* nature of innovation, not productivity growth per se, that drives both local inflation and income inequality.

SDF and Financial Assets

Figure 7 plots impulse responses for the (log) stochastic discount factor (average marginal utility), consumption, the incumbent wealth share b , and stock returns. The two types of shocks have sharply different asset-pricing implications. A positive displacement shock lowers b —non-

innovating households experience declines in relative wealth—and raises the SDF, because the convexity of marginal utility means that the losses of the many outweigh the gains of the few. A positive neutral shock, by contrast, lowers the SDF through higher aggregate consumption shared equally across households. Both shocks raise output, but their risk-premium implications are opposite: displacement shocks carry a *negative* risk premium (high marginal utility states for the representative household), while neutral shocks carry a positive one.

The stock-return responses are equally revealing. A positive u -shock *lowers* returns in both sectors despite raising aggregate dividends, because displacement dilutes incumbent firms' market share—an investor at time t receives only a $1 - (1 - e^{-u})\eta$ fraction of dividends at $t + 1$. A positive ε_T -shock, by contrast, raises returns in both sectors through higher dividends and valuation effects, with no dilution of existing claims.

4 Testable Predictions

A central prediction of the model is that the local growth-minus-value portfolio, rather than the broad local market, is the mimicking portfolio for local inflation risk. We test this prediction and its implications for portfolio choice.

4.1 Inflation and stock returns of local firms

Table 1 shows that the correlation between the returns of local firms and local inflation is negative. Our model does not yield an unambiguous prediction for this sign. A positive displacement shock raises the value of new growth opportunities (wealth effect) but simultaneously dilutes existing firms' market share (displacement effect). The net effect on the local market portfolio therefore depends on whether most of firms' market value comes from assets in place or from growth opportunities (see Appendix C.2 for more details).

By contrast, the model predicts that a portfolio long local growth firms and short local value firms—a *growth-minus-value* (GMV) portfolio—is heavily exposed to the displacement shock u . Value firms derive their value solely from assets in place, while growth firms also receive new projects; thus, the return on the GMV portfolio is highly correlated with u_{t+1} in the calibrated model. Because displacement shocks raise local inflation, the GMV portfolio provides a hedge against increases in the local price level.

Figure 8 (right column) illustrates this prediction in the model. A positive displacement shock generates a large positive GMV return (Panel A): growth firms gain value as they receive new projects, while value firms lose market share. In contrast, neutral productivity shocks in the tradable sector (Panel B) or the non-tradable sector (Panel C) produce little differential response between growth and value firms, since these shocks raise output symmetrically across all existing projects. The sharp contrast between displacement and neutral shocks underscores why the GMV portfolio,

rather than the broad local market, is the natural mimicking portfolio for local inflation risk.

We test these predictions by constructing empirical equivalents of the local market and GMV portfolios for each state (including neighboring states). The local market return is the value-weighted portfolio of all local firms. The GMV return is the difference between value-weighted returns on the top and bottom halves of local firms sorted on intangible-capital-adjusted market-to-book (Eisfeldt and Papanikolaou, 2013; Eisfeldt, Kim, and Papanikolaou, 2022), following the sector classifications in Mian and Sufi (2014); see Appendix A for details.

We first examine empirically whether GMV returns are indeed linked to local innovation intensity by regressing local portfolio returns on local innovation intensity over a five-year horizon:

$$R_{i,t \rightarrow t+5}^p = \beta \ln(\text{innov}_{\{i\},t \rightarrow t+5}) + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad (33)$$

for $p \in \{GMV, MKT\}$, controlling for state and year fixed effects and lagged five-year portfolio returns. Table 5 shows that local GMV returns are positively correlated with local innovation intensity, while the local market portfolio has no significant relation.

We next examine whether GMV returns are correlated with local inflation:

$$\pi_{i,t \rightarrow t+5} = \beta R_{i,t \rightarrow t+5}^p + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad (34)$$

for $p \in \{GMV, MKT\}$, controlling for state and year fixed effects, lagged price level, unemployment, and lagged portfolio returns. Table 6 confirms a positive and significant correlation between local GMV returns and inflation. In contrast, the correlation between the local market portfolio and inflation is weak and negative. Taken together, the local GMV portfolio is a better mimicking portfolio for displacement shocks—and therefore local inflation—than the broad local market.

4.2 Portfolio risk exposure and home bias

Empirically, Coval and Moskowitz (1999) document a strong local preference for locally headquartered firms, particularly firms that produce non-tradable goods. Our model generates this pattern. Although financial markets are integrated—households in both regions can trade the same equity claims—households in different regions optimally hold different portfolios. Because claims to new projects (innovation rents) are not fully contractable ex ante, displacement shocks concentrate wealth among local innovators, bidding up local prices. Local households therefore have an incentive to hedge this displacement risk by tilting their portfolios toward local growth firms and away from local value firms.

To quantify the degree of home bias in the model, we simulate the model from the symmetric steady state (10,000 draws) and regress the return differential between home and foreign households'

portfolios on differentials in local MKT and GMV returns:

$$\bar{R}_{t,t+1}^H - \bar{R}_{t,t+1}^F = \sum_{p \in \{MKT, GMV\}} \beta^p (R_{t,t+1}^{H,p} - R_{t,t+1}^{F,p}) + \varepsilon_{t+1}. \quad (35)$$

The coefficient β^p measures the degree to which home households tilt toward local MKT or GMV returns relative to foreign households. Table 7 reports the results. Panel A shows that households' portfolios exhibit home bias—positive loadings on the local market portfolio—and a tilt toward local growth stocks, generating a positive net exposure to the local GMV factor. Panel B decomposes the risk exposures by primitive shocks: households seek positive exposure to the local displacement shock and the local neutral tradable shock (which amplifies future displacement), and negative exposure to the local neutral non-tradable shock. Panel C confirms that the GMV portfolio is the primary vehicle for hedging displacement risk, while the non-tradable sector provides exposure to neutral non-tradable shocks.

That said, we emphasize that our model's implications for portfolio composition depend on the available investment menu. In practice, investors can also hedge local inflation risk through local real estate or other assets. The robust prediction of our model is therefore not about asset holdings per se, but rather about household risk exposures. That is, our model implies that local households seek positive exposure to local displacement shocks to hedge against local inflation—regardless of how they implement these exposures. Having established these asset-pricing implications, we turn to the welfare consequences of displacement risk.

5 Welfare Implications of Innovation Shocks

We quantify the welfare cost of market incompleteness by comparing the incomplete-markets economy to a complete-markets counterfactual in which innovation rents are fully shared ex ante ($\eta = 0$).

Consumption-equivalent welfare

Let $C_{r,t}$ denote the consumption bundle of region $r \in \{H, F\}$. Under incomplete markets, households within a region have heterogeneous consumption levels. We summarize within-region inequality through the function

$$\phi_{r,t} \equiv (1 - \zeta) b_{r,t}^{1-\gamma} + \zeta \bar{b}_{r,t}^{1-\gamma}, \quad (36)$$

where $b_{r,t}$ is the relative consumption share of the $(1 - \zeta)$ households that do not receive new projects and $\bar{b}_{r,t} = (1 - (1 - \zeta)b_{r,t})/\zeta$ is the corresponding share of innovators (see Appendix C.10 for the derivation). Utilitarian welfare within region r can be expressed in consumption-equivalent

(CE) units as

$$C_{r,t}^{CE} = C_{r,t} \cdot \phi_{r,t}^{1/(1-\gamma)}. \quad (37)$$

Under complete markets all households share innovation rents equally, so $b_{r,t} = \bar{b}_{r,t} = 1$ and $\phi_{r,t} = 1$. Because we impose the same shock on both regions, the equilibrium is symmetric and aggregate consumption is identical across the two economies (in general, inter-regional trade could imply different regional consumption across the two regimes, but symmetry ensures it holds here). The welfare gap between the incomplete- and complete-markets economies therefore reduces entirely to the inequality term:

$$\Delta \ln C_{r,t}^{CE} = \frac{1}{1-\gamma} \ln \phi_{r,t}, \quad (38)$$

where Δ denotes the difference between the incomplete- and complete-markets economies. When $\gamma > 1$, Jensen's inequality implies $\phi_{r,t} > 1$ whenever there is consumption dispersion ($b_{r,t} \neq 1$), so the welfare gap is unambiguously negative: market incompleteness lowers welfare whenever displacement shocks concentrate gains among innovators.

Quantitative Results

Figure 4 reports CE welfare under complete and incomplete markets, as well as the welfare wedge. Both economies start from the symmetric steady state and are subject to the same aggregate displacement shock ($u_H = u_F = u_n$). Because the shock is symmetric, the two regions are affected identically and the analysis reduces to a representative-region problem. Given our calibration, a simultaneous displacement shock raises aggregate output by approximately 1.7%. To aid interpretation, we rescale all responses so that they correspond to a 1% increase in aggregate output relative to the pre-shock steady state.

Under complete markets, welfare rises by the same 1% in CE terms, since all households share the productivity gains equally. Under incomplete markets, however, the same shock lowers welfare. In our calibrated model, the welfare gap in (38) is roughly -7% in CE terms—a permanent reduction in lifetime consumption relative to the complete-markets benchmark. The mechanism is the unequal distribution of gains: innovators capture the bulk of the new output and bid up local non-tradable prices, lowering $b_{r,t}$ and eroding the real purchasing power of non-innovating households.

6 Additional Evidence and Robustness Checks

We conduct several additional analyses to confirm the robustness of our main findings; detailed results are in Appendix D.

MSA-level rent-based inflation

After 1983 the BLS shifted to rent-based housing CPI, which addresses potential concerns about the measurement of housing inflation in our state-level analysis. Using CPI-U data for 27 metropolitan areas (1983–2017), we construct MSA-level innovation measures using patents and re-estimate the effect of innovation on inflation. A one-standard-deviation increase in innovation is associated with a 0.9 to 2.1 pp increase in headline inflation (Columns 1 to 3 of Table A.7). To identify exogenous variation, we aggregate state-level R&D tax credits to the MSA level and estimate reduced-form regressions; a one-standard-deviation increase in R&D tax credits leads to an approximately 0.7 pp increase in local inflation (Columns 4 and 5 of Table A.7). These results confirm that the relationship between innovation and inflation is robust to an alternative method of measuring housing costs and to a finer geographic scale of aggregation.

Evidence from the Eurozone

We replicate the analysis in a very different institutional setting: the Eurozone. Countries in the euro area share a common currency and trade frequently with one another, so interactions across these countries resemble those across U.S. states. Using country-level CPI data (1999–2017) and patent data from the European Patent Office (PATSTAT), we estimate the same specification as in our baseline analysis. A one-standard-deviation increase in regional innovation raises local prices by 4 to 5 pp over a three-year horizon (Table A.8), with quantitatively similar results at the five-year horizon.

IPOs as an alternative displacement proxy

New firms create illiquid wealth that becomes liquid only upon going public; the IPO event can therefore be interpreted as a realized displacement shock. Using IPO data from SDC Platinum, we examine whether IPO activity is associated with both local inflation and income inequality. We find that IPO intensity is positively and significantly correlated with growth in top income shares—with effects that increase toward the right tail of the distribution—and with local inflation, driven primarily by the non-tradable component (Tables A.9 and A.10).

Additional robustness checks

We examine whether a small number of highly innovative states drive our results. Excluding the five most innovative states (California, Illinois, Texas, Massachusetts, and New York), yields estimates that are comparable in magnitude and significance to the baseline (Table A.11). Our results are also robust to population-weighted regressions (Table A.12) and GDP-weighted regressions (Table A.13), confirming that the findings are not driven by a few outlier states or by a particular weighting scheme.

7 Conclusion

Innovation raises productivity, yet it can also raise local prices. This paper documents a causal link between innovation and local inflation, proposes a mechanism, quantifies its welfare costs, and derives testable asset-pricing predictions that are borne out in the data.

Using R&D tax credits as a source of exogenous variation, we show that innovation raises local inflation (especially in non-tradables and housing) while gains concentrate among top earners rather than raising wages broadly—consistent with incomplete markets rather than a standard wage channel. A multi-region model in which displacive shocks redistribute output to new entrants rationalizes these facts: concentrated gains bid up local non-tradable prices. The calibrated model matches the key moments and predicts that a local growth-minus-value portfolio hedges local inflation and that households tilt toward local growth stocks; both implications are confirmed in the data.

Our calibrated model implies that welfare under incomplete markets is roughly 7% lower in consumption-equivalent terms than under full risk-sharing for an innovation shock that raises aggregate output by 1%. Understanding these distributional effects is therefore relevant for policy that promotes innovation. The interaction between innovation-driven displacement and monetary policy is a natural extension for future research.

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Tables

Table 1: Inflation and Productivity

	Headline Inflation					
	(1)	(2)	(3)	(4)	(5)	(6)
Productivity growth	1.21*** (0.44)	1.08** (0.49)	0.91 (0.54)	1.57*** (0.40)	1.51*** (0.42)	1.55*** (0.40)
lagged price level			-20.11*** (3.35)	-21.61*** (3.44)	-22.50*** (3.47)	-22.56*** (3.60)
Lagged productivity level				4.01* (2.00)	4.07** (1.99)	3.74** (1.83)
Unemployment					-0.82 (0.67)	-0.73 (0.64)
Local market portfolio returns						-0.76 (0.53)
State FE	NO	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951
Adj R2	0.67	0.70	0.78	0.79	0.79	0.79

Notes: This table reports the regression coefficients (times 100) of (1): $\pi_{i,t,t+5} = \beta_1 g_{\{i\},t,t+5} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. The independent variable $g_{i,t,t+5} = \ln X_{i,t+5} - \ln X_{i,t}$ is the growth of productivity of state i (column 1) from t to $t+5$. Column 1 controls for year fixed effects γ_t . Column 2 adds state fixed effects as additional control α_i . Columns 3-4 report estimates with controls including lagged price level $\ln Y_{i,t}$, lagged productivity level $\ln X_{i,t}$. Column 5 controls for unemployment rate at t . Column 6 adds market return of the vicinity of state i from t to $t+5$ as additional control. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 2: Innovation, Inflation and Unemployment

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	3.42*** (0.75)	3.61*** (0.80)				
Citations (top 20%)			3.73*** (0.57)	3.94*** (0.58)		
KPSS (top 20%)					3.69*** (0.85)	3.74*** (0.87)
Unemployment		-2.03** (0.79)		-1.31* (0.67)		-1.15* (0.61)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	663	663	951	951	951	951
Adj R2	0.76	0.78	0.81	0.81	0.80	0.81

Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. The vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+5}$ is the flow of innovation measure at the *vicinity* of state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 3: R&D Tax Credits, Productivity and Inflation (Headline and Components)

	Productivity		Inflation		Inflation Decomposition							
	(1)	(2)	(Headline)		Tradables (T)		Non-Trad. (NT)		w.o. Housing		Housing (H)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
RDTC	0.58** (0.26)	0.64** (0.26)	1.17*** (0.39)	1.33*** (0.42)	0.61* (0.34)	0.64* (0.34)	1.46*** (0.49)	1.67*** (0.54)	0.57 (0.35)	0.62* (0.35)	2.01** (0.74)	2.65*** (0.79)
Unemployment		YES		YES		YES		YES		YES		
State FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951	951	951	951	951	951	951
Adj R2	0.65	0.66	0.79	0.79	0.80	0.80	0.75	0.76	0.87	0.87	0.71	0.73

Notes: We report the estimated coefficients (times 100) to the R&D tax credit in the *vicinity* of state i from the regression coefficients (times 100) of (4) in the main text: $g_{i,t,t+5} = \beta_1 RDTC_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$, where $g_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. **Productivity (Cols 1–2):** Dependent variable is the growth in real GDP per worker in state i (Source: BEA). **Inflation (Cols 3–12):** Dependent variable is growth in the (log) price level. Columns 3–4: headline inflation. Columns 5–12 decompose inflation into tradables (T), non-tradables (NT), headline excluding housing (w.o. H), and housing (H). The vector of controls $\mathbf{X}_{i,t}$ includes the (log) level of the dependent variable Y at time t . Even-numbered columns add the unemployment rate at time t as an additional control. All specifications include state and year fixed effects. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. First-stage estimates for the RDTC instrument, together with Kleibergen–Paap rk Wald F -statistics, are reported in Appendix Table A.3. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 4: R&D Tax Credits and Local Wealth

	Wage Earnings						Income				Population	
	Median		Mean				Median		Top 1% Share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
RDTC	-2.74*** (0.44)	-2.65*** (0.49)	-1.51 (1.03)	-1.63 (1.00)	0.62*** (0.14)	0.68*** (0.13)	-0.52 (0.39)	-0.40 (0.36)	1.77*** (0.38)	1.86*** (0.38)	-0.36*** (0.08)	-0.33*** (0.07)
Data Source	PSID	PSID	PSID	PSID	BEA	BEA	Census	Census	IRS	IRS	Census	Census
Unemployment		YES		YES		YES		YES		YES		YES
State FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951	837	837	951	951	951	951
Adj R2	0.57	0.57	0.50	0.50	0.89	0.89	0.66	0.67	0.81	0.81	0.39	0.39

Notes: Columns of this table report the regression coefficients (times 100) of (4): $g_{i,t,t+5} = \beta_1 RDTC_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$, where $g_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$ for Y equal to wages, median income, Top 1% income share, or population. **Wages:** Columns 1–4: growth in median/mean nominal wages from the Panel Study of Income Dynamics (PSID) Core Sample. Columns 5–6: growth in average (mean) nominal wages from Bureau of Economic Analysis (BEA) wage and salary data. **Income:** Columns 7–8: growth in state-level median income from U.S. Census Bureau. **Inequality:** Columns 9–10: growth in the Top 1% income share. **Population:** Columns 11–12: growth in state population. The main regressor $RDTC_{\{i\},t}$ is the R&D tax credit rate in the *vicinity* of state i . The vector of controls $\mathbf{X}_{i,t}$ includes the (log) level of the dependent variable at time t and, in even-numbered columns, the unemployment rate at time t . All specifications include state and year fixed effects. Because the PSID becomes biennial after 1998, constructing wage growth over 5-year horizons requires linearly interpolating missing state-level observations between adjacent survey years. All independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Innovation and Stock Returns: Partial Correlations

Dep. Variable: Stock Return (5-Year)	Local Market (MKT)		Growth-Value (GMV)	
	(1)	(2)	(3)	(4)
KPST (top 20%)	1.09 (5.99)	0.93 (6.05)	8.31** (3.84)	8.35** (3.91)
Citations (top 20%)	-2.04 (5.26)	-2.20 (5.28)	11.15** (5.32)	11.17** (5.29)
KPSS (top 20%)	0.90 (3.53)	1.27 (3.66)	12.78* (6.81)	12.78* (6.86)
Unemployment at t	No	YES	No	YES
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

Notes: This table reports OLS regression coefficients (times 100) of local stock returns on local innovation measures. Each row corresponds to a separate regression. Dependent variable $R_{\{i\},t,t+5}^x$, $x \in \{GMV, MKT\}$, is the 5-year portfolio return (*local market* or *growth-minus-value*) of state i . $R_{\{i\},t,t+5}^x = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$, where $\text{innov}_{\{i\},t,t+5}$ is the flow of innovation at the *vicinity* of state i . Each row reports the coefficient on a different innovation measure (KPST8005, Citations(20%), KPSS(20%)). Columns 2 and 4 add unemployment rate at t as a control. All regressions include lagged 5-year cumulative portfolio return $R_{i,t-5,t}$, state and year fixed effects. Independent variables are standardized to unit standard deviation. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Inflation and Local Stock Market

Dep. Variable: Headline Inflation (5-yr)	(1)	(2)	(3)	(4)	(5)
GMV	0.72** (0.27)		0.70** (0.29)	0.74** (0.27)	0.73** (0.29)
MKT		-0.92 (0.63)	-0.85 (0.63)		-0.76 (0.63)
Unemployment				-0.94 (0.65)	-0.90 (0.63)
State FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Observations	951	951	951	951	951
Adj R2	0.77	0.76	0.77	0.77	0.77

Notes: This table reports the regression coefficients (times 100) of (34): $\pi_{i,t,t+5} = \beta_1 R_{i,t,t+5}^x + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. $R_{i,t,t+5}^x$, $x \in \{GMV, MKT\}$ are the *growth-minus-value* and *local market* portfolio returns of state i from time t to $t+5$. Control variables $\mathbf{X}_{i,t}$ include past cumulative portfolio returns $R_{i,t-5,t}^x$ and price levels $\ln Y_{i,t}$. α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 7: Portfolio Risk Exposures — Home Relative to Foreign

Panel A. Portfolio home-bias			
	Local MKT	GMV	
$\bar{R}_{t,t+1}^H - \bar{R}_{t,t+1}^F$	0.39	0.51	

Panel B. Portfolio exposure to primitive risks			
	Displacement	Neutral, T	Neutral, NT
$\bar{R}_{t,t+1}^H - \bar{R}_{t,t+1}^F$	0.35	0.20	-0.89

Panel C. Stock exposure on primitive risks			
	Displacement	Neutral, T	Neutral, NT
$R_{t,t+1}^{H,GMV} - R_{t,t+1}^{F,GMV}$	0.78	0.00	0.00
$R_{t,t+1}^{H,T} - R_{t,t+1}^{F,T}$	-0.54	0.90	0.00
$R_{t,t+1}^{H,NT} - R_{t,t+1}^{F,NT}$	0.08	0.37	-0.69

Notes: Data is from 10,000 one-period simulations from the symmetric steady state. Panel A of the table reports the regression coefficients of: $\bar{R}_{t,t+1}^H - \bar{R}_{t,t+1}^F = \sum_{p \in \{MKT, GMV\}} \beta_c^p (R_{t,t+1}^{H,p} - R_{t,t+1}^{F,p}) + \varepsilon_{i,t+1}$. Panel B of the table reports the regression coefficients of: $\bar{R}_{t,t+1}^H - \bar{R}_{t,t+1}^F = \sum_{x \in \{u, e_T, e_{NT}\}} \beta_c^p (x_{H,t+1} - x_{F,t+1}) + \varepsilon_{i,t+1}$. Panel C of the table reports the regression coefficients of: $R_{t,t+1}^{H,p} - R_{t,t+1}^{F,p} = \sum_{x \in \{u, e_T, e_{NT}\}} \beta_c^p (x_{H,t+1} - x_{F,t+1}) + \varepsilon_{i,t+1}$ for $p \in \{GMV, T, NT\}$. Dependent variable in Panel A and B is equal to the return differential between home and foreign households excluding the new projects. Dependent variables in Panel C are the stock return differential between home and foreign. Independent variables in Panel A are portfolio return differentials on MKT and GMV at home and foreign region, respectively. Independent variables in Panel B and C are the differences in the primitive shocks in the model.

Table 8: Model Parameters

Description	Symbol	Value
<i>Calibrated Parameters</i>		
<i>Preferences</i>		
Subjective discount factor	β	0.919
<i>Endowments and Shocks</i>		
Displacement shock productivity	δ	0.060
Mean of output growth	μ	0.015
Measure of project receivers	ζ	0.045
Displacement shock (high state)	u_h	0.278
Probability of displacement shock (low state)	p_l	0.825
Volatility of neutral tradable shock	σ_{e_T}	0.054
Volatility of neutral non-tradable shock	$\sigma_{e_{NT}}$	0.008
Technology spillover	ρ_u	0.543
Fraction of project value to inventors	η	0.642
Share of deceased wealth redistributed	θ	0.081
<i>Exogenously Fixed Parameters</i>		
Consumption weight on tradables	α	0.340
Risk aversion	γ	6.000
Death rate	ξ	0.025
Displacement shock (low state)	u_l	0.000
Cointegration parameter	τ	0.001
Firm transition probabilities		
– low state	q	0.232
– high state	p	0.277

Notes: This table reports the calibrated parameters of the model. See the main text and the Appendix B for details on the calibration of the model.

Table 9: Moments used in Model Calibration

	Data		Model
	Estimate	SE	
<i>Regional Quantities and Prices</i>			
Output growth, mean (%)	1.7	0.4	1.8
Output growth, volatility (%)	3.5	0.3	4.1
Consumption growth, mean (%)	1.7	0.5	1.8
Consumption growth, volatility (%)	1.6	0.3	1.5
Relative Inflation, volatility (%)	2.3	0.3	2.1
<i>Asset Prices</i>			
Equity Risk Premium ($E[R_{mkt} - R_f], \%$)	6.2	2.6	4.7
Risk-free rate, mean (%)	1.6	0.4	1.5
<i>Inequality</i>			
Top 1% income share (%)	15.4	1.2	20.8
<i>Innovation-driven passthrough</i>			
Productivity to Inflation	2.1	1.1	1.4
Productivity to Inequality	2.9	1.3	1.2
<i>Regional Correlations, 5 years</i>			
Inflation and			
—output growth (%)	48.1	5.8	43.5
—inequality growth (%)	24.3	7.2	19.2
—consumption growth (%)	20.7	5.7	19.6
<i>Pairwise Correlations, 1 year</i>			
Consumption growth (H and F, %)	76.5	9.9	88.6
Output growth (H and F, %)	44.9	7.0	73.2
Stock returns-MKT (H and F, %)	83.5	4.6	77.0

Notes: This table reports both empirical moments computed using the regional data and simulated moments from the model. All the parameters are calibrated as in Table 8.

Table 10: Non-targeted moments

	Data		Model
	Estimate	SE	
<i>Panel A: Asset Prices</i>			
Value Premium ($E[R_{value} - R_{growth}]$, %)	3.5	1.1	4.6
Price-Dividend Ratio, Non-Tradable ($E[PD_{NT}]$)	31.1	2.2	25.3
Price-Dividend Ratio, Tradable ($E[PD_T]$)	26.0	1.2	25.2
<i>Panel B: Inequality</i>			
Top 0.1% Income Share (%)	6.5	0.8	6.5
Top 10% Income Share (%)	40.5	1.6	65.6
Top 0.1% Wealth Share (%)	13.8	1.6	9.0
Top 1% Wealth Share (%)	30.1	1.8	21.3
Top 10% Wealth Share (%)	67.9	1.6	47.7
<i>Panel C: Correlation</i>			
Innovation and			
—inflation growth (%)	43.6	6.4	45.7
—inequality growth (%)	25.3	8.4	33.2
—output growth (%)	26.3	5.6	24.9

Notes: This table reports additional moments that are not explicit targets in our calibration. Inequality data is from the world inequality database. Total payout is defined as the sum of cash dividends (dvc) and stock repurchases (prstk). For each year, we calculate the aggregate P/D ratio for each sector by dividing the total sector market capitalization by the total sector payout. The reported statistic is the average of these annual sector-based ratios over the sample period. The value premium is the difference between value-weighted high and low book-to-market portfolios where book equity is adjusted to include intangible capital following [Eisfeldt et al. \(2022\)](#). Following [Bansal and Yaron \(2004\)](#), we model dividends as a levered claim on consumption, with a leverage parameter of 3.

Figures

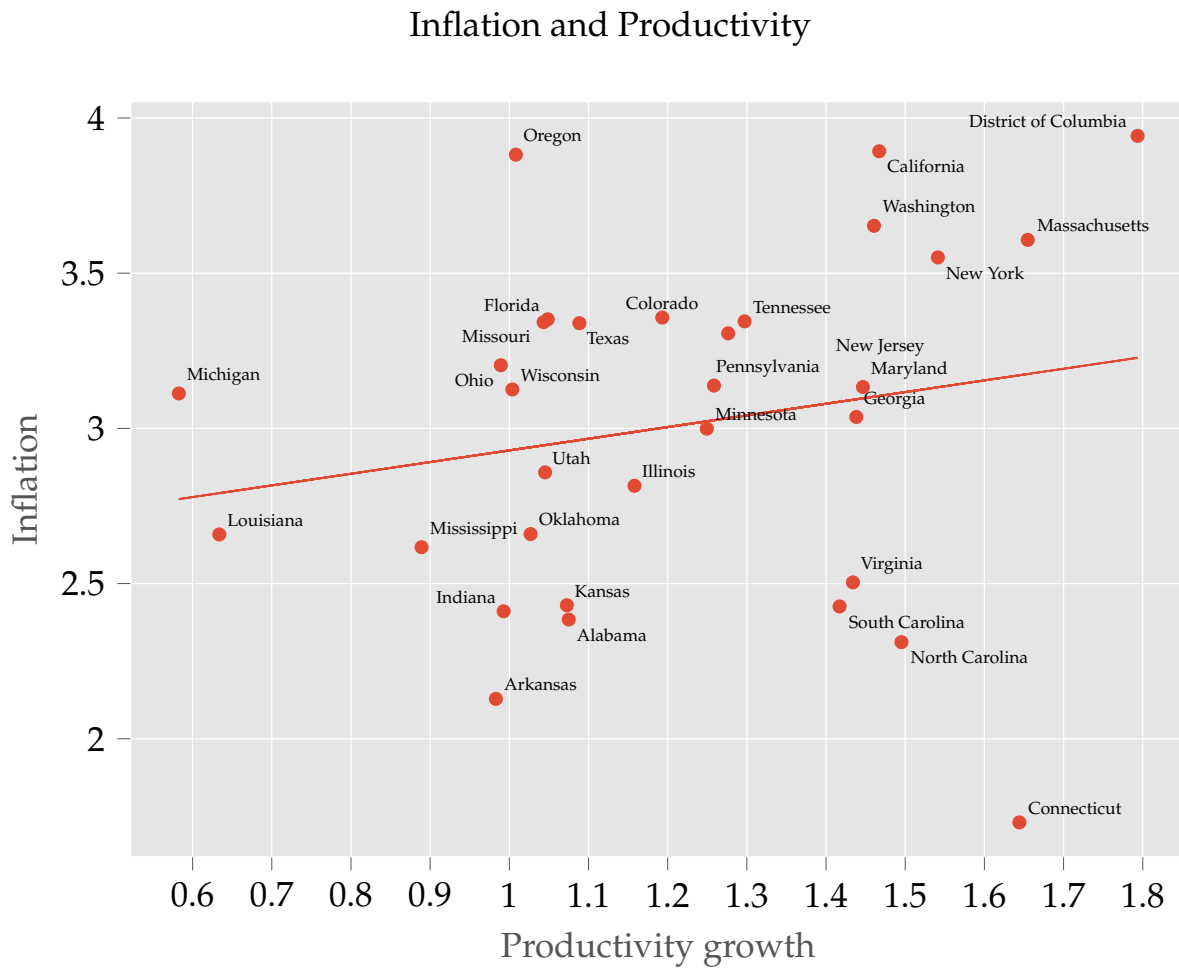


Figure 1: Inflation and Productivity Growth. This figure plots the average inflation against the average productivity growth.

Inflation and Innovation

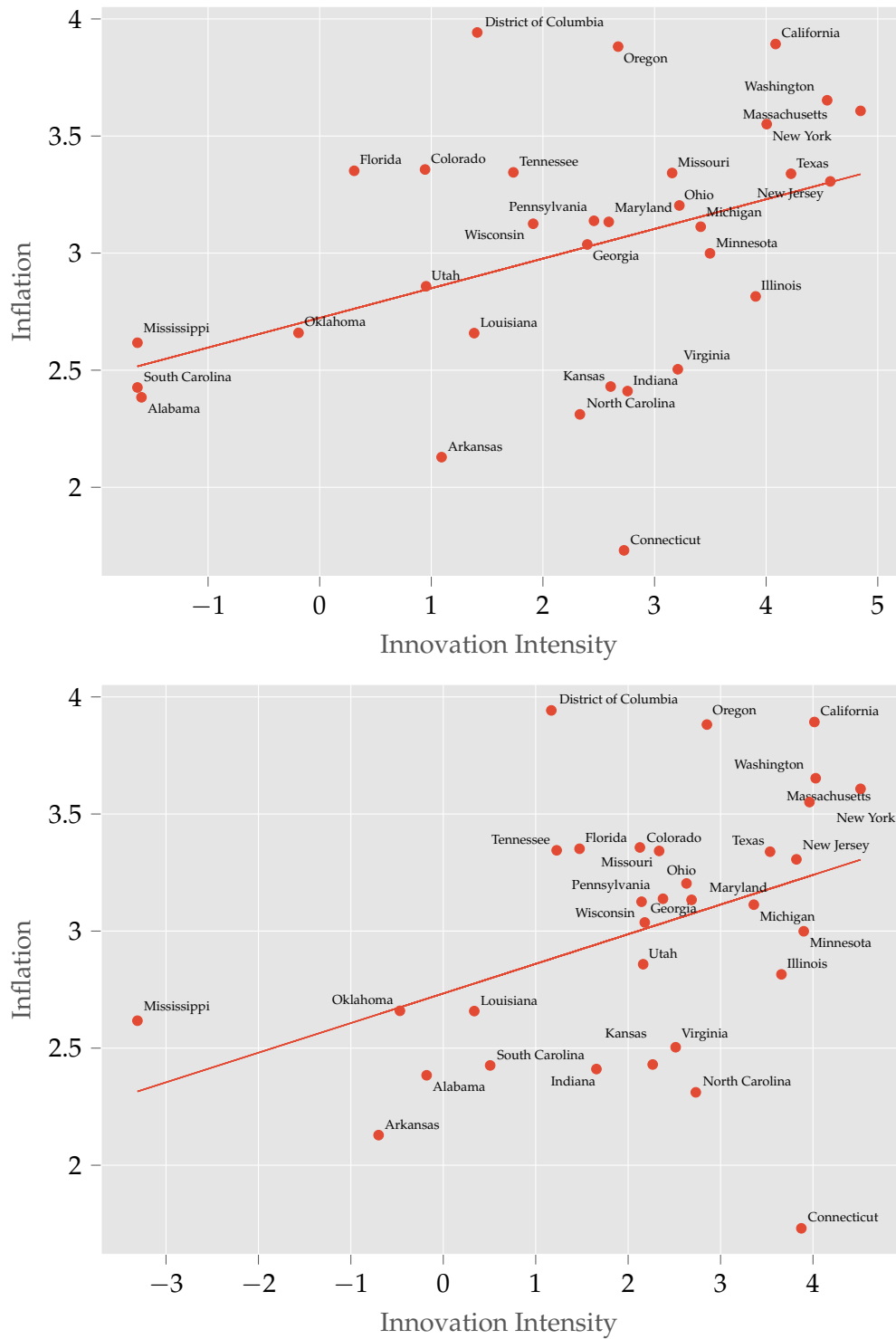


Figure 2: Inflation and Innovation. This figure plots the average state-level inflation against the average innovation intensity across U.S. states. Innovation is measured by the log number of patents in the top 20% of the annual patent value (upper panel) / citations (lower panel) distribution (as in [Kogan et al. \(2017\)](#)) per million population.

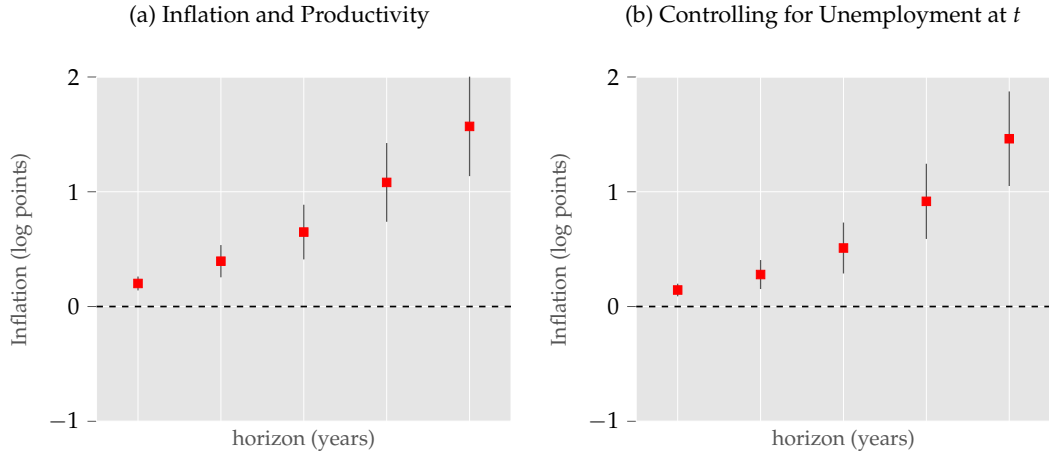


Figure 3: Inflation and Productivity Growth, the horizon ranges from 1 to 5 years. Left panel reports the coefficients β (times 100) in the contemporaneous regression of inflation on productivity, for $s = 1$ to 5.

$$\pi_{i,t,t+s} = \beta g_{i,t,t+s} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}$$

Dependent variable is the inflation of state i from t to $t + s$, $\pi_{i,t,t+s} = \ln Y_{i,t+s} - \ln Y_{i,t}$. Y is the headline price level. Right panel reports the coefficients on the contemporaneous productivity growth, controlling for the unemployment rate at t . All specifications control for state and year fixed effects, the log price level $\ln Y_{i,t}$, and the log level of independent variables $\ln X_{i,t}$ for $X =$ productivity.

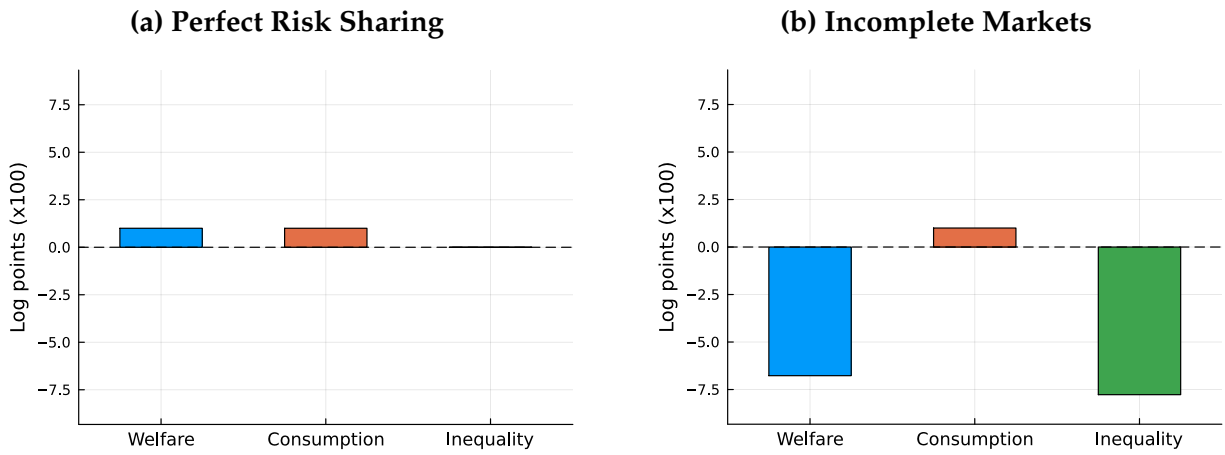


Figure 4: Decomposition of aggregate welfare effects. The figure plots the decomposition of aggregate welfare changes (average of Home and Foreign) following a positive innovation shock in both the home and foreign region. Bars are rescaled to correspond to an aggregate innovation shock that raises aggregate output by 1%. Panel (a) displays the results under the Perfect Risk Sharing benchmark, while Panel (b) shows the Incomplete Markets baseline. All quantities are reported in log points (multiplied by 100).

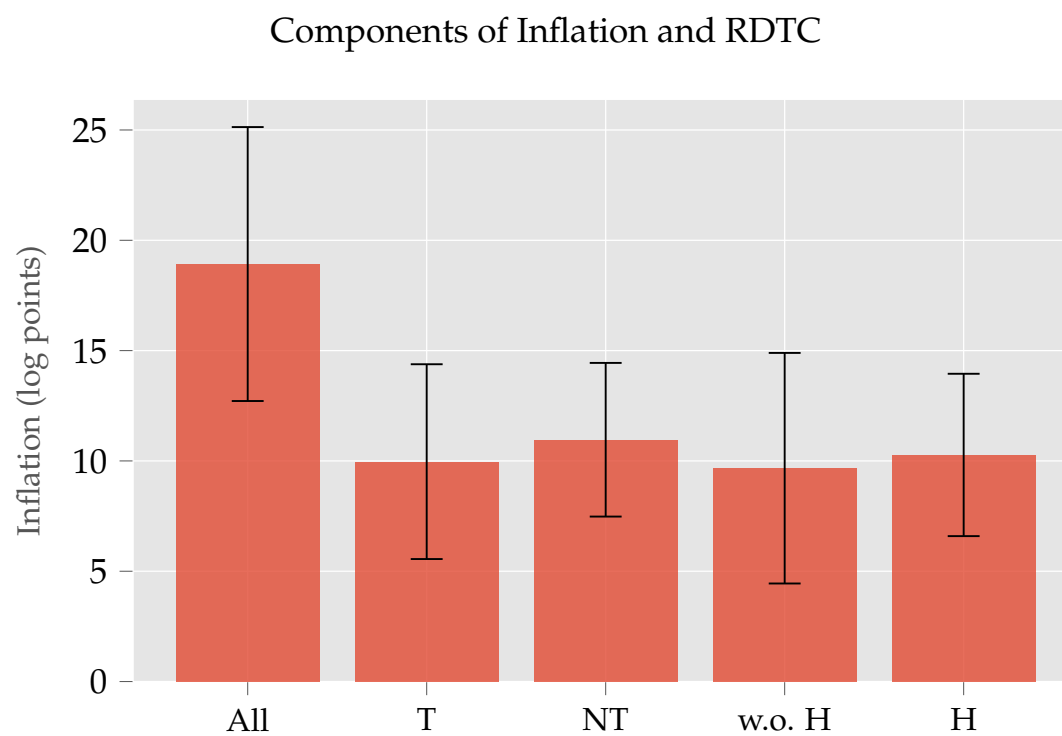
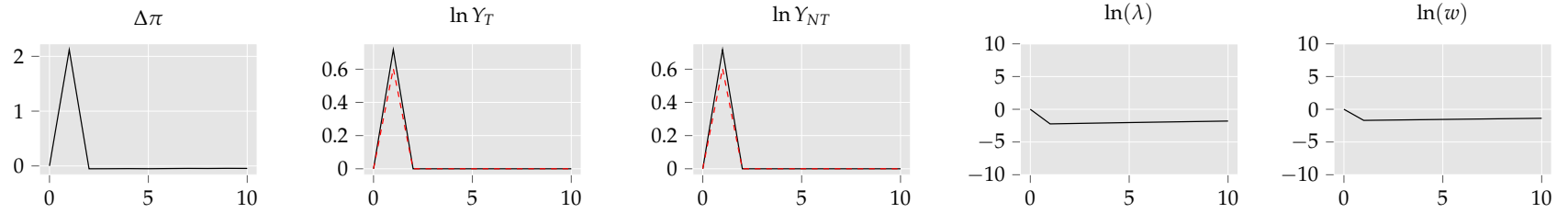


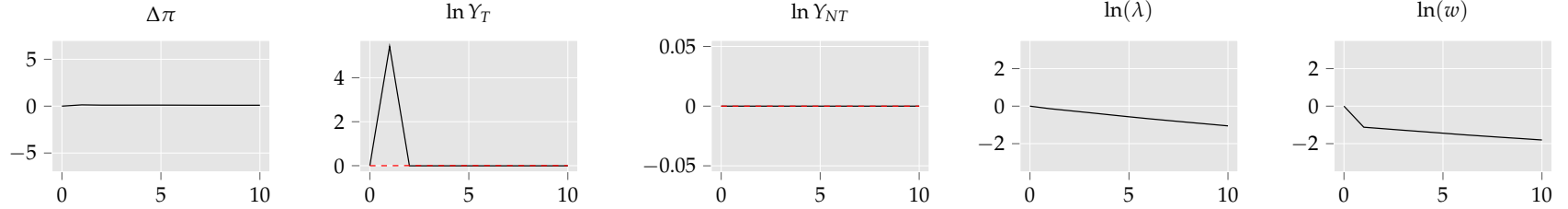
Figure 5: Components of inflation and R&D tax credits, horizon = 5 years. This figure reports the coefficient (times 100) of regression (4) where both dependent and independent variables are standardized to a unit standard deviation. All: headline inflation; T: tradables; NT: non-tradables, w.o. H: headline excluding housing; H: housing. Coefficients are reported in 100 × log points.

Model Impulse Responses

A. Response to Displacement Shock (u)



B. Response to Neutral Shock in Tradable Sector (ε_T)



C. Response to Neutral Shock in Non-Tradable Sector (ε_{NT})

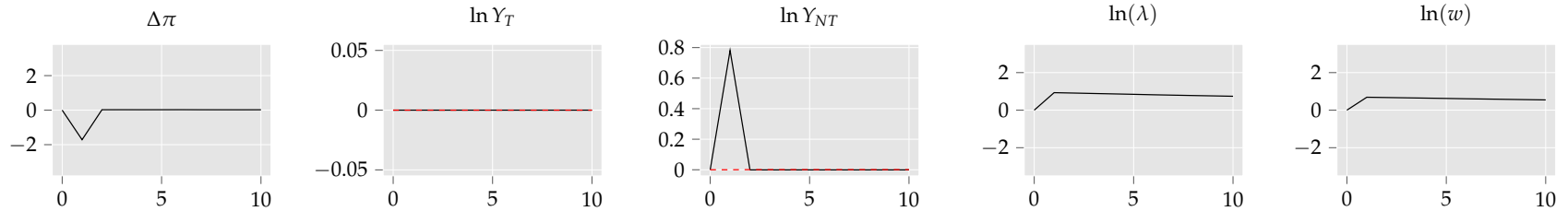
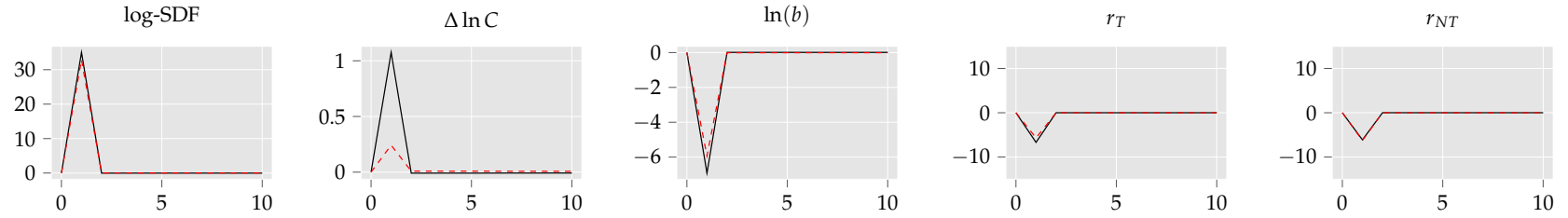
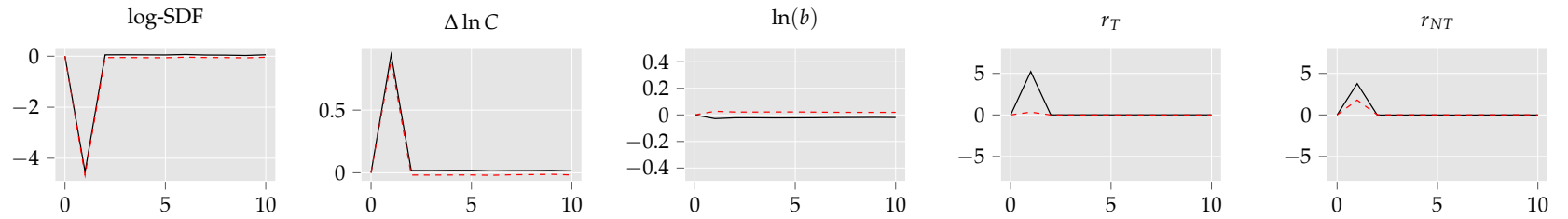


Figure 6: This figure plots the impulse responses of key variables to a shock in the home region (u in Panel A, ε_T in Panel B, and ε_{NT} in Panel C), for both the home region (solid line) and the foreign region (dashed line). All parameters are calibrated to the values reported in Table 8. In Panel A, the impulse response is constructed by simulating a transition from the low to the high state ($u = u_h$) at $t = 1$. In Panels B and C, we introduce a one-standard-deviation shock at $t = 1$. In all cases, we hold the realization of future shocks constant. The impulse responses are computed starting from the symmetric steady state.

Model Impulse Responses (cont.)
A. Response to Displacement Shock (u)



B. Response to Neutral Shock in Tradable Sector (ε_T)



C. Response to Neutral Shock in Non-Tradable Sector (ε_{NT})

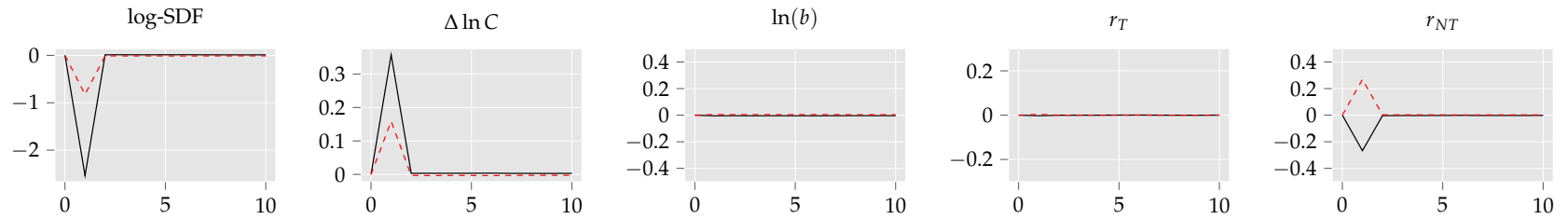
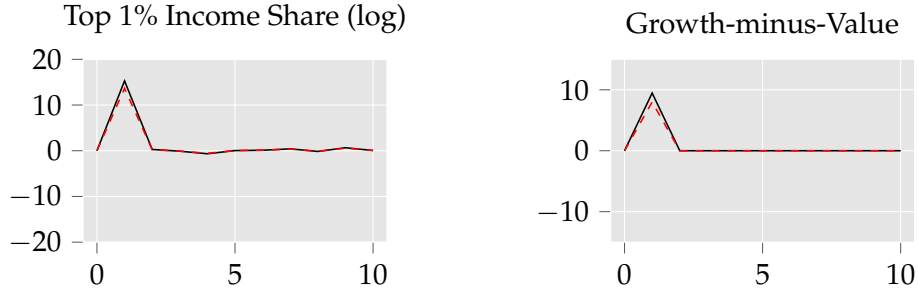


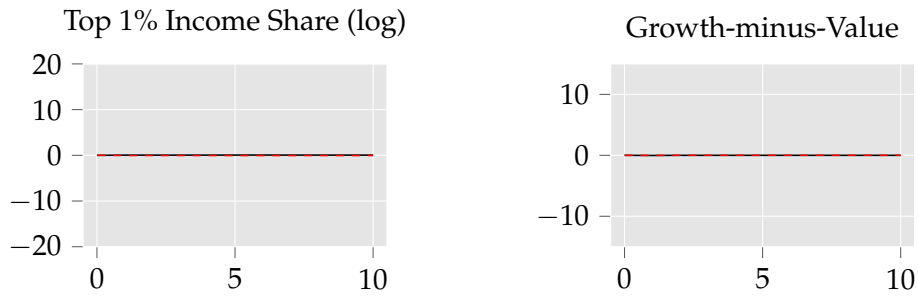
Figure 7: This figure plots the impulse responses of key variables to a shock in the home region (u in Panel A, ε_T in Panel B, and ε_{NT} in Panel C), for both the home region (solid line) and the foreign region (dashed line). All parameters are calibrated to the values reported in Table 8. In Panel A, the impulse response is constructed by simulating a transition from the low to the high state ($u = u_h$) at $t = 1$. In Panels B and C, we introduce a one-standard-deviation shock at $t = 1$. In all cases, we hold the realization of future shocks constant. The impulse responses are computed starting from the symmetric steady state.

Model Impulse Responses (cont.)

A. Response to Displacement Shock (u)



B. Response to Neutral Shock in Tradable Sector (ε_T)



C. Response to Neutral Shock in Non-Tradable Sector (ε_{NT})

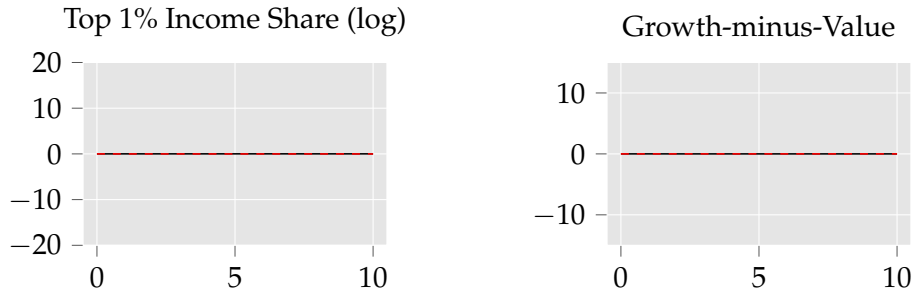


Figure 8: This figure plots the impulse responses of key variables to a shock in the home region (u in Panel A, ε_T in Panel B, and ε_{NT} in Panel C), for both the home region (solid line) and the foreign region (dashed line). All parameters are calibrated to the values reported in Table 8. In Panel A, the impulse response is constructed by simulating a transition from the low to the high state ($u = u_h$) at $t = 1$. In Panels B and C, we introduce a one-standard-deviation shock at $t = 1$. In all cases, we hold the realization of future shocks constant. The impulse responses are computed starting from the symmetric steady state.

Appendix

Section A describes the data sources and the construction for the inflation series. In Section B, we describe the construction of estimation targets and our methodology for estimation. Detailed derivations are provided in Section C.

A Data Appendix

Section A.1 includes variable definitions. Section A.2 discusses the measurement of effective changes in state-level R&D tax credit policies.

A.1 Definition of Variables

We describe the definitions of variables.

State-level inflation

Hazell et al. (2022) constructed state-level price indexes for the U.S. using data collected by the BLS for the purpose of computing the CPI. However, the dataset does not include rent prices, which are used to construct the shelter component of the CPI and account for approximately one-third of the expenditure weight. Therefore, we compute state-level headline inflation as a weighted average of housing price appreciation and the non-housing inflation measures. The housing price data comes from the Federal Housing Finance Agency (FHFA), which produces nominal state-level price indices for all states.

In constructing price indices, we closely follow the steps outlined in Hazell et al. (2022), which provides a simplified version of the procedure used by the BLS to calculate the CPI. Specifically, the first step is to compute price ratios for all-items excluding housing and for housing itself, as follows:

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} \quad i \in HHNS, H, \quad (\text{A.1})$$

where *HHNS* refers to the inflation data for all-items excluding housing from Hazell et al. (2022), and *H* refers to housing price appreciation. In the second step, we aggregate the price ratios using a weighted geometric average:

$$R = \prod r_i^{w_i / \sum w_i} \quad (\text{A.2})$$

As in Hazell et al. (2022), we use fixed CPI expenditure weights for housing, specifically assigning a weight of 32.9% to housing.

[Hazell et al. \(2022\)](#) also provides two components of inflation: tradables and non-tradables. Non-tradables, as defined in their work, primarily consist of services, with tradables being the complement of non-tradables. We construct the non-tradable category by aggregating their non-tradable inflation with housing price data using equation (A.2). In total, we consider the following categories:

- Headline: a combination of all-items inflation from [Hazell et al. \(2022\)](#) and housing price data.
- Tradables: tradables inflation from [Hazell et al. \(2022\)](#).
- Non-tradables: a combination of non-tradable inflation from [Hazell et al. \(2022\)](#) and housing price data.
- W.O. housing (all items): all-items inflation from [Hazell et al. \(2022\)](#).
- Housing: housing price appreciation.

Measures of innovation

- Citations 20%: Important patents per capita in the vicinity of state i (top 20% forward citations within the issuing year).
- KPST8005: Total number of breakthrough patents divided by total population of the *vicinity* of state i . Breakthrough patents are from [Kelly et al. \(2021\)](#) using top 20% cutoff and a 5-year window.
- KPSS (20%): Total number of breakthrough patents divided by total population of the *vicinity* of state i . Breakthrough patents are from [Kogan et al. \(2017\)](#) using top 20% cutoff and a 5-year window.

Measures of stock market return

- GMV: value-weighted growth-minus-value using median book-to-market cutoff of each sector. The set of stocks are firms with headquarters located in the *vicinity* of state i . The book value is adjusted according to [Eisfeldt and Papanikolaou \(2013\)](#).
 - Tradable: Division A - Agriculture, Forestry, And Fishing, Division B - Mining, Division D - Manufacturing
 - Non-tradable: Division G - Retail and Restaurants
 - Construction: Division C - Construction
 - Others: other divisions

- Local MKT: value-weighted market return. The set of stocks are firms with headquarters located in the *vicinity* of state i .

Measures of median wages from PSID

The PSID, which began in 1968, collected data from a sample of approximately 5,000 households. The PSID includes a variety of socioeconomic characteristics of the household, including education, food spending, and the income of household members. Notably, questions about labor income are retrospective; for example, those asked in 1993 refer to the 1992 calendar year.

Our analysis focuses on individuals aged 20 to 59 during the period from 1978 to 2017, with complete geographical information. After imposing these filters, we have 393,228 observations in the sample.

It is important to note that different states are represented unequally in our sample. In the initial year, 1978, states such as Connecticut, Kansas, Oklahoma, and Utah had fewer than 50 observations. Over time, the number of observations increased steadily. By the end of the sample period in 2017, only Connecticut still had fewer than 50 observations, while 27 out of 32 states had more than 100 observations.

Measures of median wages from BLS-OEWS

The data is sourced from the Bureau of Labor Statistics, Occupational Employment and Wage Statistics tables. For our analysis, we use the median wage across all occupations. The sample period spans from 2001 to 2017, as data on the median wage across all occupations is not available prior to 2001.

Measures of median income from Census

Annual median income at the state-level is obtained from the Census Bureau and it is adjusted by annual aggregate CPI. The data is obtained from Fred. For instance, the data for Alabama state can be downloaded from [source](#).

Measures of wages from CPS

We use data from the Current Population Survey (CPS). We restrict our sample to individuals aged 18 to 60. We classify workers into two skill groups based on educational attainment. High-skill workers are defined as individuals with a bachelor's degree or at least four years of college education, while low-skill workers include the remainder of the sample.

Other variables

We obtain the income inequality measure from [Frank \(2009\)](#), which is based on IRS records and has been updated to cover our sample period ([source](#)). Local economic indicators such as GDP, population, consumption are sourced from BEA regional economic accounts. GDP implicit deflator (BEA account code: A191RD) is used to calculate real variables. IPO data is obtained from the SDC Platinum new issuance database, and stock market data comes from the CRSP database.

- Top x%: Share of income owned by the top x% (x being equal to 1,10,0.01,0.1) of the income distribution of state i . We take 5-year average of this series.
- unemployment: unemployment rate of state i .
- Employment: employment of state i .
- GDP: real GDP of state i .
- Productivity: GDP/Employment of state i

A.2 Description of state-level R&D tax credit policy

We follow [Bloom et al. \(2013\)](#) and consider the Hall-Jorgenson user cost of R&D capital for firms in state i , denoted by $\rho_{i,t}$:

$$\rho_{i,t} = \frac{(1 - D_{i,t})}{(1 - \tau_{i,t})} \left[r_t + \delta + \frac{\Delta p_t}{p_{t-1}} \right] \quad (\text{A.3})$$

where r_t is the real interest rate, δ is the depreciation rate of R&D capital, $\frac{\Delta p_t}{p_{t-1}}$ is the growth of R&D asset price, $D_{i,t}$ is the effective R&D credit rate, $\tau_{i,t}$ is the corporate tax rate. Since $r_t + \delta + \frac{\Delta p_t}{p_{t-1}}$ does not vary between firms, we focus on the tax price component of the user cost. That is, we focus on the variation in $D_{i,t}$ across states and time. Tax credits apply to R&D performed within the state that can be offset against state-level corporate tax liabilities.

We use the state-by-year R&D tax price data, which is compiled in [Wilson \(2009\)](#) and extended by [Bloom et al. \(2013\)](#). [Wilson \(2009\)](#) quantifies the impact of state-level tax credits. In particular, the effective tax credit rate $D_{i,t}$ varies over time and across states depending on the design of the policy. There are three primary designs for R&D tax credits (i) non-incremental, where all qualified R&D is eligible for credit (ii) incremental with a fixed-period base, where eligibility is limited to R&D expenditures exceeding a baseline determined by a company's activity (e.g., R&D to sales ratio) over a fixed period and (iii) incremental with a moving-average base, where the base level is determined by the company's recent activities. For instance, Ohio implements an incremental approach with a moving-average base calculated from the past three years, Pennsylvania sets its moving-average window at four years, West Virginia offers a non-incremental tax credit, and Arizona aligns its criteria with the federal guidelines for R&D tax credits.

For both non-incremental credits and incremental credits with a fixed period base, the effective credit rate on a marginal unit of R&D, assuming that the current R&D is above the base, is $D_{i,t} = \tilde{D}_{i,t}$, where $\tilde{D}_{i,t}$ represents the statutory credit rate. For incremental credits with a moving-average base, the base is current sales multiplied by the company's average R&D to sales ratio over n previous years. The effective credit rate in this case is $D_{i,t} = \tilde{D}_{i,t}(1 - \frac{1}{n}(1 + r_{t-s})^{-s})$, where r_{t-s} is the real interest rate. In our calculation, we use $r = 2\%$ as an average real interest rate during the periods of study. Again, it is assuming that current R&D is above the base. As is well-known, the moving-average formula significantly reduces the value of the credit, as current R&D spending lowers the amount of R&D that qualified for future credits. Note that several states piggyback on the federal definition of R&D base amount, and this was changed in 1990 from a moving-average base of 3 years to a fixed-period base. This increases the effective credit rates in these states for years after 1990. Therefore, we use the moving average formula to calculate the tax credit rate for these states before 1990 and the fixed base formula afterwards.

B Calibration Details

Section B.1 describes the target moments and Section B.2 describes how we fit the model to the data.

B.1 Moments

- Output growth: Output is real gross domestic product per capita, from the BEA regional account. To calculate this variable, we obtain GDP and population series from the BEA regional account, and deflate the series using the GDP implicit deflator (BEA account code: A191RD).
- Consumption growth: Consumption is real consumption per capita, sourced from the BEA regional account. To calculate this variable, we obtain the Personal Consumption Expenditures series from the BEA regional account and deflate the series using the Consumption implicit deflator (BEA account code: DPCERD).
- Inflation volatility: The inflation series for each state is constructed as described in Section A.1. We use the volatility of the inflation series as our target moment, as it corresponds to our regression setup.
- Innovation-driven pass-through: This moment is defined as the ratio of the estimated inflation response to the estimated productivity response following a one-standard-deviation RDTC shock. Let $\hat{\beta}_\pi$ denote the slope estimate from the inflation response regression reported in Table 3, column (4), and let $\hat{\beta}_A$ denote the slope estimate from the productivity response regression reported in Table 3, column (2). We measure pass-through as $\frac{\hat{\beta}_\pi}{\hat{\beta}_A}$.

- Asset Prices
 - Risk-free rate: The risk-free rate during the sample period is obtained from the Fed (1-year US Treasury yield). The nominal interest rate is adjusted by subtracting the corresponding headline inflation rate to obtain the real interest rate. The headline inflation data is sourced from the Bureau of Labor Statistics.
 - Equity Risk Premium: The equity risk premium is estimated as the sample mean of annual excess returns (market return minus the real risk-free rate).
 - Value Premium: The value premium is estimated as the sample mean of annual portfolio return of growth-minus-value portfolio.
- Top 1% income share is estimated as the sample mean and the data is from the world inequality database.
- To examine the correlations between variables X and Y , we adopt a residualization approach whereby both X and Y are residualized with respect to state and year fixed effects, as well as the lagged levels of X and Y . This approach is consistent with the regression specification outlined in the paper. In particular,
 - Inflation and innovation: the correlation is calculated after both innovation and inflation are residualized with respect to fixed effects and their lagged levels.
 - Inflation and inequality growth: the correlation is calculated after both inequality and inflation are residualized with respect to fixed effects and their lagged levels
 - Inflation and consumption/output growth: the correlation is calculated after both consumption/output and inflation are residualized with respect to fixed effects and their lagged levels
- Pairwise Correlations: The correlation is calculated as the average of all the state-pairs in our sample. The computing horizon is one year.
 - Consumption/Output growth: Constructed from the BEA regional account as described above. We calculate the correlation using each pair of states and then compute the average.
 - Stock market return: Defined as the return of the value-weighted market portfolio of the vicinity of state i . Data is from CRSP. We calculate the correlation using each pair of states and then compute the average.

We report empirical standard errors for all moments. Our implementation expresses each moment as an OLS coefficient (either an intercept or a slope), and then uses heteroskedasticity- and autocorrelation-robust covariance estimators that allow for dependence in panel data.

For moments that are simple averages (“first moments”), we estimate by running a constant-only regression on the corresponding state-year panel series. Standard errors are computed using (Driscoll and Kraay, 1998) standard errors that are consistent with our empirical implementation.

For the innovation-driven pass-through moments, we estimate their standard errors using a 2SLS specification that treats the pass-through as the causal response of inflation or inequality to productivity, instrumenting productivity with the one-standard-deviation RDTC shock. Its standard error is the 2SLS standard error from this IV regression.

For moments that are correlations between two variables in the state-year panel, we estimate them as regression slopes. Following the residualization approach (essentially FWL), we first partial out state fixed effects, year fixed effects, and lagged levels of the relevant variables. Let $\tilde{x}_{it}$ and $\tilde{y}_{it}$ denote the resulting residualized series. We then standardize each residualized series by its sample standard deviation and run $\tilde{y}_{it}^s = \alpha + \rho \tilde{x}_{it}^s + \varepsilon_{it}$, where the superscript s denotes standardization to unit variance. Under this normalization, the slope $\hat{\rho}$ is the empirical (partial) correlation between the two residualized series. Standard errors for $\hat{\rho}$ are computed using (Driscoll and Kraay, 1998) adjustments, consistent with the panel dependence structure in our main regressions.

The aggregate moments such as equity risk premium, value premium, price-dividend ratio of tradable and nontradable sector, are estimated as the sample mean. We report the corresponding standard error of the mean for this time-series estimate. For income inequality moments, standard errors are computed using Newey–West adjustments to allow for serial correlation in the time series.

For pairwise correlations across states (e.g., output growth and consumption growth pairwise correlations), we implement the correlation as a stacked pair-year regression. We first construct a year-by-state panel of the relevant variable and standardize it within each state. We then stack the resulting pair-year observations and estimate the regression slope. We compute standard errors using a three-way clustered covariance estimator that clusters by the first state index i , the second state index j , and year t .

B.2 Calibration Procedure

We set the weight on tradable goods to 0.34, consistent with Hazell et al. (2022). We set the degree of co-integration to 0.1% and the death rate to 0.025. We calibrate the firm’s transition probability p, q directly from the data. In particular, each year we sort firms into growth and value categories based on the median book-to-market breakpoint at the local market (the vicinity) of each state. Then we estimate the transition probability over the sample period across all the states. We set the relative risk aversion $\gamma = 6$, consistent with the asset pricing literature.

We estimate the remaining parameters of the model using a simulated minimum distance method Lee and Ingram (1991). Specifically, given a vector X of target statistics in the data, we

obtain parameter estimates by

$$\hat{p} = \arg \min_{p \in \mathcal{P}} \left(X - \frac{1}{S} \sum_{i=1}^S \hat{X}_i(p) \right)' W \left(X - \frac{1}{S} \sum_{i=1}^S \hat{X}_i(p) \right), \quad (\text{B.1})$$

where $\hat{X}_i(p)$ is the vector of statistics computed in one simulation of the model. Our choice of weighting matrix $W = \text{diag}(XX')^{-1} I_W$ penalizes proportional deviations of the model statistics from their empirical counterparts. I_W is the identity matrix.

Our estimation targets are reported in the first column of Table 9. They include a combination of first and second moments of aggregate quantities, risk-free rates and inflation. In addition to these standard moments in the literature, we also target a set of correlations at the five-year horizon. The neutral shock and displacement shock have different implications for the correlation between real quantities, innovation and inflation. Thus, the set of correlation between inflation, output and innovation, are informative about the relative magnitude of these two sets of shocks.

In addition, we target the correlation between inequality, output and inflation. An important distinction between these two shocks is that displacement shock drives inequality dynamics whereas neutral shock does not. These sets of correlations help determine the size of displacement shocks.

We simulate the model at an annual frequency. For each simulation, we simulate the data for 50 years – roughly the same length as our empirical sample. The simulation starts with the symmetric steady state where $\lambda = 1, Y_T = Y_T^*, Y_{NT} = Y_{NT}^*$. In each iteration we simulate $S = 1000$ samples, and simulate pseudo-random variables using the same seed in each iteration.

We compute standard errors for the vector of parameter estimates $\hat{p}$ as

$$V(\hat{p}) = \left(1 + \frac{1}{S} \right) \left(\frac{\partial}{\partial p} \mathcal{X}(p)' W \frac{\partial}{\partial p} \mathcal{X}(p) \right)^{-1} \frac{\partial}{\partial p} \mathcal{X}(p)' W' V_X(\hat{p}) W \frac{\partial}{\partial p} \mathcal{X}(p) \left(\frac{\partial}{\partial p} \mathcal{X}(p)' W \frac{\partial}{\partial p} \mathcal{X}(p) \right)^{-1} \quad (\text{B.2})$$

where

$$V_X(\hat{p}) = \frac{1}{S} \sum_{i=1}^S (\hat{X}_i(p) - \mathcal{X}(\hat{p})) (\hat{X}_i(p) - \mathcal{X}(\hat{p}))'$$

is the estimate of the sampling variation of the statistics in X computed across simulations. The standard errors are computed using the sampling variation of the target statistics across simulations (B.2).

Solving each iteration of the model is costly, and thus computing the minimum (B.1) using standard methods is infeasible. We therefore use the Radial Basis Function (RBF) algorithm in Björkman and Holmström (2000). The Björkman and Holmström (2000) algorithm first fits a response surface to data by evaluating the objective function at a few points. Then, it searches for a

minimum by balancing between local and global search in an iterative fashion. We use a commercial implementation of the RBF algorithm that is available through the TOMLAB optimization package.

C Detailed Model Derivations

In this section, we provide detailed derivations. We begin in Sections C.1 to C.5 by focusing on the log utility case, describing how quantities are aggregated in each region. Next, we calculate the equilibrium allocation across regions and the SDF in each region. We then discuss the specific form of the solution in three cases: when displacement shocks occur only in the non-tradable sector, only in the tradable sector, and the general case where displacement shocks occur in both sectors. Section C.6 provides a microfoundation for our reduced form assumptions on output. Section C.8 provides derivations for the CRRA case.

C.1 Representative agents

The representative agent in each region

First, we show that it is possible to construct a representative agent within each region. Although agents within each region are heterogeneous in terms of their wealth, their consumption-wealth ratios are equalized due to the homotheticity of their preferences.

Consider the home region. We define the representative agent as

$$U_t = \int_{h \in [0,1]} U_{h,t} w_t^h \quad (\text{C.1})$$

where $U_{h,t}$ and w_t^h denote the utility and the wealth share of household h . That is, the representative agent takes the region-level endowment and the wealth distribution as given and maximizes the wealth-weighted utility.

Because all agents within a region are solving the same optimization problem (up to scaling determined by their wealth), the wealth-weighted representative agent does so as well. Put differently, the representative agent behaves in the same way as any individual agent but is scaled up to a level of wealth equal to the region's total wealth. Thus, solving for the heterogeneous-agent equilibrium within a region is equivalent to finding the optimal solution for the representative agent. Consequently, the equilibrium allocation problem reduces to a problem with two (representative) agents and incomplete markets.

Aggregation

The home's representative agent's utility can be written as

$$U_t = \sum_{s=t}^{\infty} \beta^{s-t} \ln C_s \quad (\text{C.2})$$

With incomplete markets, the usual construction of a planner's utility as a weighted sum, with constant weights, of individual representative utility function is not possible. Instead, we will employ a fictitious planner with stochastic weights maximizing his utility subject to the resource constraints:

$$\begin{aligned} \max_{\{C_{T,t}, C_{T,t}^*, C_{NT,t}, C_{NT,t}^*\}_{t=0,1,2,\dots}} & \sum_t \beta^t (\ln C_t + \lambda_t \ln C_t^*) \\ \text{s.t.} \quad & C_{T,t} + C_{T,t}^* = \bar{Y}_{T,t} \\ & C_{NT,t} = Y_{NT,t} \\ & C_{NT,t}^* = Y_{NT,t}^* \\ & C_t = (C_{T,t})^\alpha (C_{NT,t})^{1-\alpha} \\ & C_t^* = (C_{T,t}^*)^\alpha (C_{NT,t}^*)^{1-\alpha} \end{aligned} \quad (\text{C.3})$$

where we have normalized the weight on the home representative agent to be equal to one and assigned the weight λ_t to the foreign representative agent.

C.2 Allocations and inflation

Allocations

For concreteness, consider the home consumer. First, at each t , we derive the consumer's demands for tradable and nontradable goods, keeping overall consumption expenditure $\mathcal{C}_t$ fixed:

$$\max_{\{C_{T,t}, C_{NT,t}\}} \alpha \ln C_{T,t} + (1 - \alpha) \ln C_{NT,t} \quad (\text{C.4})$$

$$\text{s.t.} \quad C_{T,t} + P_{NT,t} C_{NT,t} = \mathcal{C}_t. \quad (\text{C.5})$$

The problem for the foreign household is defined analogously. Given the above, we obtain the following demands:

$$C_{T,t} = \alpha \mathcal{C}_t, \quad C_{NT,t} = (1 - \alpha) \frac{\mathcal{C}_t}{p_{NT,t}} \quad (\text{C.6})$$

$$C_{T,t}^* = \alpha \mathcal{C}_t^*, \quad C_{NT,t}^* = (1 - \alpha) \frac{\mathcal{C}_t^*}{p_{NT,t}^*} \quad (\text{C.7})$$

Denote by V_t the utility of the home-region's representative agent. Denoting the partial deriva-

tives with respect to consumption basket by $V_{C,t}$, we obtain

$$\frac{\partial V_t}{\partial C_t} = \frac{1}{C_t}$$

The intertemporal marginal rate of substitution of the representative agent of region c is

$$M_{t,t+1} = \beta \frac{V_{C,t+1}}{V_{C,t}} \quad (\text{C.8})$$

Trade implies that the marginal utility of the tradable good T for $t = 1, 2, \dots$ in each possible state is

$$\beta^t V_{C,t} C_t \frac{\alpha}{C_{T,t}} = \frac{\alpha}{C_{T,t}^*} C_t^* V_{C,t}^* \beta^t \quad (\text{C.9})$$

Define the date- t Pareto weights Λ_t as

$$\begin{aligned} \Lambda_t &= \Lambda_0 \beta^t V_{C,t} C_t \\ &= \Lambda_{t-1} \beta \frac{V_{C,t}}{V_{C,t-1}} \frac{C_t}{C_{t-1}} = \Lambda_{t-1} M_{t-1,t} \exp(\Delta c_t) \end{aligned} \quad (\text{C.10})$$

Since the economy starts with a symmetric setup, $\Lambda_0 = \Lambda_0^*$. We can rewrite (C.9) as

$$\Lambda_t \frac{\alpha}{C_{T,t}} = \frac{\alpha}{C_{T,t}^*} \Lambda_t^*$$

Recall that the ratio of Pareto weights by $\lambda_t = \frac{\Lambda_t^*}{\Lambda_t}$. Then the optimality condition can be written as

$$\lambda_t = \frac{C_{T,t}^*}{C_{T,t}} \quad (\text{C.11})$$

Equation (C.11) shows that λ_t is also the ratio of consumption expenditures between the foreign and home regions:

$$\lambda_t = \frac{P_{NT,t}^* C_{NT,t}^*}{P_{NT,t} C_{NT,t}} = \frac{C_t^*}{C_t}. \quad (\text{C.12})$$

Also, from (C.10) and the definition of $\lambda_t = \frac{\Lambda_t^*}{\Lambda_t}$, we infer that

$$\lambda_{t+1} = \lambda_t \frac{M_{t,t+1}^* e^{\Delta c_{t+1}^*}}{M_{t,t+1} e^{\Delta c_{t+1}}} \quad (\text{C.13})$$

Substituting the demand functions into the budget constraints, we find the optimal allocations (C.14)-(C.17):

$$C_{T,t} = \frac{1}{1 + \lambda_t} \bar{Y}_{T,t} \quad (\text{C.14})$$

$$C_{T,t}^* = \frac{\lambda_t}{1 + \lambda_t} \bar{Y}_{T,t} \quad (\text{C.15})$$

$$C_{NT,t} = Y_{NT,t} \quad (\text{C.16})$$

$$C_{NT,t}^* = Y_{NT,t}^* \quad (\text{C.17})$$

where $\bar{Y}_{T,t} = Y_{T,t} + Y_{T,t}^*$ is the total output of tradable goods.

Given the optimal allocations above, we calculate the consumption bundles:

$$C_t = (C_{T,t})^\alpha (C_{NT,t})^{1-\alpha} \quad (\text{C.18})$$

$$C_t^* = (C_{T,t}^*)^\alpha (C_{NT,t}^*)^{1-\alpha} \quad (\text{C.19})$$

We also compute the prices of consumption bundles in both regions:

$$P_t = \frac{\mathcal{C}_t}{C_t} = \frac{C_{T,t} + C_{NT,t} P_{NT,t}}{C_t} \quad (\text{C.20})$$

$$P_t^* = \frac{\mathcal{C}_t^*}{C_t^*} = \frac{C_{T,t}^* + C_{NT,t}^* P_{NT,t}^*}{C_t^*} \quad (\text{C.21})$$

The relative price level across two regions is then equal to

$$e_t \equiv \frac{P_t}{P_t^*} = \frac{C_t^*}{C_t} \frac{1}{\lambda_t} \quad (\text{C.22})$$

where the equality follows directly from (C.11). As a result, the growth in relative price levels (relative inflation) is equal to

$$\pi_{t+1} = \Delta c_{t+1}^* - \Delta c_{t+1} - \Delta \ln \lambda_{t+1}. \quad (\text{C.23})$$

Note that (C.13) and (C.23) shows that in our model relative inflation is equal to the growth of SDF, as the model has an integrated financial markets.

Moreover, the intratemporal problem with fixed expenditure $\mathcal{C}_t$ implies the indirect utility

$$U_t = \ln \mathcal{C}_t - (1 - \alpha) \ln p_{NT,t} + \alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha), \quad (\text{C.24})$$

where $p_{NT,t}$ denotes the relative price of nontradables (in units of tradables). From the viewpoint of the consumer, $p_{NT,t}$ is taken as given. Thus, it enters U_t additively and therefore drops out the portfolio-choice problem. Hence, the consumer's intertemporal optimization is equivalent to a single-good consumption-investment problem, with consumption expenditure $\mathcal{C}_t$ replacing the consumption bundle, similar to (Pavlova and Rigobon, 2007, 2015).

With log utility, the optimal consumption-expenditure-to-wealth ratio is constant. Combining this with the planner's FOC for tradable consumption (which implies $\lambda_t = C_t^* / C_t$), we obtain that

the Pareto–Negishi weight ratio λ_t equals the wealth ratio between two regions,

$$\lambda_t = \frac{W_t^*}{W_t}. \quad (\text{C.25})$$

SDF

We focus on the home region, the derivation for the foreign region is similar. Because preferences are homothetic, consumption is proportional to wealth. To calculate the SDF of the representative agent, we need to consider two population groups: the population that receives new firms in the current period (with measure ζ , denoted by N); and the population that does not receive new firms in the current period (with measure $1 - \zeta$, denoted by O).

Recall that b_{t+1} is the fraction of wealth of agents in the home region excluding the value of new projects created at time $t + 1$ at home. The time- $(t + 1)$ wealth shares of the two groups of agents within the home region are

$$b_{t+1}(1 - \zeta) \text{ and } b_{t+1}\zeta + 1 - b_{t+1}$$

respectively. Accordingly, the consumption growth for group N is $\frac{C_{t+1}}{C_t} b_{t+1}$, and the consumption growth of group O is $\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \frac{C_{t+1}}{C_t}$. The intertemporal marginal rates of substitution of the above two groups of agents can be written as

$$\begin{aligned} M_{N,t,t+1} &= \beta \left(\frac{C_{t+1}}{C_t} b_{t+1} \right)^{-1} \\ M_{O,t,t+1} &= \beta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \frac{C_{t+1}}{C_t} \right)^{-1} \end{aligned}$$

Note that the time- $t + 1$ state used to define the intertemporal marginal rates of substitution above includes the investor type, N or O, which is idiosyncratic and becomes known only at time $t + 1$. Because each agent is assigned new projects in proportion to their current wealth level and becoming an innovator in each period is independent of all other shocks in the economy, the equilibrium SDF in this economy (which prices claims contingent on *aggregate* states) can be expressed as the conditional expectation of the inter-temporal marginal rate of substitution of any agent, which is conditioned on the aggregate state at time $t + 1$. This equals a weighted average of $M_{N,t,t+1}$ and $M_{O,t,t+1}$:

$$\begin{aligned} M_{t,t+1} &= (1 - \zeta)M_{O,t,t+1} + \zeta M_{N,t,t+1} \\ &= \beta \left(\frac{C_{t+1}}{C_t} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta)b_{t+1}^{-1} \right) \end{aligned}$$

Stock Market

Stock prices of non-tradable firms are equal to

$$S_{NT} = p_{NT,t} Y_{NT,t} + E_t[M_{t,t+1}^T (S_{NT,t+1} e^{-u_{t+1}})] = p_{NT,t} Y_{NT,t} (pd_{NT,t}) \quad (C.26)$$

$$S_{NT}^* = p_{NT,t}^* Y_{NT,t}^* + E_t[M_{t,t+1}^T (S_{NT,t+1}^* e^{-u_{t+1}^*})] = p_{NT,t}^* Y_{NT,t}^* (pd_{NT,t}^*) \quad (C.27)$$

Similarly, for tradable firms, their stock prices are equal to

$$S_T = Y_{T,t} + E_t[M_{t,t+1}^T (S_{T,t+1})] = Y_{T,t} (pd_{T,t}) \quad (C.28)$$

$$S_T^* = Y_{T,t}^* + E_t[M_{t,t+1}^T (S_{T,t+1}^*)] = Y_{T,t}^* (pd_{T,t}^*) \quad (C.29)$$

where pd_T, pd_{NT} are price-dividend ratios. Hence, the total market value of the tradable (T) sector is

$$\bar{S}_{T,t} = S_T + S_T^*$$

We have normalized the price of tradable goods to be one. The total output of tradable goods is $\bar{Y}_{T,t} = Y_{T,t} + Y_{T,t}^*$. We denote $M_{t,t+1}^T$ as the SDF using the tradable good as a numeraire. Similarly, $M_{t,t+1}^{NT}$ and $M_{t,t+1}^{NT,*}$ are the SDFs using non-tradable goods as numeraire. By definition, they have the following relationship:

$$M_{t,t+1}^T = M_{t,t+1}^{NT} \frac{p_{NT,t}}{p_{NT,t+1}} = M_{t,t+1}^{NT,*} \frac{p_{NT,t}^*}{p_{NT,t+1}^*} = M_{t,t+1} \frac{p_t}{p_{t+1}} \quad (C.30)$$

Consequently,

$$M_{t,t+1}^T = \beta \left(\frac{C_{T,t+1}}{C_{T,t}} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta) b_{t+1}^{-1} \right) \quad (C.31)$$

$$M_{t,t+1}^{NT} = \beta \left(\frac{C_{NT,t+1}}{C_{NT,t}} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta) b_{t+1}^{-1} \right) \quad (C.32)$$

$$M_{t,t+1}^{NT,*} = \beta \left(\frac{C_{NT,t+1}^*}{C_{NT,t}^*} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-1} + (1 - \zeta) (b_{t+1}^*)^{-1} \right) \quad (C.33)$$

where p, p_{NT} are the prices of consumption bundles and non-tradable goods, respectively. The wealth shares of non-innovating households (households that do not receive any shares in new projects) are described in the following expressions:

$$b_{t+1} = 1 - \frac{\text{value of new projects at home region}}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{1}{1+\lambda_{t+1}} \right)} \quad (C.34)$$

$$b_{t+1}^* = 1 - \frac{\text{value of new projects at foreign region}}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{\lambda_{t+1}}{1+\lambda_{t+1}} \right)} \quad (\text{C.35})$$

To save space and describe the dynamics of SDFs, we define the following quantities:

$$\Omega_{t+1} = \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta)b_{t+1}^{-1} \right) \quad (\text{C.36})$$

$$\Omega_{t+1}^* = \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-1} + (1 - \zeta)(b_{t+1}^*)^{-1} \right) \quad (\text{C.37})$$

Given the trade of T-goods, the marginal utility of tradable goods needs to be equalized across regions. Therefore, from (C.11), (C.31), (C.36), and (C.37), we have:

$$\Delta \ln \lambda_{t+1} = \ln \Omega_{t+1}^* - \ln \Omega_{t+1} \quad (\text{C.38})$$

Local Market Portfolio Return

The realized return on the local market portfolio is approximately equal to

$$\begin{aligned} R_{t+1}^{mkt} \approx & w_{T,t} \left(\underbrace{\mu + \delta u_{t+1} + \varepsilon_{t+1}}_{\Delta \ln Y_{T,t+1}} - \underbrace{\eta u_{t+1}}_{\text{displacement on T}} + \ln(pd_{T,t+1}) - \ln(pd_{T,t} - 1) \right) \\ & + (1 - w_{T,t}) \left(\Delta \ln \tilde{Y}_{T,t+1} + \underbrace{\frac{\lambda_t}{1 + \lambda_t} (\Delta \ln \Omega_{t+1} - \Delta \ln \Omega_{t+1}^*)}_{\text{wealth effect}} - \underbrace{\eta u_{t+1}}_{\text{displacement on NT}} + \ln \frac{pd_{NT,t+1}}{pd_{NT,t} - 1} \right), \end{aligned} \quad (\text{C.39})$$

where w_T is the share of total market capitalization due to firms producing the tradable good in the home region. A positive neutral shock raises the value of the local market portfolio. A positive displacement shock, however, has two offsetting effects: it increases the value of local non-tradable firms (wealth effect) but dilutes existing firms' share (displacement effect). The net effect on market returns is therefore ambiguous.

C.3 Displacement shocks only in the non-tradable good sector

Case when $\eta = 1$.

Displacement Effects

Recall that the displacement effects are summarized by terms b_t, b_t^* , which are the wealth shares of households that do not receive profitable projects within their respective regions. In this case,

these shares are equal to

$$b_{t+1} = 1 - \frac{S_{NT,t+1}(1 - e^{-u_{t+1}})}{(S_{T,t+1} + S_{T,t+1}^* + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{1}{1+\lambda_{t+1}} \right)} \quad (\text{C.40})$$

$$b_{t+1}^* = 1 - \frac{S_{NT,t+1}^*(1 - e^{-u_{t+1}^*})}{(S_{T,t+1} + S_{T,t+1}^* + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{\lambda_{t+1}}{1+\lambda_{t+1}} \right)} \quad (\text{C.41})$$

Recall that tradable firms cater to the national demand, and their price-dividend ratio is

$$\begin{aligned} pd_{T,t} &= 1 + E_t \left[M_{t,t+1}^T \frac{\tilde{Y}_{T,t+1}}{\tilde{Y}_{T,t}} pd_{T,t+1} \right] \\ &= 1 + E_t \left[\beta \left(\frac{\tilde{Y}_{T,t+1} \frac{1}{1+\lambda_{t+1}}}{\tilde{Y}_{T,t} \frac{1}{1+\lambda_t}} \right)^{-1} \frac{\tilde{Y}_{T,t+1}}{\tilde{Y}_{T,t}} \Omega_{t+1} pd_{T,t+1} \right] \\ &= 1 + E_t \left[\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \right] \end{aligned} \quad (\text{C.42})$$

and the price-dividend ratio for NT firms is

$$\begin{aligned} pd_{NT,t} &= 1 + E_t \left[M_{t,t+1}^T \frac{p_{t+1} Y_{NT,t+1}}{p_t Y_{NT,t}} (pd_{NT,t+1}) e^{-u_{t+1}} \right] \\ &= 1 + E_t \left[M_{t,t+1}^T \frac{\tilde{Y}_{T,t+1} \frac{1}{1+\lambda_{t+1}}}{\tilde{Y}_{T,t} \frac{1}{1+\lambda_t}} (pd_{NT,t+1}) e^{-u_{t+1}} \right] \\ &= 1 + E_t \left[\beta \left(\frac{\tilde{Y}_{T,t+1} \frac{1}{1+\lambda_{t+1}}}{\tilde{Y}_{T,t} \frac{1}{1+\lambda_t}} \right)^{-1} \frac{\tilde{Y}_{T,t+1}}{\tilde{Y}_{T,t}} \frac{1}{1+\lambda_t} (pd_{NT,t+1}) e^{-u_{t+1}} \Omega_{t+1} \right] \\ &= 1 + E_t \left[\beta (pd_{NT,t+1}) e^{-u_{t+1}} \Omega_{t+1} \right]. \end{aligned} \quad (\text{C.43})$$

Similarly, for foreign non-tradable firms,

$$pd_{NT,t}^* = 1 + E_t \left[\beta (pd_{NT,t+1}^*) e^{-u_{t+1}^*} \Omega_{t+1}^* \right]. \quad (\text{C.44})$$

To obtain a closed-form solution, we conjecture and then subsequently verify that pd_T, pd_{NT}, pd_{NT}^* are constant and we denote them as $C_1 = pd_T, C_2 = \frac{1-\alpha}{\alpha} pd_{NT} = \frac{1-\alpha}{\alpha} pd_{NT}^*$. Then

$$b_{t+1} = \frac{C_1 + C_2 e^{-u_{t+1}}}{C_1 + C_2} \quad (\text{C.45})$$

$$b_{t+1}^* = \frac{C_1 + C_2 e^{-u_{t+1}^*}}{C_1 + C_2}. \quad (\text{C.46})$$

As a result, from (C.36) and (C.37), the expectation of $\Omega_{t+1}, \Omega_{t+1}^*$ are also constant. Given that u

and u^* are symmetric, we can denote them as

$$E_t \Omega_{t+1} = E_t \Omega_{t+1}^* = \chi_0 \quad (\text{C.47})$$

Next, we verify our conjecture that the price-dividend ratios are constant. First, note that

$$\begin{aligned} & E_t \left[\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \right] \\ &= \beta \frac{1}{1 + \lambda_t} E_t \left[\left(1 + \lambda_t \frac{\Omega_{t+1}^*}{\Omega_{t+1}} \right) \Omega_{t+1} \right] \\ &= \beta \frac{1}{1 + \lambda_t} E_t [\Omega_{t+1} + \Omega_{t+1}^*]. \end{aligned} \quad (\text{C.48})$$

Solving for the price-dividend ratios recursively as

$$\begin{aligned} pd_T &= 1 + E_t \left[\left(\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \right) + \left(\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \right)^2 + \dots \right] \\ &= 1 + \frac{\beta \chi_0}{1 - \beta \chi_0}. \end{aligned} \quad (\text{C.49})$$

In the above calculation, we use the fact that

$$\frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} = \frac{1 + \lambda_t \frac{\Omega_{t+1}^*}{\Omega_{t+1}}}{1 + \lambda_t} \Omega_{t+1} = \frac{\Omega_{t+1} + \lambda_t \Omega_{t+1}^*}{1 + \lambda_t} = \chi_0. \quad (\text{C.50})$$

For the price-dividend ratio in the NT sector

$$\begin{aligned} E_t [\beta \Omega_{t+1} e^{-u_{t+1}}] &= \beta \left(\zeta E_t \left(\frac{b_{t+1} \zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} e^{-u_{t+1}} + (1 - \zeta) E_t (b_{t+1}^{-1} e^{-u_{t+1}}) \right) \\ &= \beta \chi_1 \end{aligned} \quad (\text{C.51})$$

where χ_1 is a constant, defined as

$$\chi_1 = \left(\zeta E_t \left(\frac{b_{t+1} \zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} e^{-u_{t+1}} + (1 - \zeta) E_t (b_{t+1}^{-1} e^{-u_{t+1}}) \right). \quad (\text{C.52})$$

Therefore,

$$pd_{NT} = pd_{NT}^* = 1 + \frac{\beta \chi_1}{1 - \beta \chi_1}. \quad (\text{C.53})$$

Thus, we have verified our conjecture that the price-dividend ratios pd_T and pd_{NT} are constant.

Note that for small values of u , b can be approximated as

$$b = \frac{C_1 + C_2 e^{-u}}{C_1 + C_2} \approx \frac{C_1 + C_2 - C_2 u}{C_1 + C_2} = 1 - \frac{C_2}{C_1 + C_2} u. \quad (\text{C.54})$$

As a result, in the limiting case $\zeta \rightarrow 0$, the dynamics of λ can be approximated as

$$\begin{aligned} \Delta \ln \lambda_{t+1} &= \ln b - \ln b^* \\ &= \ln \frac{C_1 + C_2 e^{-u}}{C_1 + C_2} - \ln \frac{C_1 + C_2 e^{-u^*}}{C_1 + C_2} \\ &\approx \frac{C_2}{C_1 + C_2} (u^* - u) \\ &= \frac{(1 - \alpha) p d_{NT}}{\alpha p d_T + (1 - \alpha) p d_{NT}} (u^* - u). \end{aligned} \quad (\text{C.55})$$

Case when $\eta < 1$

In this case, new projects are allocated not only to the inventors but also to the existing firms. Assume there are two types of firms: those that can receive these new projects and those that cannot. The former type trades at a higher valuation multiple in equilibrium, and we denote such firms by H –“growth”. We denote the type that can not receive projects as L –“value”. Non-tradables firms can transit between these two states according to the following transition probability:

$$\Sigma = \begin{bmatrix} 1 - p & p \\ q & 1 - q \end{bmatrix}. \quad (\text{C.56})$$

Suppose that the fraction of new projects that go to H type is ω , so the L type get $1 - \omega$ of the total $1 - \eta$ projects. In our case, $\omega = 1$ and value firms do not receive projects. Among the all the NT firms, a fraction of m_H is the H type, and the remaining fraction $1 - m_H$ is the L type. In the steady state, we have

$$\begin{aligned} m_H(1 - p) + q(1 - m_H) &= m_H \\ p m_H + (1 - q)(1 - m_H) &= 1 - m_H \end{aligned}$$

$$m_H = \frac{q}{p+q}, 1 - m_H = \frac{p}{p+q}.$$

When $\eta < 1$, the wealth share (C.40) and (C.41) become

$$b_{t+1} = 1 - \frac{S_{NT,t+1}(1 - e^{-u_{t+1}})\eta}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*)(\frac{1}{1+\lambda_{t+1}})} \quad (\text{C.57})$$

$$b_{t+1}^* = 1 - \frac{S_{NT,t+1}^*(1 - e^{-u_{t+1}^*})\eta}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*)(\frac{\lambda_{t+1}}{1+\lambda_{t+1}})} \quad (\text{C.58})$$

So we have

$$\begin{aligned} b &= \frac{C_1 + C_2 - C_2\eta(1 - e^{-u})}{C_1 + C_2} \\ &\approx \frac{C_1 + C_2 - C_2\eta u}{C_1 + C_2} = 1 - \frac{C_2}{C_1 + C_2}\eta u \end{aligned} \quad (C.59)$$

The valuation of NT trees consists of two parts: asset in place (A) and growth opportunities (G). Asset in place are the same in both types of firms,

$$pd_{NT} = pd_{NT}^* = 1 + E_t \left[\beta (pd_{NT,t+1}) e^{-u_{t+1}} \Omega_{t+1} \right] \quad (C.60)$$

$$= 1 + \frac{\beta \chi_1}{1 - \beta \chi_1} \quad (C.61)$$

The value of growth opportunities, expressed as a multiple of dividends, satisfies

$$\begin{aligned} G_{H,t} &= E_t \left(M_{t,t+1}^T \frac{p_{NT,t+1} Y_{NT,t+1}}{p_{NT,t} Y_{NT,t}} \left[(1 - e^{-u_{t+1}}) pd_{NT} (1 - \eta) \left(\frac{\omega}{m_H} \right) \right. \right. \\ &\quad \left. \left. + ((1 - p)G_{H,t+1} + (p)G_{L,t+1})(1 - (1 - e^{-u_{t+1}})\eta) \right] \right) \\ G_{L,t} &= E_t \left(M_{t,t+1}^T \frac{p_{NT,t+1} Y_{NT,t+1}}{p_{NT,t} Y_{NT,t}} \left[(1 - e^{-u_{t+1}}) pd_{NT} (1 - \eta) \left(\frac{1 - \omega}{1 - m_H} \right) \right. \right. \\ &\quad \left. \left. + ((q)G_{H,t+1} + (1 - q)G_{L,t+1})(1 - (1 - e^{-u_{t+1}})\eta) \right] \right) \end{aligned}$$

From the expression of marginal utility we know that

$$M_{t,t+1}^T = \beta \left(\frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \right)^{-1} \frac{\Omega_{t+1} + \lambda_t \Omega_{t+1}^*}{(1 + \lambda_t)} \quad (C.62)$$

Similarly,

$$\frac{p_{NT,t+1} Y_{NT,t+1}}{p_{NT,t} Y_{NT,t}} = \frac{1 + \lambda_t}{1 + \lambda_{t+1}} \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \quad (C.63)$$

Substituting the above equations into the valuation of the new projects, one can see that the total value (as a multiple of dividends) of all new projects is equal to

$$\chi_G = E_t \left[M_{t,t+1}^T \frac{p_{NT,t+1} Y_{NT,t+1}}{p_{NT,t} Y_{NT,t}} (1 - e^{-u_{t+1}}) \chi_G \right] \quad (C.64)$$

$$= E_t \beta \Omega_{t+1} (1 - e^{-u_{t+1}}) \chi_G \quad (C.65)$$

The relative inflation is given by

$$\begin{aligned}
\pi_{t+1} &= \ln M_{t,t+1} - \ln M_{t,t+1}^* \\
&= -\Delta c_{t+1} + \Delta c_{t+1}^* + \ln \Omega_{t+1} - \ln \Omega_{t+1}^* \\
&= (1 - \alpha) (\ln \Omega_{t+1} - \ln \Omega_{t+1}^* - \delta u_{t+1} - \varepsilon_{t+1} + \delta u_{t+1}^* + \varepsilon_{t+1}^*)
\end{aligned} \tag{C.66}$$

Without displacement effects, relative inflation is purely determined by the change in the relative supply of non-tradable goods, $\delta u^* + \varepsilon_{t+1}^* - \delta u - \varepsilon_{t+1}$. However, displacement effects lead to a reallocation of wealth and consequently alter the demand for non-tradable goods, which is characterized by $\ln \Omega_{t+1} - \ln \Omega_{t+1}^*$. When $\zeta \rightarrow 0$, we have

$$\ln \Omega_{t+1} - \ln \Omega_{t+1}^* = \left(\frac{(1 - \alpha)pd_{NT}}{\alpha pd_T + (1 - \alpha)pd_{NT}} \right) (u_{t+1} - u_{t+1}^*) \tag{C.67}$$

So

$$\pi_{t+1} = (1 - \alpha) \left(\left(\frac{(1 - \alpha)pd_{NT}}{\alpha pd_T + (1 - \alpha)pd_{NT}} - \delta \right) (u_{t+1} - u_{t+1}^*) + \varepsilon_{t+1}^* - \varepsilon_{t+1} \right). \tag{C.68}$$

C.4 Displacement shocks only in the tradable good sector

Case when $\eta \leq 1$

Most of the equations that describe the equilibrium conditions remain unchanged. They are as follows:

$$pd_{T,t} = 1 + E_t \left[\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \frac{Y_{T,t+1}/\bar{Y}_{T,t+1}}{Y_{T,t}/\bar{Y}_{T,t}} (pd_{T,t+1}) \right] \tag{C.69}$$

$$pd_{T,t}^* = 1 + E_t \left[\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \frac{Y_{T,t+1}^*/\bar{Y}_{T,t+1}}{Y_{T,t}^*/\bar{Y}_{T,t}} (pd_{T,t+1}^*) \right] \tag{C.70}$$

$$pd_{NT,t} = 1 + E_t \left[\beta (pd_{NT,t+1}) \Omega_{t+1} \right] \tag{C.71}$$

$$pd_{NT,t}^* = 1 + E_t \left[\beta (pd_{NT,t+1}^*) \Omega_{t+1}^* \right] \tag{C.72}$$

$$b_{t+1} = 1 - \frac{\frac{Y_{T,t+1}}{\bar{Y}_{T,t+1}} (pd_{T,t+1}) (1 - e^{-u_{t+1}}) \eta}{\left(\widetilde{pd}_{T,t+1} + \frac{1-\alpha}{\alpha} (\widetilde{pd}_{NT,t+1}) \right) \left(\frac{1}{1+\lambda_{t+1}} \right)} \tag{C.73}$$

$$b_{t+1}^* = 1 - \frac{\frac{Y_{T,t+1}^*}{\bar{Y}_{T,t+1}}(pd_{T,t+1}^*)(1 - e^{-u_{t+1}^*})\eta}{\left(\widetilde{pd}_{T,t+1} + \frac{1-\alpha}{\alpha}(\widetilde{pd}_{NT,t+1})\right)\left(\frac{\lambda_{t+1}}{1+\lambda_{t+1}}\right)} \quad (\text{C.74})$$

$$\Omega_{t+1} = \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta)b_{t+1}^{-1} \right) \quad (\text{C.75})$$

$$\Omega_{t+1}^* = \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-1} + (1 - \zeta)(b_{t+1}^*)^{-1} \right) \quad (\text{C.76})$$

$$\Delta \ln \lambda_{t+1} = \ln \Omega_{t+1}^* - \ln \Omega_{t+1} \quad (\text{C.77})$$

where

$$\widetilde{pd}_{T,t+1} = \frac{Y_{T,t+1}}{\bar{Y}_{T,t+1}}pd_{T,t+1} + \frac{Y_{T,t+1}^*}{\bar{Y}_{T,t+1}}pd_{T,t+1}^* \quad (\text{C.78})$$

$$\widetilde{pd}_{NT,t+1} = \frac{1}{1 + \lambda_{t+1}}pd_{NT,t+1} + \frac{\lambda_{t+1}}{1 + \lambda_{t+1}}pd_{NT,t+1}^*. \quad (\text{C.79})$$

When $\zeta \rightarrow 0$ and u shocks are small, around the symmetric steady state, we have the following approximation:

$$\Delta \ln \lambda_{t+1} \approx \ln b_{t+1} - \ln b_{t+1}^* \quad (\text{C.80})$$

$$\approx \frac{\alpha pd_{T,s}}{\alpha pd_{T,s} + (1 - \alpha)pd_{NT,s}}(u_{t+1} - u_{t+1}^*), \quad (\text{C.81})$$

where $pd_{T,s}, pd_{NT,s}$ are the price-dividend ratio at the steady states for T and NT firms. The value of growth opportunities, expressed as a multiple of dividends, satisfies

$$G_{H,t} = E_t \left(M_{t,t+1}^T \frac{Y_{T,t+1}}{Y_{T,t}} \left[(1 - e^{-u_{t+1}})pd_{t+1}(1 - \eta) \left(\frac{\omega}{m_H} \right) + ((1 - p)G_{H,t+1} + (p)G_{L,t+1})(1 - (1 - e^{-u_{t+1}})\eta) \right] \right) \quad (\text{C.82})$$

$$G_{L,t} = E_t \left(M_{t,t+1}^T \frac{Y_{T,t+1}}{Y_{T,t}} \left[(1 - e^{-u_{t+1}})pd_{t+1}(1 - \eta) \left(\frac{1 - \omega}{1 - m_H} \right) + ((q)G_{H,t+1} + (1 - q)G_{L,t+1})(1 - (1 - e^{-u_{t+1}})\eta) \right] \right) \quad (\text{C.83})$$

where

$$\begin{aligned} M_{t,t+1}^T &= \beta \left(\frac{C_{T,t+1}}{C_{T,t}} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta)b_{t+1}^{-1} \right) \\ &= \beta \left(\frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \right)^{-1} \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \end{aligned} \quad (\text{C.84})$$

The relative inflation is given by

$$\begin{aligned}
\pi_{t+1} &= -\Delta c_{t+1} + \Delta c_{t+1}^* + \ln \Omega_{t+1} - \ln \Omega_{t+1}^* \\
&= (1 - \alpha) \left(\varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} + \ln \Omega_{t+1} - \ln \Omega_{t+1}^* \right) \\
&\approx (1 - \alpha) \left(\frac{\alpha p d_{T,s}}{\alpha p d_{T,s} + (1 - \alpha) p d_{NT,s}} (u_{t+1} - u_{t+1}^*) + \varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} \right). \tag{C.85}
\end{aligned}$$

Without displacement effects, the relative inflation is purely determined by the change in the relative supply of non-tradable goods, $\varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1}$. However, displacement effects lead to a reallocation of wealth and consequently alter the demand for non-tradable goods, which is characterized by $\ln \Omega_{t+1} - \ln \Omega_{t+1}^*$.

A positive u shock at home would always lead to a larger displacement effect at home—smaller b (the wealth share of non-innovators), larger Ω —and an increase in the relative price of home goods. The intuition is straightforward: the displacement T shock reallocates wealth to the home but does not increase the supply of non-tradables, hence the price of non-tradables increases. Therefore, a positive T displacement shock always increases local price levels.

C.5 Displacement shocks in both sectors

Case when $\eta \leq 1$.

The equilibrium conditions become:

$$p d_{T,t} = 1 + E_t \left[\beta \frac{1 + \lambda_{t+1}}{1 + \lambda_t} \Omega_{t+1} \frac{Y_{T,t+1} / \bar{Y}_{T,t+1}}{Y_{T,t} / \bar{Y}_{T,t}} (p d_{T,t+1}) e^{-u_{T,t+1}} \right] \tag{C.86}$$

$$p d_{NT,t} = 1 + E_t \left[\beta (p d_{NT,t+1}) \Omega_{t+1} e^{-u_{NT,t+1}} \right] \tag{C.87}$$

$$\begin{aligned}
b_{t+1} &= 1 - \frac{\frac{Y_{T,t+1}}{\bar{Y}_{T,t+1}} \eta \left((p d_{T,t+1}) (1 - e^{-u_{T,t+1}}) \right) + \frac{1-\alpha}{\alpha} p d_{NT} \frac{1}{1+\lambda_{t+1}} \eta \left(1 - e^{-u_{NT,t+1}} \right)}{\left(\widetilde{p d}_{T,t+1} + \frac{1-\alpha}{\alpha} \widetilde{p d}_{NT,t+1} \right) \left(\frac{1}{1+\lambda_{t+1}} \right)} \tag{C.88}
\end{aligned}$$

$$\begin{aligned}
b_{t+1}^* &= 1 - \frac{\frac{Y_{T,t+1}^*}{\bar{Y}_{T,t+1}^*} \eta (p d_{T,t+1}^*) (1 - e^{-u_{T,t+1}^*}) + \frac{1-\alpha}{\alpha} \frac{\lambda_{t+1}}{1+\lambda_{t+1}} \eta (p d_{NT}^*) (1 - e^{-u_{NT,t+1}^*})}{\left(\widetilde{p d}_{T,t+1} + \frac{1-\alpha}{\alpha} \widetilde{p d}_{NT,t+1} \right) \left(\frac{\lambda_{t+1}}{1+\lambda_{t+1}} \right)} \tag{C.89}
\end{aligned}$$

$$\Omega_{t+1} = \left(\zeta \left(\frac{b_{t+1} \zeta + 1 - b_{t+1}}{\zeta} \right)^{-1} + (1 - \zeta) b_{t+1}^{-1} \right) \tag{C.90}$$

$$\Omega_{t+1}^* = \left(\zeta \left(\frac{b_{t+1}^* \zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-1} + (1 - \zeta)(b_{t+1}^*)^{-1} \right) \quad (\text{C.91})$$

$$\Delta \ln \lambda_{t+1} = \ln \Omega_{t+1}^* - \ln \Omega_{t+1} \quad (\text{C.92})$$

where

$$\widetilde{pd}_{T,t+1} = \frac{Y_{T,t+1}}{\bar{Y}_{T,t+1}} pd_{T,t+1} + \frac{Y_{T,t+1}^*}{\bar{Y}_{T,t+1}} pd_{T,t+1}^* \quad (\text{C.93})$$

$$\widetilde{pd}_{NT,t+1} = \frac{1}{1 + \lambda_{t+1}} pd_{NT,t+1} + \frac{\lambda_{t+1}}{1 + \lambda_{t+1}} pd_{NT,t+1}^*. \quad (\text{C.94})$$

When $\zeta \rightarrow 0$ and u shocks are small, around the symmetric steady state, we obtain:

$$\Delta \ln \lambda_{t+1} \approx \ln b_{t+1} - \ln b_{t+1}^* \approx (u_{t+1} - u_{t+1}^*), \quad (\text{C.95})$$

where we use the approximation that around the symmetric steady state $Y_T \approx Y_T^*$, $u_T = u_{NT}$ and $\lambda \approx 1$. The relative inflation is equal to

$$\begin{aligned} \pi_{t+1} &= \ln M_{t,t+1} - \ln M_{t,t+1}^* \\ &= -\Delta c_{t+1} + \Delta c_{t+1}^* + \ln \Omega_{t+1} - \ln \Omega_{t+1}^* \\ &= (1 - \alpha)(\Delta \ln Y_{NT,t+1}^* - \Delta \ln Y_{NT,t+1} + \ln \Omega_{t+1} - \ln \Omega_{t+1}^*) \\ &= (1 - \alpha) \left(\varepsilon_{t+1}^* - \varepsilon_{NT,t+1} + \delta(u_{t+1}^* - u_{NT,t+1}) + \ln \Omega_{t+1} - \ln \Omega_{t+1}^* \right) \\ &\approx (1 - \alpha) \left((1 - \delta)(u_{t+1} - u_{t+1}^*) + \varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} \right). \end{aligned} \quad (\text{C.96})$$

Without displacement effects, relative inflation is purely determined by the change in the relative supply of non-tradable goods, $\varepsilon_{NT,t+1}^* - \varepsilon_{NT,t+1} + \delta(u_{t+1}^* - u_{t+1})$. However, displacement effects lead to a reallocation of wealth and consequently alter the demand for non-tradable goods, which is characterized by $\ln \Omega_{t+1} - \ln \Omega_{t+1}^*$.

Examining equilibrium conditions, a positive u_T shock at home would lead to a larger displacement effect at home—smaller b (the wealth share of non-innovators), larger Ω —and an increase in the relative price of home goods. A positive u_{NT} shock would lead to the same result. The relative magnitude of these two shocks is determined by the valuation of these firms and the relative variance of these shocks.

C.6 Microfoundation and Alternative Spatial Mechanisms

We present a two-region, two-sector production microfoundation with Schumpeterian creative destruction in the spirit of [Aghion and Howitt \(1992\)](#), locally supplied land, and (potentially) mobile labor. The purpose of this appendix is to provide a spatial environment in which the

reduced-form displacement shock used in the paper (i) raises aggregate output through higher effective productivity and (ii) reallocates profit flows from incumbents to new entrants (innovators). This reallocation generates unspanned risk and a wealth effect on local non-tradable prices. The appendix also allows us to contrast the predictions of the baseline endowment economy with those of a spatial production model, and to clarify which data moments can help distinguish our mechanism from alternative spatial channels.

To keep the exposition focused, we assume creative destruction occurs only in the non-tradable sector. The more general case in which creative destruction occurs in both sectors leads to similar insights at the cost of some additional analytic complexity.

Roadmap. Part A sets up the environment (aggregation, production, factors, households, and markets) and provides the microfoundation for the reduced-form endowment processes and profit reallocation used in the main text. Part B derives price indices and highlights the wage versus land-rent components of non-tradable prices in a spatial production economy. Part C introduces an inter-regional labor-mobility block that generates a standard spatial wage wedge while preserving tractability. Part D discusses how the mobility-cost parameter is disciplined (or bounded) using wage-gap and migration moments. Part E concludes by contrasting the predictions of this spatial model with alternative spatial mechanisms.

A. Environment and Mapping to Reduced-Form Endowments

Regions, sectors, and varieties. There are two regions $c \in \{H, F\}$ and two sectors $S \in \{T, NT\}$. Each sector is a CES aggregate of a continuum of varieties $j \in [0, 1]$. Each variety is produced by a monopolist facing CES demand.

Final goods. Non-tradable output is aggregated locally:

$$Y_{NT,t}^c = \left(\int_0^1 x_{NT,t}^c(j)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1. \quad (\text{C.97})$$

Tradable goods are aggregated at the national level:

$$Y_{T,t} = \left(\int_0^1 x_{T,t}(j)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (\text{C.98})$$

where total output of each tradable variety is the sum of quantities produced in both regions:

$$x_{T,t}(j) = x_{T,t}^H(j) + x_{T,t}^F(j). \quad (\text{C.99})$$

Intermediate production. Each variety is produced with Cobb–Douglas technology using local land and local labor:

$$x_{T,t}^c(j) = Z_{T,t}^c A_{T,t}^c(j) (K_{T,t}^c(j))^\vartheta (L_{T,t}^c(j))^{1-\vartheta}, \quad (\text{C.100})$$

$$x_{NT,t}^c(j) = Z_{NT,t}^c A_{NT,t}^c(j) (K_{NT,t}^c(j))^\vartheta (L_{NT,t}^c(j))^{1-\vartheta}, \quad (\text{C.101})$$

where $\vartheta \in (0,1)$ is the land share, and $A_{S,t}^c(j)$ captures variety-level productivity. Exogenous productivity enters at two levels:

- Sector-region-wide productivity shifters $Z_{S,t}^c$ follow:

$$\Delta \ln Z_{S,t}^c = \mu + \varepsilon_{S,t}^c, \quad (\text{C.102})$$

where $\varepsilon_{S,t}^c$ is a region-sector-specific shock.

- Variety-specific productivity $A_{S,t}^c(j)$ evolves through innovation shocks, as described in later sections.

Factor supplies and clearing. Land is fixed and local. Each region has fixed land supply $\bar{K}^c$, used in both sectors:

$$\sum_{S \in \{T, NT\}} \int_0^1 K_{S,t}^c(j) dj = \bar{K}^c, \quad c \in \{H, F\}. \quad (\text{C.103})$$

Labor is supplied inelastically within each region. Let $N_{c,t}$ denote the mass of workers residing (and working) in region c at time t . Intra-regional sector mobility is frictionless, so a single local wage $w_{c,t}$ prevails across sectors within c . Local labor market clearing is

$$\sum_{S \in \{T, NT\}} \int_0^1 L_{S,t}^c(j) dj = N_{c,t}, \quad c \in \{H, F\}. \quad (\text{C.104})$$

Total population is fixed: $N_{H,t} + N_{F,t} = \bar{N}$.

Households, location, and financial markets. Each region hosts a unit mass of households with homothetic preferences over a Cobb–Douglas bundle,

$$C_{c,t} = (C_{T,c,t})^\alpha (C_{NT,c,t})^{1-\alpha}, \quad \alpha \in (0,1), \quad (\text{C.105})$$

where $C_{NT,c,t}$ must be consumed locally and $C_{T,c,t}$ is nationally tradable.

Households can trade (i) riskless bonds, (ii) equity claims on incumbent firms in both regions and sectors, and (iii) Arrow–Debreu claims contingent on aggregate shocks $\{\varepsilon_{S,t}^c, u_{NT,t}^c\}$. However, markets are incomplete with respect to the *identity* of who receives claims to the newly created firms (or displaced product lines). Specifically, in each period a small subset of households

(innovators) receives ownership of new entrant profits; this risk cannot be diversified ex ante. This incompleteness is the key ingredient that creates wealth redistribution and a wedge between aggregate consumption growth and the SDF.

Goods-market clearing. Non-tradables clear locally, and tradables clear nationally:

$$C_{NT,c,t} = Y_{NT,t}^c, \quad C_{T,H,t} + C_{T,F,t} = Y_{T,t}. \quad (\text{C.106})$$

This implies that shifts in regional expenditures translate into local non-tradable price adjustments, which is the key spatial link to inflation.

Displacement shock and creative destruction (quality ladder). At the beginning of each period $t + 1$, a displacement (or innovation) shock $u_{NT,t+1}^c$ is realized in region c . This shock raises the probability that each product line in the non-tradable sector is challenged by a new entrant:

$$m_{NT,t+1}^c = 1 - e^{-u_{NT,t+1}^c}, \quad \text{with } 0 < m_{NT,t+1}^c < 1. \quad (\text{C.107})$$

If an entrant arrives at line j , it competes with the incumbent under Bertrand pricing. The entrant operates with a productivity advantage: it can produce at productivity $e^\delta A_{NT,t}^c(j)$, where $\delta > 0$ and $A_{NT,t}^c(j)$ is the incumbent's productivity. As a result, the entrant undercuts the incumbent and takes over the product line. Thus, productivity at line j in region c evolves as:

$$A_{NT,t+1}^c(j) = \begin{cases} e^\delta A_{NT,t}^c(j), & \text{with probability } m_{NT,t+1}^c, \\ A_{NT,t}^c(j), & \text{with probability } 1 - m_{NT,t+1}^c. \end{cases} \quad (\text{C.108})$$

A fraction $m_{NT,t+1}^c$ of incumbents are therefore displaced and replaced by more productive entrants. In each displaced line, the old firm's profits fall to zero, and the entrant captures the full profit stream.

Aggregate Productivity and Output We define the effective productivity in region c as the CES aggregator:

$$\tilde{A}_{NT,t}^c = Z_{NT,t}^c \left(\int_0^1 A_{NT,t}^c(j)^{\sigma-1} dj \right)^{1/(\sigma-1)}. \quad (\text{C.109})$$

Using the updating rule and noting that displacement occurs independently across product lines, the evolution of aggregate productivity is:

$$\begin{aligned} (\tilde{A}_{NT,t+1}^c)^{\sigma-1} &= (Z_{NT,t+1}^c)^{\sigma-1} \left[(1 - m_{NT,t+1}^c) + m_{NT,t+1}^c e^{\delta(\sigma-1)} \right] (\tilde{A}_{NT,t}^c / Z_{NT,t}^c)^{\sigma-1}, \\ \tilde{A}_{NT,t+1}^c &= \tilde{A}_{NT,t}^c \left(\frac{Z_{NT,t+1}^c}{Z_{NT,t}^c} \right) \left[(1 - m_{NT,t+1}^c) + m_{NT,t+1}^c e^{\delta(\sigma-1)} \right]^{1/(\sigma-1)}. \end{aligned}$$

Taking logs and linearizing for small $u_{NT,t+1}^c$ and $\Delta \ln Z_{NT,t+1}^c$ yields:

$$\Delta \ln \tilde{A}_{NT,t+1}^c \approx \mu + \varepsilon_{NT,t+1}^c + \delta u_{NT,t+1}^c, \quad (\text{C.110})$$

where μ is the drift term and $\varepsilon_{NT,t+1}^c$ is a productivity shock.

Summary. This setup captures how a displacement shock $u_{NT,t+1}^c$ can simultaneously:

1. Increase aggregate productivity in region c by δ for the fraction $m_{NT,t+1}^c$ of replaced varieties.
2. Reallocate profits from displaced incumbents to new entrants.

with the following implication to the reduced form endowment economy:

- **Creative destruction:** A fraction $m_{NT,t+1}^c$ of varieties in the non-tradable sector is displaced each period, representing Schumpeterian creative destruction. New, more productive firms replace old incumbents.
- **Reallocation:** Ownership of displaced product lines switches to new entrants. Incumbents lose the revenue from those lines; new entrepreneurs gain it.
- **Unspanned risk:** If households cannot insure ex ante over “who will be the successful innovator,” markets are incomplete. This introduces idiosyncratic risk over who gains or loses from each period’s displacement.

B. Prices and the Spatial Wage vs Land-Rent Channels

Each intermediate producer minimizes costs and charges a constant markup over marginal cost.

Unit costs and monopoly pricing. Define the Cobb–Douglas unit-cost constant

$$\Omega(\vartheta) \equiv \vartheta^{-\vartheta} (1 - \vartheta)^{-(1-\vartheta)}. \quad (\text{C.111})$$

Marginal cost in region c is

$$MC_{S,t}^c(j) = \frac{\Omega(\vartheta)}{Z_{S,t}^c A_{S,t}^c(j)} (R_t^c)^\vartheta (w_{c,t})^{1-\vartheta}, \quad S \in \{T, NT\}, \quad (\text{C.112})$$

where R_t^c is the local land rental rate and $w_{c,t}$ is the local wage.

With CES demand elasticity σ , the optimal monopoly price is

$$p_{S,t}^c(j) = \frac{\sigma}{\sigma - 1} MC_{S,t}^c(j) = \kappa \frac{(R_t^c)^\vartheta (w_{c,t})^{1-\vartheta}}{Z_{S,t}^c A_{S,t}^c(j)}, \quad \kappa \equiv \frac{\sigma}{\sigma - 1} \Omega(\vartheta). \quad (\text{C.113})$$

Tradables (national market). With zero trade costs, varieties are priced at the lowest delivered cost:

$$p_{T,t}(j) = \min_{c \in \{H,F\}} \{p_{T,t}^c(j)\}. \quad (\text{C.114})$$

Define the cost-adjusted (effective) productivity

$$\hat{A}_{T,t}(j) \equiv \max_{c \in \{H,F\}} \left\{ \frac{Z_{T,t}^c A_{T,t}^c(j)}{(R_t^c)^\theta (w_{c,t})^{1-\theta}} \right\}, \quad (\text{C.115})$$

so that $p_{T,t}(j) = \kappa [\hat{A}_{T,t}(j)]^{-1}$ and

$$P_{T,t} = \left(\int_0^1 p_{T,t}(j)^{1-\sigma} dj \right)^{\frac{1}{1-\sigma}} = \kappa \left(\int_0^1 \hat{A}_{T,t}(j)^{\sigma-1} dj \right)^{-\frac{1}{\sigma-1}} \equiv \kappa (\tilde{A}_{T,t})^{-1}. \quad (\text{C.116})$$

Non-tradables (local markets). Non-tradables are priced locally. The local NT price index is

$$P_{NT,t}^c = \left(\int_0^1 (p_{NT,t}^c(j))^{1-\sigma} dj \right)^{\frac{1}{1-\sigma}} = \kappa (R_t^c)^\theta (w_{c,t})^{1-\theta} (\tilde{A}_{NT,t}^c)^{-1}, \quad (\text{C.117})$$

where $\tilde{A}_{NT,t}^c$ is defined as:

$$\tilde{A}_{NT,t}^c \equiv Z_{NT,t}^c \left(\int_0^1 A_{NT,t}^c(j)^{\sigma-1} dj \right)^{\frac{1}{\sigma-1}}. \quad (\text{C.118})$$

Relative prices and the two spatial channels. Combining (C.116) and (C.117), we obtain the relative price level,

$$\frac{P_{NT,t}^c}{P_{T,t}} = (R_t^c)^\theta (w_{c,t})^{1-\theta} \cdot \frac{\tilde{A}_{T,t}}{\tilde{A}_{NT,t}^c}. \quad (\text{C.119})$$

Taking differences across regions delivers a transparent decomposition:

$$\Delta \ln \frac{P_{NT,t}^H}{P_{NT,t}^F} = \theta \Delta \ln \frac{R_t^H}{R_t^F} + (1 - \theta) \Delta \ln \frac{w_{H,t}}{w_{F,t}} - \Delta \ln \frac{\tilde{A}_{NT,t}^H}{\tilde{A}_{NT,t}^F}. \quad (\text{C.120})$$

The second term is the standard spatial wage (or labor-cost) channel emphasized in the spatial-economics literature (and in Balassa–Samuelson-type arguments). The first term is the *local land-rent channel*: because land is fixed locally, shifts in local demand—including demand shifts induced by innovation rents accruing to local innovators—are absorbed in land rents and hence local non-tradable prices.

Equation (C.120) highlights that relative inflation can be driven by either the land-rent channel (first term) or the wage channel (second term). As we document in Table 4, the empirical response of relative median wages to innovation is quantitatively small. This evidence implies that the

wage term in (C.120) is limited, leaving the land-rent term as the dominant driver of the observed non-tradable inflation. We return to the set of empirical moments that discipline this conclusion in Part E.

C. Labor Mobility, Wage Wedges, and the Roundabout Economy

A standard migration block generates regional wage wedges while preserving tractability.

Population stocks and flows. Let $\mathcal{M}_t$ be the net migration flow into the Home region. Labor stocks evolve as

$$N_{H,t+1} = N_{H,t} + \mathcal{M}_t, \quad (\text{C.121})$$

$$N_{F,t+1} = \bar{N} - N_{H,t+1}. \quad (\text{C.122})$$

Mobility costs and the wage wedge. A competitive mobility sector supplies migration services with convex cost

$$\Phi(\mathcal{M}_t) = \varphi \mathcal{M}_t^2, \quad \varphi \geq 0. \quad (\text{C.123})$$

Taking the wage gap as the output price, optimality implies

$$w_{H,t} - w_{F,t} = \Phi'(\mathcal{M}_t) = 2\varphi \mathcal{M}_t. \quad (\text{C.124})$$

As $\varphi \rightarrow 0$, wages equalize across regions, $w_{H,t} = w_{F,t}$, which is the “near-perfect mobility” benchmark used to interpret the weak empirical wage response.

Resource-cost versus roundabout interpretations. There are two accounting conventions for migration costs:

(i) *Real resource cost.* If $\Phi(\mathcal{M}_t)$ consumes tradable output, then feasibility includes an extra term,

$$C_{T,H,t} + C_{T,F,t} + \Phi(\mathcal{M}_t) = Y_{T,t}. \quad (\text{C.125})$$

This adds a small wedge to aggregate feasibility. For small $\Phi(\mathcal{M}_t)$, it does not overturn the mechanism but complicates notation.

(ii) *Roundabout (pecuniary) wedge with proportional rebates.* Alternatively, interpret $\Phi(\mathcal{M}_t)$ as a fee paid to the mobility sector and rebated lump-sum to households (e.g., in proportion to wealth). In this case, the fee affects migration incentives via (C.124) but does not alter aggregate feasibility or the distribution of wealth shares across regions. Under homothetic preferences, proportional rebates scale all households’ budgets in the same way and therefore leave relative wealth shares—and the associated stochastic Negishi/Pareto weight that governs the cross-region allocation—unchanged.

This “roundabout” interpretation is useful because it allows the spatial model to generate a wage wedge (and thus nest the classic spatial wage channel) while preserving the reduced-form allocation and SDF structure emphasized in the main text.

D. Identification and Calibration of the Mobility Cost Parameter φ

In the baseline quantitative model, the output processes $\{Y_{S,t}^c\}$ are treated as reduced-form endowments and the state vector does not explicitly include $(N_{c,t}, \mathcal{M}_t, w_{c,t})$. Under the roundabout interpretation above, φ affects the mapping between two objects (wage gaps and migration flows) that are not directly used as moments in the SMM procedure. Consequently, φ is typically weakly identified (or not identified) by the model moments matched in the main text.

A practical approach is therefore to discipline φ outside SMM using external spatial evidence (migration and wage-gap responses). A simple bound follows from combining the stock-flow identity with (C.124). Over an h -period window,

$$\sum_{s=0}^{h-1} \mathcal{M}_{t+s} = N_{H,t+h} - N_{H,t} \equiv \Delta N_{H,t \rightarrow t+h}. \quad (\text{C.126})$$

Hence $\frac{1}{h} \sum_{s=0}^{h-1} |\mathcal{M}_{t+s}| \geq |\Delta N_{H,t \rightarrow t+h}|/h$. If the empirical impulse response of the regional wage gap satisfies $|w_{H,t+s} - w_{F,t+s}| \leq \varepsilon_h$ for $s \leq h$ around an innovation shock, then (C.124) implies the conservative bound

$$\varphi \leq \frac{\varepsilon_h}{2} \cdot \frac{h}{|\Delta N_{H,t \rightarrow t+h}|}. \quad (\text{C.127})$$

This bound uses only population responses and the observed near-zero relative wage response; it does not require embedding $(\mathcal{M}_t, w_{c,t})$ as explicit state variables in the main quantitative solution.

E. Discussion: Comparing Spatial Mechanisms and Model Predictions

In the spatial models above, local productivity shocks can raise local non-tradable prices through multiple channels. The microfoundation above nests these channels and clarifies how their predictions differ from the limited risk sharing of innovation rents mechanism.

(1) Wage/marginal-cost channel (Balassa–Samuelson-type). In spatial models with limited mobility (high φ) or sectoral labor immobility, a positive local productivity shock can raise local wages relative to other regions. Through (C.119), higher relative wages increase local non-tradable marginal costs and thus raise $P_{NT,t}^c$.

Prediction vs Data: This channel predicts that local non-tradable inflation should increase with relative wage growth. However, as shown in Table 4, relative median wages do not rise significantly following innovation shocks, implying this channel is quantitatively small.

(2) Migration/population-demand channel. Higher local wages can attract migrants ($\mathcal{M}_t > 0$), raising local housing and services demand. If housing supply is inelastic (fixed land), this pushes up rents and non-tradable prices.

Prediction vs Data: This channel requires significant net migration inflows to drive prices. While migration occurs, our bounds in Section D suggest it is insufficient to generate the observed price wedge solely through labor market clearing without the rent-driven wealth effect. Moreover, we see that in Table 4 that population coefficients are negative.

(3) Capital-income / local-portfolio channel. If local residents hold undiversified claims on local firms, local capital gains can raise local demand for non-tradables. This channel requires that the recipients of gains spend locally (or face frictions that keep expenditures local).

Prediction vs Data: local stock market returns should correlate positively with local non-tradable prices. However, this is not consistent with the evidence presented in Table 1. In contrast, the return of a local growth-minus-value portfolio is positively correlated with local relative inflation.

(4) Our mechanism: uninsurable innovation rents interacting with fixed land. The key friction in the main text is that the gains from displacement (new entrant profits) are not perfectly shared ex ante. In the microfoundation, displacement reallocates monopoly profit streams from incumbents to new entrants. Because innovators capture a non-diversifiable share of these profits, regional wealth rises, increasing local expenditures and therefore local non-tradable prices $P_{NT,t}^c$.

The spatial production model clarifies that relative non-tradable prices are driven by the sum of a wage term and a land-rent term. Since the empirical evidence indicates a muted relative median wage response, the main margin is the local factor prices, i.e., land rents. Therefore, the baseline endowment economy allows us to characterize the asset-pricing implications of unspanned innovation risk in a tractable way, nesting the key spatial transmission mechanism identified in the above environment.

C.7 Income Inequality in the Model

The value of new projects $\eta S(1 - e^{-u_t})$ accrues to a measure ζ of the population. For tractability, we assume the gains from innovation are proportional to household wealth. A successful innovator with wealth w_i experiences a positive capital income shock equal to

$$I_i = \frac{\eta S_t(1 - e^{-u_t})}{\zeta} \frac{w_i}{\bar{W}}. \quad (\text{C.128})$$

The smaller the value of ζ , the more concentrated the benefits of new projects to innovators.

To calculate the level of top income inequality in the model, we compute the share of total income that accrues to the top 1% of households in each region. This income is a combination of

capital gains from creating new projects and dividends from incumbent firms. Households at the top of the income distribution belong to one of two groups. The first includes households that received a share of newly created firms (the entrepreneurs). The total capital gains from the creation of new projects in the home region, as a fraction of total income at time t , is

$$I_{E,t}^{capital} = \frac{\eta S_t (1 - e^{-u_t})}{\xi W_t + \bar{D}'_t W'_t / \bar{W}'_t + \eta S_t (1 - e^{-u_t})}, \quad (\text{C.129})$$

where $S_t = S_{T,t} + S_{NT,t}$ is the total market capitalization of local firms, W'_t and W'^{*}_t are the total wealth of the two regions excluding new projects:

$$W'_t = W_t - \eta S_t (1 - e^{-u_t}), \quad (\text{C.130})$$

and $\bar{W}'_t = W'_t + W'^{*}_t$ is the total wealth across both regions excluding the market value of new projects. The share of the total dividend $\bar{D}'_t$ that accrues to local households equals $\bar{D}'_t W'_t / \bar{W}'_t$.

The realization of the u -shock at time t determines the amount of wealth transferred from existing shareholders to new entrepreneurs. The value of these new projects constitutes a capital gain distributed to a small fraction of the population, pushing some entrepreneurs into the top 1% of the income distribution.

The second group of households at the top of the income distribution are those sufficiently rich before the current period—households who have received shares in new projects and consequently earn large dividend income. These households derive capital income proportional to their wealth share, receiving a fraction of

$$\xi W_t + \bar{D}'_t W'_t / \bar{W}'_t. \quad (\text{C.131})$$

Households in the region share the dividend and annuity income in (C.131) in direct proportion to their wealth. Greater wealth inequality implies that dividend and annuity payments account for a larger portion of the top income share.

C.8 Allowing for CRRA utility

In this section, we derive the solution for CRRA utility. We follow the sequence outlined in the previous section on log utility. The aggregation result still applies because it relies solely on the homotheticity of preferences. The sections on allocations, the SDF, and stock markets closely resemble those with log utility. The key difference, as derived in Section C.8, is that the consumption-wealth ratio is no longer constant.

Allocations and inflation

Allocations First, at each t , we derive the consumer's demands for tradable and nontradable goods, keeping overall consumption expenditure $\mathcal{C}_t$ fixed.

$$\max_{\{C_{T,t}, C_{NT,t}\}} C_{T,t}^\alpha C_{NT,t}^{1-\alpha} \quad (\text{C.132})$$

$$\text{s.t. } C_{T,t} + P_{NT,t} C_{NT,t} = \mathcal{C}_t. \quad (\text{C.133})$$

The problem for the foreign household is defined analogously. Given the above, we obtain the following demands

$$C_{T,t} = \alpha \mathcal{C}_t, \quad C_{NT,t} = (1 - \alpha) \frac{\mathcal{C}_t}{p_{NT,t}} \quad (\text{C.134})$$

$$C_{T,t}^* = \alpha \mathcal{C}_t^*, \quad C_{NT,t}^* = (1 - \alpha) \frac{\mathcal{C}_t^*}{P_{NT,t}^*} \quad (\text{C.135})$$

Dynamics of the consumption ratio Let V_t denote the utility of the representative agent in the home region. Denote the partial derivative of V_t with respect to the consumption basket by $V_{C,t}$, which is given by

$$\frac{\partial V_t}{\partial \mathcal{C}_t} = \frac{1}{C_t^\gamma}.$$

The intertemporal marginal rate of substitution for the representative agent in region c is then

$$M_{t,t+1} = \beta \frac{V_{C,t+1}}{V_{C,t}}. \quad (\text{C.136})$$

Trade implies that the marginal utilities of tradable good T for $t = 1, 2, \dots$ in each possible state is

$$\beta^t V_{C,t} C_t \frac{\alpha}{c_{T,t}} = \frac{\alpha}{c_{T,t}^*} C_t^* V_{C,t}^* \beta^t \quad (\text{C.137})$$

Define the Pareto weights at period t as

$$\begin{aligned} \Lambda_t &= \Lambda_0 \beta^t V_{C,t} C_t \\ &= \Lambda_{t-1} \beta \frac{V_{C,t}}{V_{C,t-1}} \frac{C_t}{C_{t-1}} = \Lambda_{t-1} M_{t-1,t} \exp(\Delta c_t) \end{aligned} \quad (\text{C.138})$$

Since the economy starts with a symmetric setup $\Lambda_0 = \Lambda_0^*$. We can rewrite (C.137) as

$$\Lambda_t \frac{\alpha}{C_{T,t}} = \frac{\alpha}{C_{T,t}^*} \Lambda_t^*$$

Define $\lambda_t = \frac{\Lambda_t^*}{\Lambda_t}$ as the ratio of Pareto weights. The optimality condition can be written as

$$\lambda_t = \frac{C_{T,t}^*}{C_{T,t}} \quad (\text{C.139})$$

Similar to the log utility case, note that with Cobb-Douglas preferences over different goods, households' consumption expenditure shares for each good are fixed. Therefore, (C.139) shows that λ_t is also the ratio of consumption expenditures between foreign and home regions:

$$\lambda_t = \frac{P_{NT,t}^* C_{NT,t}^*}{P_{NT,t} C_{NT,t}} = \frac{C_t^*}{C_t}. \quad (\text{C.140})$$

Additionally, from the definition of $\lambda_t = \frac{\Lambda_t^*}{\Lambda_t}$, we have

$$\lambda_{t+1} = \lambda_t \frac{M_{t,t+1}^* e^{\Delta c_{t+1}^*}}{M_{t,t+1} e^{\Delta c_{t+1}}}. \quad (\text{C.141})$$

Substituting the demand functions into the budget constraints, we obtain the allocations:

$$C_{T,t} = \frac{1}{1 + \lambda_t} \bar{Y}_{T,t}, \quad (\text{C.142})$$

$$C_{T,t}^* = \frac{\lambda_t}{1 + \lambda_t} \bar{Y}_{T,t}, \quad (\text{C.143})$$

$$C_{NT,t} = Y_{NT,t}, \quad (\text{C.144})$$

$$C_{NT,t}^* = Y_{NT,t}^*, \quad (\text{C.145})$$

where $\bar{Y}_{T,t} = Y_{T,t} + Y_{T,t}^*$ is the total output of tradable goods.

Given the above allocations, we calculate the consumption bundles:

$$C_t = (C_{T,t})^\alpha (C_{NT,t})^{1-\alpha} \quad (\text{C.146})$$

$$C_t^* = (C_{T,t}^*)^\alpha (C_{NT,t}^*)^{1-\alpha} \quad (\text{C.147})$$

We also compute the price of consumption bundles in home and foreign regions:

$$P_t = \frac{C_t}{C_t} = \frac{C_{T,t} + C_{NT,t} P_{NT,t}}{C_t} \quad (\text{C.148})$$

$$P_t^* = \frac{C_t^*}{C_t^*} = \frac{C_{T,t}^* + C_{NT,t}^* P_{NT,t}^*}{C_t^*} \quad (\text{C.149})$$

The relative price level across the two regions is then equal to

$$e_t \equiv \frac{P_t}{P_t^*} = \frac{C_t^*}{C_t} \frac{1}{\lambda_t} \quad (\text{C.150})$$

where the equality follows directly from (C.139). As a result, the growth in relative price levels (relative inflation) is equal to

$$\pi_{t+1} = \Delta c_{t+1}^* - \Delta c_{t+1} - \Delta \ln \lambda_{t+1}. \quad (\text{C.151})$$

SDF

We follow the same argument in the log utility case and consider two population groups: the population that receives the new firms in the current period (with measure ζ , denoted as N); and the population that does not receive the new firms in the current period (with measure $1 - \zeta$, denoted as O).

The SDF of these two groups can be written as

$$\begin{aligned} M_{O,t,t+1} &= \beta \left(\frac{C_{t+1}}{C_t} b_{t+1} \right)^{-\gamma} \\ M_{N,t,t+1} &= \beta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \frac{C_{t+1}}{C_t} \right)^{-\gamma} \end{aligned}$$

In this economy, the conditional expectation of investors' inter-temporal marginal rates of substitution is a valid stochastic discount factor. That is,

$$\begin{aligned} M_{t,t+1} &= (1 - \zeta)M_{O,t,t+1} + \zeta M_{N,t,t+1} \\ &= \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1 - \zeta)b_{t+1}^{-\gamma} \right) \end{aligned}$$

Stock Market

The value of non-tradable firms and tradable firms are equal to

$$S_{NT} = p_{NT,t} Y_{NT,t} + E_t[M_{t,t+1}^T (S_{NT,t+1} e^{-u_{t+1}})] = p_{NT,t} Y_{NT,t} (pd_{NT,t}) \quad (\text{C.152})$$

$$S_{NT}^* = p_{NT,t}^* Y_{NT,t}^* + E_t[M_{t,t+1}^T (S_{NT,t+1}^* e^{-u_{t+1}^*})] = p_{NT,t}^* Y_{NT,t}^* (pd_{NT,t}^*) \quad (\text{C.153})$$

$$S_T = Y_{T,t} + E_t[M_{t,t+1}^T (S_{T,t+1})] = Y_{T,t} (pd_{T,t}) \quad (\text{C.154})$$

$$S_T^* = Y_{T,t}^* + E_t[M_{t,t+1}^T (S_{T,t+1}^*)] = Y_{T,t}^* (pd_{T,t}^*) \quad (\text{C.155})$$

And the total value of T sector is equal to

$$\bar{S}_{T,t} = S_T + S_T^*$$

We have normalized the price of tradable goods to be one. The total output of tradable goods is $\bar{Y}_{T,t} = Y_{T,t} + Y_{T,t}^*$. We denote $M_{t,t+1}^T$ as the SDF using the tradable good as a numeraire. Similarly, $M_{t,t+1}^{NT}$ and $M_{t,t+1}^{NT,*}$ are the SDFs using non-tradable goods as numeraire. By definition, they have

the following relationship:

$$M_{t,t+1}^T = M_{t,t+1}^{NT} \frac{p_{NT,t}}{p_{NT,t+1}} = M_{t,t+1}^{NT,*} \frac{p_{NT,t}^*}{p_{NT,t+1}^*} = M_{t,t+1} \frac{p_t}{p_{t+1}} \quad (\text{C.156})$$

and

$$M_{t,t+1}^T = \beta \left(\frac{C_{t+1}}{C_t} \right)^{1-\gamma} \left(\frac{C_{T,t+1}}{C_{T,t}} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1-\zeta)b_{t+1}^{-\gamma} \right) \quad (\text{C.157})$$

$$M_{t,t+1}^{T,*} = \beta \left(\frac{C_{t+1}^*}{C_t^*} \right)^{1-\gamma} \left(\frac{C_{T,t+1}^*}{C_{T,t}^*} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-\gamma} + (1-\zeta)(b_{t+1}^*)^{-\gamma} \right) \quad (\text{C.158})$$

$$M_{t,t+1}^{NT} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{1-\gamma} \left(\frac{C_{NT,t+1}}{C_{NT,t}} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1-\zeta)b_{t+1}^{-\gamma} \right) \quad (\text{C.159})$$

$$M_{t,t+1}^{NT,*} = \beta \left(\frac{C_{t+1}^*}{C_t^*} \right)^{1-\gamma} \left(\frac{C_{NT,t+1}^*}{C_{NT,t}^*} \right)^{-1} \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-\gamma} + (1-\zeta)(b_{t+1}^*)^{-\gamma} \right) \quad (\text{C.160})$$

where p, p_{NT} are the prices of consumption bundles and nontradable goods, respectively. The wealth share of non-innovators is equal to

$$b_{t+1} = 1 - \frac{\text{new projects at home}}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{1}{1+w_{t+1}} \right)} \quad (\text{C.161})$$

$$b_{t+1}^* = 1 - \frac{\text{new projects at foreign}}{(\bar{S}_{T,t+1} + S_{NT,t+1} + S_{NT,t+1}^*) \left(\frac{w_{t+1}}{1+w_{t+1}} \right)} \quad (\text{C.162})$$

We define the following quantities as in the log utility case:

$$\Omega_{t+1} = \left(\zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1-\zeta)b_{t+1}^{-\gamma} \right) \quad (\text{C.163})$$

$$\Omega_{t+1}^* = \left(\zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-\gamma} + (1-\zeta)(b_{t+1}^*)^{-\gamma} \right) \quad (\text{C.164})$$

From the equalization of marginal utility of tradable goods, we obtain

$$\begin{aligned} 0 &= \ln \Omega_{t+1}^* - \ln \Omega_{t+1} - \Delta \ln \lambda_{t+1} + (1-\gamma) \left(\alpha \ln \frac{C_{T,t+1}^*/C_{T,t+1}}{C_{T,t}^*/C_{T,t}} + (1-\alpha) \ln \frac{Y_{NT,t+1}^*/Y_{NT,t+1}}{Y_{NT,t}^*/Y_{NT,t}} \right) \\ &= \ln \Omega_{t+1}^* - \ln \Omega_{t+1} - \Delta \ln \lambda_{t+1} + (1-\gamma) \left(\alpha \Delta \ln \lambda_{t+1} + (1-\alpha) \ln \frac{Y_{NT,t+1}^*/Y_{NT,t+1}}{Y_{NT,t}^*/Y_{NT,t}} \right) \end{aligned} \quad (\text{C.165})$$

Price dividend ratios The price-dividend ratio of tradable firms is equal to

$$pd_{T,t} = 1 + E_t \left[M_{t,t+1}^T \frac{Y_{T,t+1}}{Y_{T,t}} (pd_{T,t+1}) (1 - (1 - e^{-u_{T,t+1}}) \eta) \right]. \quad (C.166)$$

Similarly for foreign tradable firms,

$$pd_{T,t}^* = 1 + E_t \left[M_{t,t+1}^T \frac{Y_{T,t+1}^*}{Y_{T,t}^*} (pd_{T,t+1}^*) (1 - (1 - e^{-u_{T,t+1}^*}) \eta) \right]. \quad (C.167)$$

The price-dividend ratio of non-tradable firms is equal to

$$\begin{aligned} pd_{NT,t} &= 1 + E_t \left[M_{t,t+1}^T \frac{p_{t+1} Y_{NT,t+1}}{p_t Y_{NT,t}} (pd_{NT,t+1}) (1 - (1 - e^{-u_{t+1}}) \eta) \right] \\ &= 1 + E_t \left[M_{t,t+1}^T \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \frac{1}{1+\lambda_{t+1}} (pd_{NT,t+1}) (1 - (1 - e^{-u_{NT,t+1}}) \eta) \right] \end{aligned} \quad (C.168)$$

Similarly for foreign non-tradable firms,

$$pd_{NT,t}^* = 1 + E_t \left[M_{t,t+1}^T \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \frac{\frac{\lambda_{t+1}}{1+\lambda_{t+1}}}{\frac{\lambda_t}{1+\lambda_t}} (pd_{NT,t+1}^*) (1 - (1 - e^{-u_{NT,t+1}^*}) \eta) \right] \quad (C.169)$$

In what follows, we derive the expressions for sector stock returns and the returns for various portfolios.

Sector Returns The total return of non-tradable firms is

$$\begin{aligned} r_{NT,t+1} &= \ln \frac{p_{NT,t+1} Y_{NT,t+1} (1 - (1 - e^{-u_{NT,t+1}}) \eta)}{p_{NT,t} Y_{NT,t}} \frac{pd_{NT,t+1}}{pd_{NT,t} - 1} \\ &\approx \ln \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} + \frac{\lambda_t}{1 + \lambda_t} (\Delta \ln \Omega_{t+1} - \Delta \ln \Omega_{t+1}^*) - u_{NT,t+1} \eta + \ln \frac{pd_{NT,t+1}}{pd_{NT,t} - 1} \end{aligned} \quad (C.170)$$

A positive u_{NT} shock at home leads to both an increase in Ω and an increase in $\frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}}$, but also causes a displacement effect $u_{NT} \eta$.

The total return of tradable firms at home is

$$\begin{aligned} r_{T,t+1} &= \ln \left(\frac{Y_{T,t+1} (1 - (1 - e^{-u_{T,t+1}}) \eta)}{Y_{T,t}} \frac{pd_{T,t+1}}{pd_{T,t} - 1} \right) \\ &\approx \ln \left(\frac{Y_{T,t+1} (1 - u_{T,t+1} \eta)}{Y_{T,t}} \frac{pd_{T,t+1}}{pd_{T,t} - 1} \right) \\ &\approx \Delta \ln(Y_{T,t+1}) - \Delta \ln(Y_{T,t}) - u_{T,t+1} \eta + \ln(pd_{T,t+1}) - \ln(pd_{T,t} - 1) \\ &= \mu + (\delta - \eta) u_{T,t+1} + \varepsilon_{t+1} + \ln(pd_{T,t+1}) - \ln(pd_{T,t} - 1) \end{aligned} \quad (C.171)$$

Market Returns The market return is the weighted average of T and NT firms. The weights of these

firms are given by $w_T = \frac{\alpha(pd_{T,t}-1)}{\alpha(pd_{T,t}-1)+(1-\alpha)(pd_{NT,t}-1)}$ and $w_{NT} = \frac{(1-\alpha)(pd_{NT,t}-1)}{\alpha(pd_{T,t}-1)+(1-\alpha)(pd_{NT,t}-1)}$. Therefore, the value-weighted return is:

$$r_{MKT,t} = w_T r_{T,t+1} + w_{NT} r_{NT,t+1} \quad (C.172)$$

$$\begin{aligned} &\approx w_T \left(\underbrace{\mu + (\delta - \eta)u_{T,t+1} + \varepsilon_{t+1}}_{\frac{\ln \bar{Y}_{T,t+1}}{\bar{Y}_{T,t}}} + \ln(pd_{T,t+1}) - \ln(pd_{T,t} - 1) \right) \\ &+ w_{NT} \left(\ln \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} + \frac{\lambda_t}{1 + \lambda_t} (\Delta \ln \Omega_{t+1} - \Delta \ln \Omega_{t+1}^*) - u_{NT,t+1} \eta + \ln \frac{pd_{NT,t+1}}{pd_{NT,t} - 1} \right) \end{aligned} \quad (C.173)$$

GMV Returns We focus on the home region in our derivation. Recall that value firms are those that do not receive new projects, while growth firms are those that do. Note that pd_t^V and pd_t^G denote the price-dividend ratios of firms that were sorted into value and growth categories, respectively, at period t .

For tradable firms that are sorted into the value group, their return is equal to

$$r_{V,t+1}^T = q r_{V \rightarrow G,t+1}^T + (1 - q) r_{V \rightarrow V,t+1}^T \quad (C.174)$$

where

$$r_{V \rightarrow V,t+1}^T = \ln \left(\frac{Y_{T,t+1} e^{-u_{t+1}}}{\bar{Y}_{T,t}} \frac{pd_{T,t+1}}{pd_{T,t} - 1} \right) = \mu + (\delta - 1)u_{T,t+1} + \varepsilon_{T,t+1} + \ln \frac{pd_{T,t+1}^V}{pd_{T,t}^V - 1} \quad (C.175)$$

$$r_{V \rightarrow G,t+1}^T = \mu + (\delta - 1)u_{T,t+1} + \varepsilon_{T,t+1} + \ln \frac{pd_{T,t+1}^G}{pd_{T,t}^V - 1} \quad (C.176)$$

For non-tradable firms that are sorted into the value group, their return is equal to

$$r_{V,t+1}^{NT} = q r_{V \rightarrow G,t+1}^{NT} + (1 - q) r_{V \rightarrow V,t+1}^{NT} \quad (C.177)$$

where

$$\begin{aligned} r_{V \rightarrow V,t+1}^{NT} &= \ln \left(\frac{p_{NT,t+1} Y_{NT,t+1} e^{-u_{t+1}}}{p_{NT,t} Y_{NT,t}} \frac{pd_{NT,t+1}}{pd_{NT,t}} \right) \\ &\approx \ln \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} + \frac{\lambda_t}{1 + \lambda_t} (\ln \Omega_{t+1} - \ln \Omega_{t+1}^*) - u_{NT,t+1} + \ln \frac{pd_{NT,t+1}^V}{pd_{NT,t}^V} \end{aligned} \quad (C.178)$$

$$r_{V \rightarrow G,t+1}^{NT} = \ln \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} + \frac{\lambda_t}{1 + \lambda_t} (\ln \Omega_{t+1} - \ln \Omega_{t+1}^*) - u_{NT,t+1} + \ln \frac{pd_{NT,t+1}^G}{pd_{NT,t}^V} \quad (C.179)$$

Similarly, we can derive the stock returns for growth firms. For tradable firms, we obtain

$$r_{G,t+1}^T = (1 - p) r_{G \rightarrow G,t+1}^T + p r_{G \rightarrow V,t+1}^T \quad (C.180)$$

where

$$r_{G \rightarrow V, t+1}^T = \mu + \left(\delta + \frac{1-\eta}{m_H} - 1\right)u_{T, t+1} + \varepsilon_{T, t+1} + \ln \frac{pd_{T, t+1}^V}{pd_{T, t}^G - 1} \quad (\text{C.181})$$

$$\begin{aligned} r_{G \rightarrow G, t+1}^T &= \Delta \ln X_{t+1} + \ln \frac{pd_{T, t+1}^G}{pd_{T, t}^G - 1} \\ &= \ln \left(\frac{Y_{T, t+1} e^{-u_{T, t+1}} + Y_{T, t+1} (1 - e^{-u_{T, t+1}})^{\frac{1-\eta}{m_H}}}{Y_{T, t}} \frac{pd_{T, t+1}^G}{pd_{T, t}^G - 1} \right) \\ &\approx \mu + \left(\delta + \frac{1-\eta}{m_H} - 1\right)u_{T, t+1} + \varepsilon_{T, t+1} + \ln \frac{pd_{T, t+1}^G}{pd_{T, t}^G - 1} \end{aligned} \quad (\text{C.182})$$

And for non-tradable firms, their return is equal to

$$r_{G, t+1}^{NT} = (1-p)r_{G \rightarrow G, t+1}^{NT} + pr_{G \rightarrow V, t+1}^{NT} \quad (\text{C.183})$$

where

$$\begin{aligned} r_{G \rightarrow G, t+1}^{NT} &= \ln \left(\frac{p_{NT, t+1} Y_{NT, t+1} (e^{-u_{NT, t+1}} + (1 - e^{-u_{NT, t+1}})^{\frac{1-\eta}{m_H}})}{p_{NT, t} Y_{NT, t}} \frac{pd_{NT, t+1}^G}{pd_{NT, t}^G - 1} \right) \\ &\approx \ln \frac{\tilde{Y}_{T, t+1}}{\tilde{Y}_{T, t}} + \frac{\lambda_t}{1 + \lambda_t} (\ln \Omega_{t+1} - \ln \Omega_{t+1}^*) - \left(1 - \frac{1-\eta}{m_H}\right)u_{NT, t+1} + \ln \frac{pd_{NT, t+1}^G}{pd_{NT, t}^G - 1} \end{aligned} \quad (\text{C.184})$$

$$r_{G \rightarrow V, t+1}^{NT} = \ln \frac{\tilde{Y}_{T, t+1}}{\tilde{Y}_{T, t}} + \frac{\lambda_t}{1 + \lambda_t} (\ln \Omega_{t+1} - \ln \Omega_{t+1}^*) - \left(1 - \frac{1-\eta}{m_H}\right)u_{NT, t+1} + \ln \frac{pd_{NT, t+1}^V}{pd_{NT, t}^G - 1} \quad (\text{C.185})$$

Thus, the return differential between value and growth firms is

$$\begin{aligned} r_{GMV, t+1} &= w_T(r_{G, t+1}^T - r_{V, t+1}^T) + (1 - w_T)(r_{G, t+1}^{NT} - r_{V, t+1}^{NT}) \\ &\propto w_T u_{T, t+1} + (1 - w_T) u_{NT, t+1} \end{aligned} \quad (\text{C.186})$$

Where w_T is the weight on the tradable firms in the growth portfolio. Therefore, GMV correlates positively with the linear combination of displacement shocks on both T and NT firms.

Wealth ratio

Given the recursive definition

$$V^{1-\gamma} = C^{1-\gamma} + \beta E V^{1-\gamma}, \quad (\text{C.187})$$

the marginal utility of consumption of T-good for the representative agent is equal to

$$\frac{\partial V}{\partial c_T} = \frac{\partial V}{\partial C} \frac{\partial C}{\partial c_T} = V^\gamma C^{-\gamma} \alpha \frac{C}{c_T}. \quad (\text{C.188})$$

The wealth of the representative agent (in units of T-good) is

$$W_H = \left(\frac{V}{C} \right)^{1-\gamma} c_T \frac{1}{\alpha} \quad (\text{C.189})$$

Similarly the wealth of the representative agent in the foreign region is equal to

$$W_F = \left(\frac{V^*}{C^*} \right)^{1-\gamma} c_T^* \frac{1}{\alpha} \quad (\text{C.190})$$

So the wealth ratio

$$w_t = \frac{W_F}{W_H} = \left(\frac{\left(\frac{V^*}{C^*} \right)}{\left(\frac{V}{C} \right)} \right)^{1-\gamma} \lambda_t \quad (\text{C.191})$$

The recursive definition of Utility-Consumption ratio We can express $\left(\frac{V}{C} \right)^{1-\gamma}$ as follows :

$$VC = \left(\frac{V}{C} \right)^{1-\gamma} = \left[1 + \beta E \left(\left(\frac{V}{C} \right)^{1-\gamma} \left(\frac{C_{t+1}}{C_t} \right)^{1-\gamma} \right) \right] \quad (\text{C.192})$$

The consumption growth is

$$\ln \left(\frac{C_{t+1}}{C_t} \right) = \alpha \Delta \ln c_T + (1 - \alpha) \Delta \ln Y_{NT}, \quad (\text{C.193})$$

where

$$\Delta \ln c_T = \frac{\bar{Y}_{T,t+1} \left(\frac{1}{1+\lambda_{t+1}} \right)}{\bar{Y}_{T,t} \frac{1}{1+\lambda_t}} = \frac{\bar{Y}_{T,t+1}}{\bar{Y}_{T,t}} \frac{1 + \lambda_t}{1 + \lambda_{t+1}}. \quad (\text{C.194})$$

Therefore from the recursive definition, we obtain

$$\begin{aligned} VC_t &= \left[1 + \beta E \left(VC_{t+1} \left(\frac{C_{t+1}}{C_t} \right)^{1-\gamma} \right) \right] \\ &= 1 + \beta E_t \left[(1 - \zeta) VC_{t+1} \left(\frac{C_{t+1}}{C_t} b \right)^{1-\gamma} + \zeta VC_{t+1} \left(\frac{C_{t+1}}{C_t} \frac{b\zeta + 1 - b}{\zeta} \right)^{1-\gamma} \right] \end{aligned} \quad (\text{C.195})$$

C.9 Recursive equilibrium and numerical solution

To connect to the log case, it is useful to distinguish explicitly between the Negishi weight and the wealth ratio. With Cobb–Douglas intratemporal allocation, the Negishi weight equals the ratio of consumption expenditures across regions,

$$\lambda_t = \frac{C_t^*}{C_t}. \quad (\text{C.196})$$

With log utility, the consumption-expenditure-to-wealth ratio is constant, so λ_t also equals the wealth ratio:

$$\lambda_t = \frac{C_t^*}{C_t} = \frac{W_t^*}{W_t}. \quad (\text{C.197})$$

Under CRRA utility, λ_t continues to equal the expenditure ratio, but it is no longer the wealth ratio. Define the utility–consumption ratio

$$VC_t \equiv \left(\frac{V_t}{C_t} \right)^{1-\gamma}, \quad VC_t^* \equiv \left(\frac{V_t^*}{C_t^*} \right)^{1-\gamma}, \quad (\text{C.198})$$

where C_t and C_t^* denote the regional consumption bundles. Then the wealth ratio is

$$w_t \equiv \frac{W_t^*}{W_t} = \frac{VC_t^*}{VC_t} \lambda_t. \quad (\text{C.199})$$

Hence, in the CRRA economy, λ_t is the expenditure ratio, while w_t is the corresponding wealth ratio.

The aggregate state is

$$s_t \equiv (\lambda_t, Z_t), \quad Z_t \equiv (z_{T,t}, z_{NT,t}), \quad (\text{C.200})$$

with

$$z_{T,t} \equiv \ln \left(\frac{Y_{T,t}^*}{Y_{T,t}} \right), \quad z_{NT,t} \equiv \ln \left(\frac{Y_{NT,t}^*}{Y_{NT,t}} \right). \quad (\text{C.201})$$

Using (28), each relative endowment state evolves according to

$$z_{S,t+1} = (1 - 2\tau)z_{S,t} + \delta(u_{S,t+1}^* - u_{S,t+1}) + \varepsilon_{S,t+1}^* - \varepsilon_{S,t+1}, \quad S \in \{T, NT\}. \quad (\text{C.202})$$

A recursive equilibrium consists of home-region functions

$$pd_T(s), \quad pd_{NT}(s), \quad VC(s), \quad (\text{C.203})$$

together with a policy function for next period's stochastic Pareto weight,

$$\lambda'(s_t, \xi_{t+1}), \quad (\text{C.204})$$

where ξ_{t+1} collects the aggregate shocks. In the numerical implementation, we solve on a grid for $(\ln \lambda_t, z_{T,t}, z_{NT,t})$; by symmetry, foreign-region objects are recovered from the mirrored state $(-\ln \lambda_t, -z_{T,t}, -z_{NT,t})$.

Given current state s_t and a realization of ξ_{t+1} , the endowment block implies Z_{t+1} . For a candidate value of λ_{t+1} , the planner allocation determines (C_{t+1}, C_{t+1}^*) and therefore the bundle growth rates $(C_{t+1}/C_t, C_{t+1}^*/C_t^*)$. The corresponding wealth ratio is

$$w_{t+1} = \frac{VC_{t+1}^*}{VC_{t+1}} \lambda_{t+1}. \quad (\text{C.205})$$

Let $\mathcal{N}_{H,t+1}$ and $\mathcal{N}_{F,t+1}$ denote the value of new projects allocated to innovators in the home and foreign regions, and let $\bar{W}_{t+1}$ denote the value of incumbent claims across the two regions. Then the wealth shares of non-innovators are

$$b_{t+1} = 1 - \frac{\mathcal{N}_{H,t+1}}{\bar{W}_{t+1}/(1+w_{t+1})}, \quad b_{t+1}^* = 1 - \frac{\mathcal{N}_{F,t+1}}{\bar{W}_{t+1}w_{t+1}/(1+w_{t+1})}. \quad (\text{C.206})$$

The displacement terms are

$$\Omega_{t+1} = \zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{-\gamma} + (1 - \zeta)b_{t+1}^{-\gamma}, \quad (\text{C.207})$$

$$\Omega_{t+1}^* = \zeta \left(\frac{b_{t+1}^*\zeta + 1 - b_{t+1}^*}{\zeta} \right)^{-\gamma} + (1 - \zeta)(b_{t+1}^*)^{-\gamma}, \quad (\text{C.208})$$

so the regional stochastic discount factors are

$$M_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Omega_{t+1}, \quad M_{t,t+1}^* = \beta \left(\frac{C_{t+1}^*}{C_t^*} \right)^{-\gamma} \Omega_{t+1}^*. \quad (\text{C.209})$$

With log utility, the transition law for the Negishi weight is available in closed form:

$$\Delta \ln \lambda_{t+1} = \ln \Omega_{t+1}^* - \ln \Omega_{t+1}. \quad (\text{C.210})$$

In the CRRA economy, by contrast, λ_{t+1} is defined implicitly by

$$0 = \ln \Omega_{t+1}^* - \ln \Omega_{t+1} - \Delta \ln \lambda_{t+1} + (1 - \gamma) [\alpha \Delta \ln \lambda_{t+1} + (1 - \alpha) \Delta z_{NT,t+1}]. \quad (\text{C.211})$$

This is the one-dimensional fixed-point condition solved at each state-shock node.

The utility-consumption ratio satisfies

$$VC_t = 1 + \beta E_t \left[VC_{t+1} \left(\frac{C_{t+1}}{C_t} \right)^{1-\gamma} \left((1 - \zeta)b_{t+1}^{1-\gamma} + \zeta \left(\frac{b_{t+1}\zeta + 1 - b_{t+1}}{\zeta} \right)^{1-\gamma} \right) \right], \quad (\text{C.212})$$

with the foreign counterpart defined analogously. The pricing functions $pd_T(s)$ and $pd_{NT}(s)$ then satisfy the corresponding Euler equations for tradable and non-tradable incumbent claims.

Numerically, we initialize the functions (pd_T, pd_{NT}, VC) and the policy rule for λ_{t+1} , and iterate jointly until convergence. For each current state and shock realization, we: (i) update Z_{t+1} ; (ii) solve the one-dimensional fixed-point problem for λ_{t+1} ; (iii) recover w_{t+1} , (b_{t+1}, b_{t+1}^*) , and the regional stochastic discount factors; and (iv) update (pd_T, pd_{NT}, VC) using the corresponding Euler equations. To improve stability, we use interpolation between grid points and solve on a coarse-to-fine sequence of grids for $\ln \lambda_t$.

C.10 Welfare Derivation

Within-region consumption-equivalent welfare. Consider a region $r \in \{H, F\}$ with CRRA preferences

$$U(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma \neq 1.$$

Let $C_{r,t}$ denote the region- r consumption bundle. At date t , a fraction $(1 - \zeta)$ of households do not receive a new project and have relative consumption (equivalently, wealth) share $b_{r,t}$.¹ Feasibility pins down the relative share of the remaining fraction ζ as

$$\bar{b}_{r,t} \equiv \frac{1 - (1 - \zeta)b_{r,t}}{\zeta}, \quad (1 - \zeta)b_{r,t} + \zeta\bar{b}_{r,t} = 1.$$

Utilitarian within-region welfare is

$$\bar{U}_{r,t} \equiv (1 - \zeta) U(b_{r,t} C_{r,t}) + \zeta U(\bar{b}_{r,t} C_{r,t}).$$

Define consumption-equivalent (CE) welfare $C_{r,t}^{CE}$ by $U(C_{r,t}^{CE}) = \bar{U}_{r,t}$. Under CRRA,

$$\begin{aligned} C_{r,t}^{CE} &= C_{r,t} \cdot \phi_{r,t}^{1/(1-\gamma)}, \\ \phi_{r,t} &\equiv (1 - \zeta) b_{r,t}^{1-\gamma} + \zeta \bar{b}_{r,t}^{1-\gamma}. \end{aligned} \tag{C.213}$$

By construction, $\phi_{r,t} = 1$ when $b_{r,t} = \bar{b}_{r,t} = 1$ (no within-region dispersion).

Log wedges and exact decomposition. For any object $X \in \{C, \phi, C^{CE}\}$ and region $r \in \{H, F\}$, define the log wedge of the incomplete-markets economy (IM) relative to the complete-markets

¹Within a region, the consumption-wealth ratio is common, so relative consumption shares coincide with relative wealth shares.

counterfactual (CM) as

$$\tilde{X}_{r,t} \equiv \ln X_{r,t}^{IM} - \ln X_{r,t}^{CM}. \quad (\text{C.214})$$

Using (C.213), the CE welfare wedge decomposes exactly in log points as

$$\tilde{C}_{r,t}^{CE} = \tilde{C}_{r,t} + \frac{1}{1-\gamma} \tilde{\phi}_{r,t}. \quad (\text{C.215})$$

Aggregate welfare across regions. We summarize global welfare using a utilitarian criterion with equal weights on Home and Foreign:

$$W_t \equiv \frac{1}{2} U(C_{H,t}^{CE}) + \frac{1}{2} U(C_{F,t}^{CE}). \quad (\text{C.216})$$

Our experiments start from the symmetric steady state with $C_{H,ss}^{CE} = C_{F,ss}^{CE} \equiv C_{ss}^{CE}$. A first-order expansion around this point implies

$$W_t - W_{ss} \approx \left(C_{ss}^{CE}\right)^{1-\gamma} \cdot \frac{1}{2} \left[(\ln C_{H,t}^{CE} - \ln C_{ss}^{CE}) + (\ln C_{F,t}^{CE} - \ln C_{ss}^{CE}) \right]. \quad (\text{C.217})$$

Motivated by (C.217), we report aggregate log wedges as the equal-weight average of regional log wedges:

$$\tilde{X}_t^{\text{Agg}} \equiv \frac{1}{2} \left(\tilde{X}_{H,t} + \tilde{X}_{F,t} \right), \quad X \in \{C, \phi, C^{CE}\}.$$

D Additional Evidence and Robustness Checks: Details

This appendix provides the full details underlying the results summarized in Section 6 of the main text.

D.1 Rent-Based Inflation at the MSA Level

Our main analysis shows that both the housing and non-housing components, at the state-level, respond with a similar relative magnitude to changes in local innovation. However, we acknowledge that the unconditional volatility of housing inflation is higher. To further assess the robustness of our findings, we explore alternative ways of measuring housing inflation.

In 1983, the Bureau of Labor Statistics (BLS) shifted from using a weighted average of changes in housing prices and mortgage costs to using rent changes as a proxy for overall housing inflation. We next present additional evidence using this rent-based method to examine the relationship between innovation and inflation.

To this end, we use BLS data on the Consumer Price Index (CPI) at the metropolitan level, focusing on the CPI for All Urban Consumers (CPI-U). Our sample covers 27 metropolitan areas and spans the period from 1983 to 2017.²

Building on our state-year analysis, we construct an innovation measure at the MSA-year level using patents. Consistent with the state-level approach, we assign the innovation impact of a patent attributed to MSA i to the collection of states overlapping with this MSA. Using the same methodology, we construct three measures of innovation flows that account for differences in patent quality.

We re-estimate the impact of innovation on inflation at the MSA level using (2). The results are presented in Table A.7. We see that, over the sample period, a one-standard-deviation increase in innovation is associated with a 0.9 to 2.1 percentage point increase in headline inflation.

To identify exogenous variation in local innovation levels, we construct MSA-level R&D tax credit changes using (3). For each MSA i , we aggregate state-level R&D tax credits from overlapping states, applying either equal weights or lagged GDP weights. We then examine the relationship between R&D tax credit and inflation at the MSA level by estimating (4). The results, reported in columns 4 and 5 of Table A.7, show that a one-standard-deviation increase in local R&D tax credit leads to an approximately 0.7 percentage point increase in local inflation. This finding aligns with

²Similar to our state-level analysis, we exclude Urban Alaska, AK, and Urban Hawaii, HI. Our sample therefore includes the following metropolitan areas: Atlanta-Sandy Springs-Roswell, GA; Baltimore-Columbia-Towson, MD; Boston-Cambridge-Newton, MA-NH; Chicago-Naperville-Elgin, IL-IN-WI; Cincinnati-Hamilton, OH-KY-IN; Cleveland-Akron, OH; Dallas-Fort Worth-Arlington, TX; Denver-Aurora-Lakewood, CO; Detroit-Warren-Dearborn, MI; Houston-The Woodlands-Sugar Land, TX; Kansas City, MO-KS; Los Angeles-Long Beach-Anaheim, CA; Miami-Fort Lauderdale-West Palm Beach, FL; Milwaukee-Racine, WI; Minneapolis-St. Paul-Bloomington, MN-WI; New York-Newark-Jersey City, NY-NJ-PA; Philadelphia-Camden-Wilmington, PA-NJ-DE-MD; Phoenix-Mesa-Scottsdale, AZ; Pittsburgh, PA; Riverside-San Bernardino-Ontario, CA; San Diego-Carlsbad, CA; San Francisco-Oakland-Hayward, CA; Seattle-Tacoma-Bellevue, WA; St. Louis, MO-IL; Tampa-St. Petersburg-Clearwater, FL; Washington-Arlington-Alexandria, DC-VA-MD-WV; and Washington-Baltimore, DC-MD-VA-WV.

our state-level analysis. It provides evidence that the observed relationship between innovation and inflation is robust to an alternative method of measuring the housing component and different geographic scales of aggregation.

D.2 Evidence from the Eurozone

We next conduct a similar analysis by looking at a very different setting: the EU area. Countries in the Euro-zone have adopted a common currency and trade frequently with each other. Thus, interactions in these countries likely closely resemble those of U.S. states. Building on this economic parallel, we examine the relation between innovation and inflation in the euro zone.

We obtain CPI and population data for Eurozone countries from the World Bank, while patent data is sourced from the European Patent Office. Specifically, we use the PATSTAT dataset. We exclude patents that are either not granted or have missing publication dates. To associate a patent with a country, we rely on information about the patent applicants (assignees) and inventors. For patents with N co-applicants or co-inventors, we split them evenly, allocating $1/N$ of the patent to each applicant's (or inventor's) country. To account for patent quality, we adjust based on forward citations, with the top 20% of patents in forward citations each year classified as important patents. Our sample covers the period 1999–2017, allowing patents a five-year window for citation accumulation. To calculate the flow of innovation measure, we normalize the count of significant patents by the country's population.

It is important to note that not all current Eurozone member countries adopted the euro as their official currency at the same time. Many joined the Eurozone at its inception in 1999, while others entered in subsequent years in a staggered manner. For example, Germany, France, and Italy were among the countries that adopted the euro at its inception in 1999, whereas nations like Slovenia joined in 2007, Slovakia in 2009, and Lithuania in 2015.

For many late-joining countries, their currencies were de facto pegged to the euro even prior to their formal membership. Consequently, our sample includes all these countries from 1999 to 2017, with the exceptions of Latvia prior to 2005, and Lithuania and Malta in 1999. Specifically, our sample covers the following:

- Austria, Belgium, Croatia, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, Portugal, Slovakia, Slovenia, Spain: 1999–2017.
- Latvia: 2005–2017. Prior to 2005, Latvia's currency was not well aligned with the euro.
- Malta: 2000–2017. In 1999, Malta's currency was not sufficiently aligned with the euro
- Lithuania: 2000–2017. In 1999, Lithuania's currency was not sufficiently aligned with the euro.

We then examine the effects of innovation on inflation using the same specification as in (2)

$$\pi_{i,t,t+s} = \beta \ln(\text{innov}_{\{i\},t,t+s}) + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}. \quad (\text{D.1})$$

The dependent variable is equal to the growth in the (log) CPI level in country i from t to $t + s$, where $s \in \{3, 5\}$. As before, the vector of controls includes one lag of the (log) CPI level, country and year fixed effects. The independent variable $\text{innov}_{i,t,t+s}$ is the flow of innovation measure at the country i over the same horizon, constructed in the same way as the citation-based measure of Table 2.

Table A.8 shows that the coefficient β estimate is positive and statistically significant across different innovation measures and specifications. In magnitude, the local price level rises by 4 to 5 percentage points over a three-year horizon in response to a one-standard-deviation increase in regional innovation. The estimates using the five-year horizon are quantitatively similar.

D.3 Inflation and IPOs of Local Firms

New and young companies are an important source of job creation and economic growth. However, the wealth generated by these new firms is often illiquid, and becomes liquid only once these firms become public. As a result, the IPO event can be interpreted as a realized wealth shock to the owners of these firms.

Here, we use data from the SDC Platinum and examine whether using IPO as an alternative proxy of the displacement shock u produces comparable patterns to those we have observed earlier. First, we investigate the relationship between IPO intensity and the growth of income inequality. Second, we explore whether an increase in IPO activity leads to a rise in local price levels. To this end, we estimate a variant of our main specification 2:

$$\ln I_{i,t+s}^p - \ln I_{i,t}^p = \beta (\ln \text{IPO}_{\{i\},t,t+s}) + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t} \quad (\text{D.2})$$

where the dependent variable is the growth in top income share between year t and $t + s$. We focus on 5-year horizons. The vector of controls $\mathbf{X}$ includes the lagged level of income inequality, together with state and year fixed effects. $\ln(\text{IPO}_{\{i\},t,t+s})$ is the local IPO intensity at the vicinity of state i , calculated as the ratio of the total market value of IPOs to the total market value in the vicinity of a given state. When the independent variable is equal to 0, we use a similar procedure with patent-based measure: i.e., replacing it with the minimum value in the sample and adding a dummy equal to one if IPO is equal to 0, thereby preventing the removal of the observation from the data.

Our key coefficient of interest is β , which measures the relation of an increase in IPO in (the neighborhood) of state $\{i\}$ on growth in measured income inequality. Panel B of Table A.9 reports our estimates. The IPO based measures are positively and statistically significantly correlated

with growth in top income share. Further, both the magnitude and statistical significance increase toward the right tail of the income distribution. Focusing on the very top of the distribution, a one-standard deviation increase in our IPO measure is associated with a 0.03 log point increase in income share.

We next examine its relation to the local price level. We estimate a specification directly analogous to the one in equation (2):

$$\pi_{i,t,t+5} = \beta(\ln \text{IPO}_{\{i\},t,t+5}) + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t} \quad (\text{D.3})$$

The control variables include the past log price level and state, year fixed effects. Panel A of Table A.9 reports our estimates. The IPO based measures are positively and statistically significantly correlated with inflation. In terms of magnitude, one standard deviation increase in IPO intensity is associated with 0.01 log point increase in the local price level. In addition, the increase in the local price level is mostly driven by the non-tradable component of inflation.

In addition to the above analysis, we construct an alternative measure of the local-level IPO intensity, calculated as the total number of IPOs normalized by the total population at the vicinity of a given state. Using this alternative measure, we re-examine the specifications in (D.2) and (D.3). The estimates are reported in Table A.10. This alternative measure of IPO intensity shows a significant and positive correlation with both inflation and income inequality, albeit with different magnitudes.

D.4 Additional Robustness Checks

We examine whether our results are driven by a small number of states with high intensity of innovation. To this end, we excluded the five states with the highest innovation intensity, namely California, Illinois, Texas, Massachusetts, and New York, and re-estimated our baseline specification (2). The results in Table A.11 show that both the magnitude and statistical significance align with the baseline case.

We also examine the robustness of our results to alternative weighting methodologies. Table A.12 reports the results of re-estimating (2) with population-weighted regression, while Table A.13 shows the results of the GDP-weighted regression. We observe that the magnitude and statistical significance remain comparable to the baseline case. Therefore, a few highly innovative states do not excessively influence our results.

Appendix Tables and Figures

Table A.1: Innovation, Inflation and Unemployment

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	1.95** (0.88)	2.28** (0.85)				
Citations (top 20%)			2.68*** (0.91)	2.79*** (0.91)		
KPSS (top 20%)					1.97*** (0.71)	2.09** (0.76)
Unemployment		-1.93*** (0.68)		-1.16 (0.71)		-1.20* (0.68)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	663	663	951	951	951	951
Adj R2	0.735	0.749	0.788	0.794	0.791	0.796

Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. The vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+5}$ is the flow of innovation measured *only* in state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.2: Innovation, Inflation and Unemployment

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	3.42*** (1.21)	3.61*** (1.22)				
Citations (top 20%)			3.73*** (0.89)	3.94*** (0.91)		
KPSS (top 20%)					3.69*** (0.95)	3.74*** (0.99)
Unemployment		-2.03** (0.69)		-1.31* (0.60)		-1.15* (0.58)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	663	663	951	951	951	951
Adj R2	0.78	0.80	0.82	0.82	0.82	0.82

Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. The vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+5}$ is the flow of innovation measure at the *vicinity* of state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are clustered at both the state and year levels. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.3: R&D Tax Credits and Innovation Outcomes: First Stage Regressions

	KPST (top 20%)	Citations (top 20%)	KPSS (top 20%)
RDTC	0.29*** (0.04)	0.28*** (0.06)	0.21*** (0.04)
State FE	YES	YES	YES
Year FE	YES	YES	YES
K-P F-stat	56.7	34.8	36.3
Observations	647	938	923
Adj R2	0.54	0.45	0.52

Notes: Columns 1-3 in this table report first-stage estimates of a regression of innovation outcomes on the R&D tax credit: $\ln(\text{innov}_{\{i\},t,t+5}) = \beta_1 \text{RDTC}_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the flow of innovation at the *vicinity* of state i from t to $t+5$. Independent variable $\text{RDTC}_{\{i\},t}$ is the R&D tax credit in the vicinity of state i , $\{i\}$. Control variables include (log) price levels $\ln Y_{i,t}$. α_i is state-FE, γ_t is time-FE. See Appendix A for the definition of innovation measures. Kleibergen–Paap rk Wald F statistics are reported to test for weak instruments. Both dependent variables and independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.4: R&D Tax Credits and Local Wage Earnings

Growth in	Median Wage Earnings						Mean Wage Earnings					
	High Skill		Low Skill		High - Low		High Skill		Low Skill		High - Low	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
RDTC	0.76*	0.90**	-0.75*	-0.64	3.08**	3.22**	0.49	0.53	-0.64	-0.62	2.77	2.84
	(0.42)	(0.39)	(0.41)	(0.39)	(1.41)	(1.33)	(0.61)	(0.61)	(0.38)	(0.39)	(2.11)	(2.19)
Data Source	CPS	CPS	CPS	CPS	CPS	CPS	CPS	CPS	CPS	CPS	CPS	CPS
Unemployment		YES		YES		YES		YES		YES		YES
State FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	856	856	856	856	856	856	856	856	856	856	856	856
Adj R2	0.59	0.60	0.76	0.76	0.50	0.50	0.66	0.66	0.65	0.65	0.52	0.52

Notes: This table reports estimates from regressions of local wage-earnings growth on R&D tax credits. Each column reports coefficients (multiplied by 100) from equation (4): $g_{i,t,t+5} = \beta_1 RDTC_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$, where dependent variable $g_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$ are five-year growth rates of median (Columns 1-6) and mean (Columns 7-12) wage earnings. Columns 1-2 and 7-8 report results for high-skilled workers; Columns 3-4 and 9-10 report results for low-skilled workers; Columns 5-6 and 11-12 report the difference between high- and low-skilled workers (High - Low). High-skilled workers are defined as those with at least a four-year college degree. Data is from the Current Population Survey (CPS). The independent variable *RDTC* is the R&D tax credit, standardized to unit standard deviation using unconditional moments. Control variables include the lagged level of dependent variable, and state and year fixed effects; even-numbered columns additionally control for the state unemployment rate. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.5: R&D Tax Credits and Local Wage Earnings (Alternative Sources)

	Median Wage Earnings				Mean Wage Earnings			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RDTC	-1.63** (0.64)	-1.61** (0.66)	-0.53*** (0.16)	0.07 (0.10)	-0.32 (0.68)	-0.44 (0.62)	-0.87** (0.28)	-0.47** (0.20)
Data Source	PSID Full	PSID Full	BLS	BLS	PSID Full	PSID Full	BLS	BLS
Unemployment		YES		YES		YES		YES
State FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	951	951	352	352	951	951	352	352
Adj R2	0.66	0.66	0.71	0.76	0.60	0.60	0.80	0.82

Notes: This table reports the regression coefficients (times 100) of (4): $g_{i,t,t+5} = \beta_1 RDTC_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$, where the dependent variable $g_{i,t,t+5}$ is the growth in nominal wages in state i from t to $t + 5$. **Median Wage:** Columns 1–4 report growth in the median wage. Columns 1–2 use data from the Panel Study of Income Dynamics (PSID) Full Sample. Columns 3–4 use data from the Bureau of Labor Statistics (BLS OEWS). **Mean Wage:** Columns 5–8 report growth in the mean wage. Columns 5–6 use data from the PSID Full Sample. Columns 7–8 use data from the BLS OEWS. The regressor $RDTC_{\{i\},t}$ is the R&D tax credit rate in the *vicinity* of state i . The vector of controls $\mathbf{X}_{i,t}$ includes the (log) level of the dependent variable at time t and, in even-numbered columns, the unemployment rate at time t . All specifications include state and year fixed effects. Because the PSID becomes biennial after 1998, constructing wage growth over 5-year horizons requires linearly interpolating missing state-level observations between adjacent survey years. All independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.6: R&D Tax Credits and Individual Earnings Growth

	Wage Earnings (Median)		Wage Earnings (Mean)	
	(1)	(2)	(3)	(4)
RDTC	0.27 (0.43)	0.28 (0.44)	0.00 (0.67)	0.02 (0.66)
Data Source	PSID	PSID	PSID	PSID
Unemployment		YES		YES
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	951	951	951	951
Adj R2	0.26	0.26	0.31	0.31

Notes: This table reports regression estimates (multiplied by 100) of individual wage-earnings growth on R&D tax credits from (4): $g_{i,t,t+5} = \beta_1 RDTC_{\{i\},t} + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. The dependent variable $g_{i,t,t+5}$ is constructed using PSID micro data as follows: for each individual, we compute five-year wage-earnings growth as $\ln(\text{wage}_{t+5}) - \ln(\text{wage}_t)$; we then aggregate these individual growth rates to the state-by-year level by taking the median in Columns 1–2 and the mean in Columns 3–4. The regressor $RDTC_{\{i\},t}$ is the R&D tax credit rate in the *vicinity* of state i . The control vector $\mathbf{X}_{i,t}$ includes the (log) level of lagged dependent variable and, in even-numbered columns, the state unemployment rate at time t . All specifications include state and year fixed effects. Because the PSID becomes biennial after 1998, constructing five-year horizons requires linearly interpolating wage earnings between adjacent survey waves. All independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and a small-sample adjustment. The sample period is 1978–2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.7: Innovation and Inflation, MSA Level

Dep. Variable: Headline Inflation (5yr)	(1)	(2)	(3)	(4)	(5)
KPST (top 20%)	0.93*** (0.11)				
Citations (top 20%)		1.23*** (0.24)			
KPSS (top 20%)			2.11*** (0.26)		
$\overline{RDTC}$				0.70** (0.29)	
$\overline{RDTC}_{GDP}$					0.70** (0.28)
MSA FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Observations	528	849	849	849	849
R-squared	0.79	0.81	0.83	0.80	0.80

Notes: Columns 1-3 of this table report the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$; Columns 4-5 report the reduced form with RDTC measures. The dependent variable is equal to the growth in the (log) price level in MSA i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. As before, the vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+5}$ is the flow of innovation measure at the MSA i . α_i is MSA-FE, γ_t is time-FE. See Appendix A for the definition of innovation measures. Columns 4-5 use the R&D tax credits for MSA i : in column 4, $\overline{RDTC}$ is the simple average of R&D tax credit rate of state j that overlap with MSA i ; in column 5, $\overline{RDTC}_{GDP}$ is the GDP weighted average of R&D tax credit rate of state j that overlaps with MSA i , using lagged GDP as weight. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1983–2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.8: Innovation and Inflation in Eurozone

Dep. Variable: Headline Inflation	3-Year		5-Year	
Citations (top 20%) (Assignee Location)	4.91***		7.73***	
	(1.57)		(2.08)	
Citations (top 20%) (Inventor Location)		3.90**		7.62***
		(1.40)		(2.00)
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	332	332	292	292
Adj R ²	0.77	0.75	0.84	0.83

Notes: This table reports the regression coefficients (times 100) of (2) in the euro-zone sample: $\pi_{i,t,t+s} = \beta_1 \ln(\text{innov}_{\{i\},t,t+s}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) CPI level in country i from t to $t+s$, $\pi_{i,t,t+s} = \ln Y_{i,t+s} - \ln Y_{i,t}$, for $s = 3, 5$. As before, the vector of controls $\mathbf{X}_{i,t}$ includes the (log) CPI level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+s}$ is the flow of innovation measure in country i . α_i is country-FE, γ_t is time-FE. See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1999-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.9: IPO, Inequality and Inflation

Panel A. Market Value of IPO and Inflation						
	Headline		T Inflation		NT Inflation	
IPO (mv)	1.12*** (0.41)	1.14*** (0.40)	-0.05 (0.18)	-0.04 (0.18)	1.78*** (0.64)	1.82*** (0.63)
Unemployment	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951
Adj R2	0.77	0.77	0.80	0.80	0.75	0.76

Panel B. Market Value of IPO and Inequality						
	Top 1%		Top 0.1%		Top 0.01%	
IPO (mv)	1.36** (0.61)	1.38** (0.60)	1.73 (1.08)	1.78 (1.05)	2.55 (1.61)	2.65* (1.56)
Unemployment	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951
Adj R2	0.81	0.81	0.80	0.80	0.79	0.79

Notes: Panel A of the table reports the regression coefficients (times 100) of (D.3): $\pi_{i,t,t+5} = \beta_1 \ln(\text{IPO}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$, where Y is the price level. Control variables $\mathbf{X}_{i,t}$ include (log) price level $\ln(Y_{i,t})$. $\text{IPO}_{\{i\},t,t+5}$ is the total market value of IPOs normalized by the total market of the vicinity of state i from t to $t+5$. α_i is state-FE, γ_t is time-FE. Unemployment is at t . Panel B re-estimates the above specification with the dependent variable being the inequality growth between t and $t+5$. In all specifications, the independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.10: IPO, Inequality and Inflation

Panel A. Number of IPO and Inflation						
	Headline		T Inflation		NT Inflation	
IPO (pc)	2.61*** (0.94)	2.50*** (0.91)	-0.10 (0.26)	-0.15 (0.23)	4.47*** (1.44)	4.31*** (1.39)
Unemployment	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951
Adj R2	0.77	0.78	0.80	0.80	0.76	0.76

Panel B. Number of IPO and Inequality						
	Top 1%		Top 0.1%		Top 0.01%	
IPO (pc)	2.59** (1.21)	2.52** (1.19)	4.25* (2.17)	4.12* (2.23)	6.70** (2.89)	6.39** (2.98)
Unemployment	NO	YES	NO	YES	NO	YES
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	951	951	951	951	951	951
Adj R2	0.81	0.81	0.81	0.81	0.79	0.79

Notes: Panel A of the table reports the regression coefficients (times 100) of (D.3): $\pi_{i,t,t+5} = \beta_1 \ln(\text{IPO}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$, where Y is the price level. Control variables $\mathbf{X}_{i,t}$ include (log) price level $\ln(Y_{i,t})$. $\text{IPO}_{\{i\},t,t+5}$ is the number of IPOs normalized by the total population of the vicinity of state i from t to $t+5$. α_i is state-FE, γ_t is time-FE. Unemployment is at t . Panel B re-estimates the above specification with dependent variable being the inequality growth between t and $t+5$. In all specifications, the independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.11: Innovation and Inflation, Excluding Top Innovative States

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	3.77*** (0.87)	3.79*** (0.90)				
Citations (top 20%)			3.99*** (0.72)	4.05*** (0.71)		
KPSS (top 20%)					3.19*** (0.94)	3.19*** (0.96)
Unemployment		-1.51** (0.67)		-0.90 (0.56)		-0.87 (0.53)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	543	543	786	786	786	786
Adj R2	0.717	0.726	0.776	0.779	0.760	0.763

Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. As before, the vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+s}$ is the flow of innovation measure at the *vicinity* of state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample excludes the top five states with the highly cited patents: California, New York, Illinois, Texas and Massachusetts. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.12: Innovation and Inflation, Population Weighted

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	3.43*** (0.90)	4.41*** (0.86)				
Citations (top 20%)			3.67*** (0.53)	4.45*** (0.67)		
KPSS (top 20%)					4.81*** (0.80)	5.22*** (0.93)
Unemployment		-2.78*** (0.92)		-1.94** (0.91)		-1.54* (0.85)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	663	663	951	951	951	951
Adj R2	0.781	0.803	0.842	0.850	0.847	0.853

Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. As before, the vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+s}$ is the flow of innovation measure at the *vicinity* of state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Regressions are weighted by state population at time t . Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.13: Innovation and Inflation, GDP weighted

Dep. Variable: Inflation (5-Year)	(1)	(2)	(3)	(4)	(5)	(6)
KPST (top 20%)	3.72*** (0.91)	4.66*** (0.84)				
Citations (top 20%)			3.79*** (0.55)	4.61*** (0.78)		
KPSS (top 20%)					4.92*** (0.84)	5.34*** (1.00)
Unemployment		-2.84** (1.05)		-1.89* (1.04)		-1.51 (0.95)
State FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	663	663	951	951	951	951
Adj R2	0.759	0.782	0.760	0.839	0.837	0.846

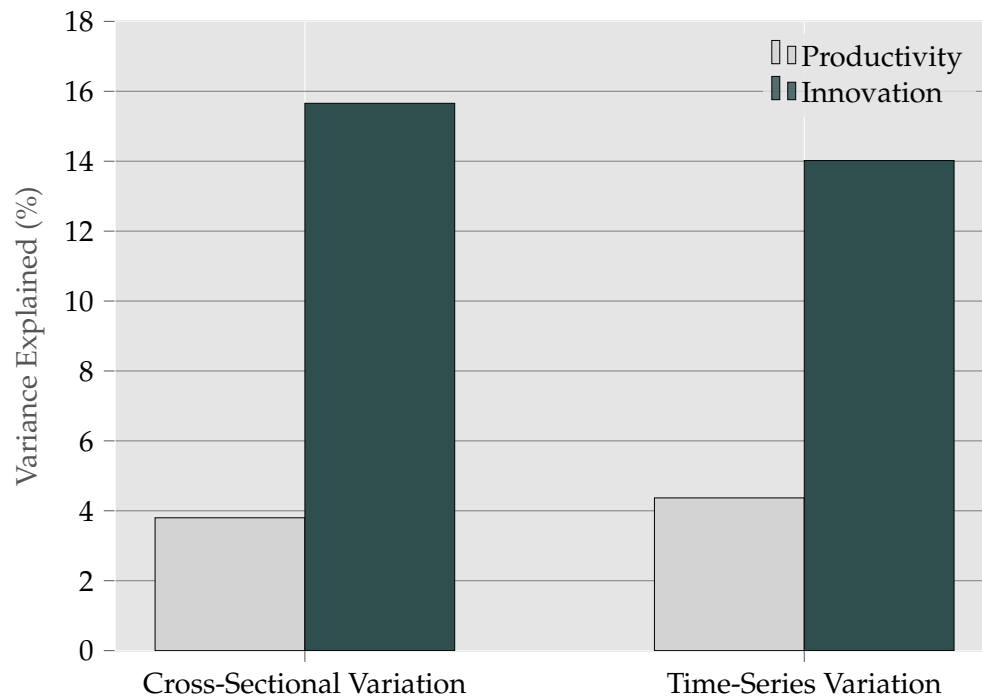
Notes: This table reports the regression coefficients (times 100) of (2): $\pi_{i,t,t+5} = \beta_1 \ln(\text{innov}_{\{i\},t,t+5}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Dependent variable is equal to the growth in the (log) price level in state i from t to $t+5$, $\pi_{i,t,t+5} = \ln Y_{i,t+5} - \ln Y_{i,t}$. As before, the vector of controls $\mathbf{X}_{i,t}$ includes the (log) price level at time t , $\ln Y_{i,t}$. $\text{innov}_{\{i\},t,t+s}$ is the flow of innovation measure at the *vicinity* of state i . α_i is state-FE, γ_t is time-FE. Unemployment rate is at t . See Appendix A for the definition of innovation measures. Regressions are weighted by state GDP at time t . Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following Driscoll and Kraay (1998) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.14: Innovation and Income Inequality

Dep. Variable: Top Income Share	Top 10%	Top 1%	Top 0.1%	Top 0.01%
KPST (top 20%)	2.08*** (0.54)	4.23*** (1.04)	6.52*** (1.63)	8.64*** (2.21)
Citations (top 20%)	2.00*** (0.52)	3.30*** (0.91)	4.94*** (1.42)	6.13*** (2.02)
KPSS (top 20%)	1.48*** (0.53)	3.21*** (0.93)	5.17*** (1.54)	7.10*** (2.22)
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

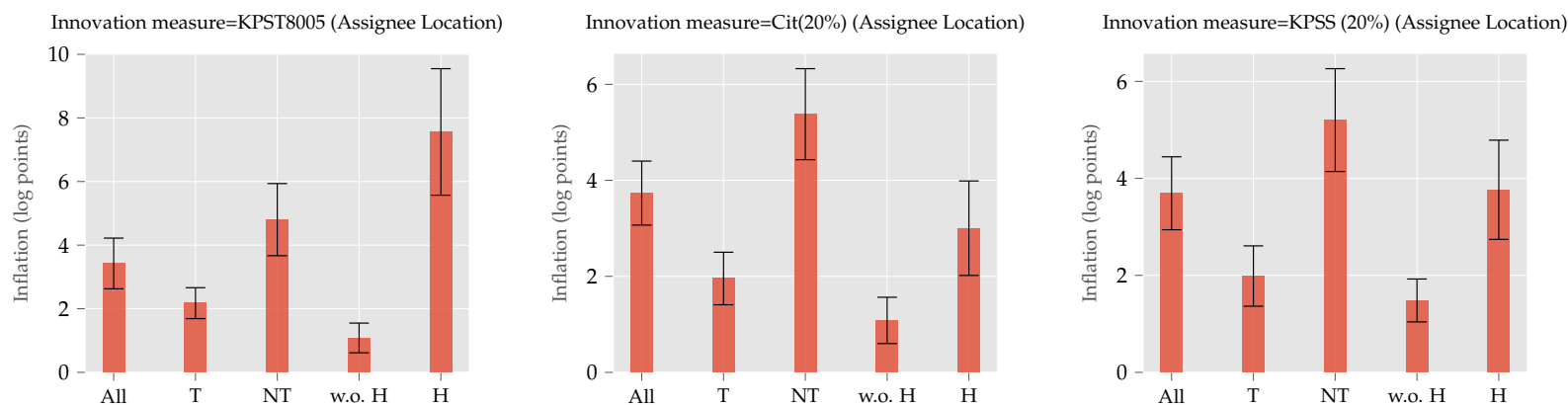
Notes: This table reports the regression coefficients (times 100) of: $\ln(I_{i,t}) = \beta_1 \ln(\text{innov}_{\{i\},t-2}) + \beta_2 \mathbf{X}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}$. Each row of the table corresponds to a different regression where the measure of innovation varies. Here, we closely follow the specification of [Aghion et al. \(2018\)](#) where $I_{i,t}$ is the top income share of state i at time t . Control variables $\mathbf{X}_{i,t}$ include GDP per capita growth, unemployment and population growth. $\text{innov}_{\{i\},t-2}$ is the lagged flow of innovation measure at the *vicinity* of state i . The specification controls for state and year fixed effects. See Appendix A for the definition of innovation measures. Independent variables are standardized to unit standard deviation using unconditional moments. Standard errors in parentheses are robust to serial correlation and cross-sectional dependence following [Driscoll and Kraay \(1998\)](#) with automatic lag length selection and small sample adjustment. The sample period is 1978-2017. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Figure A.1: Variance Explained: Innovation vs. Productivity



Notes: This figure compares the explanatory power (R^2) of standard productivity measures (gray bars) versus the innovation mechanism (dark bars) in accounting for state-level inflation. The “Cross-Sectional” group reports the variance explained in regressions of long-run state averages. The “Time-Series” group captures purely within-state fluctuations, where all variables are first residualized with respect to state and year fixed effects (Frisch-Waugh-Lovell) to remove permanent cross-sectional differences and common national trends.

Panel A. Components of Inflation and Innovation.



Panel B. Components of Inflation and Innovation (Relative Magnitude).

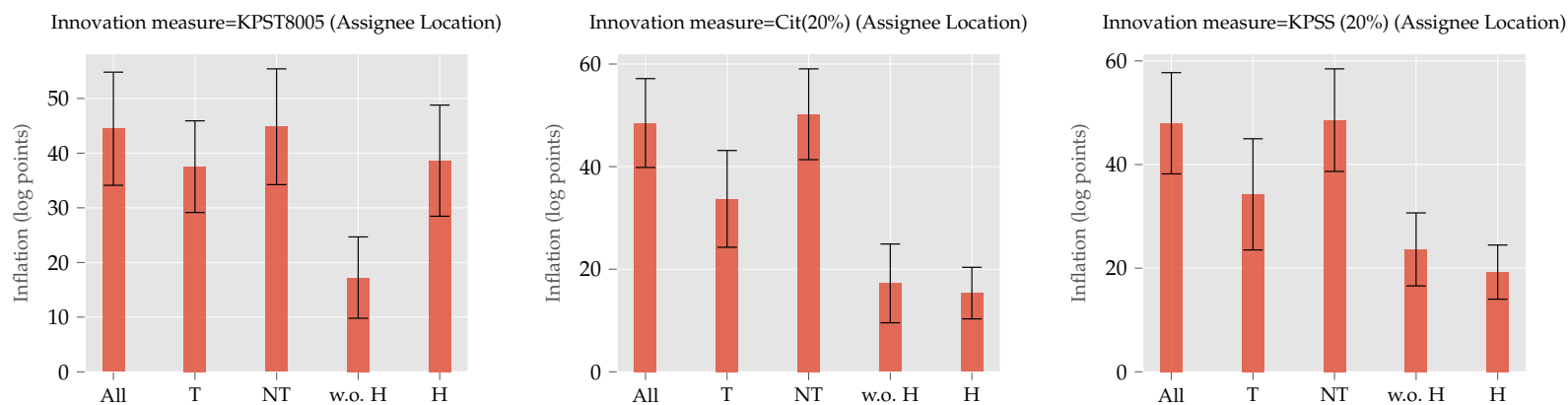


Figure A.2: Components of inflation and innovation, horizon = 5 years. Panel A of this figure reports the coefficient (times 100) of regression (2) where independent variables are standardized to unit standard deviation. Panel B reports the coefficient (times 100) of regression (2) where both dependent and independent variables are standardized to unit standard deviation. All: headline inflation; T: tradables; NT: non-tradables, w.o. H: headline excluding housing; H: housing. Coefficients are reported in 100 x log points.

